

The Link-Layer

- The IoT is made by heterogeneous links interconnected at the network (IP) layer
- Key link-layer features:
 - Framing
 - Addressing
 - Error checking
 - Length indication
 - Broadcast and unicast

HTTP, DNS, XMPP, MQTT, CoAp, TLS, SSL	Application layer
TCP, UDP	Transport Layer
IP(V4, V6), 6LowPAN	Internet Layer
Ethernet, 802.11 WiFi, 802.15.4	Link Layer
Electric, optical or RF transmission	Physical Layer

Three main approaches



Short-range multihop

- ZigBee
- Bluetooth low energy
- RFID



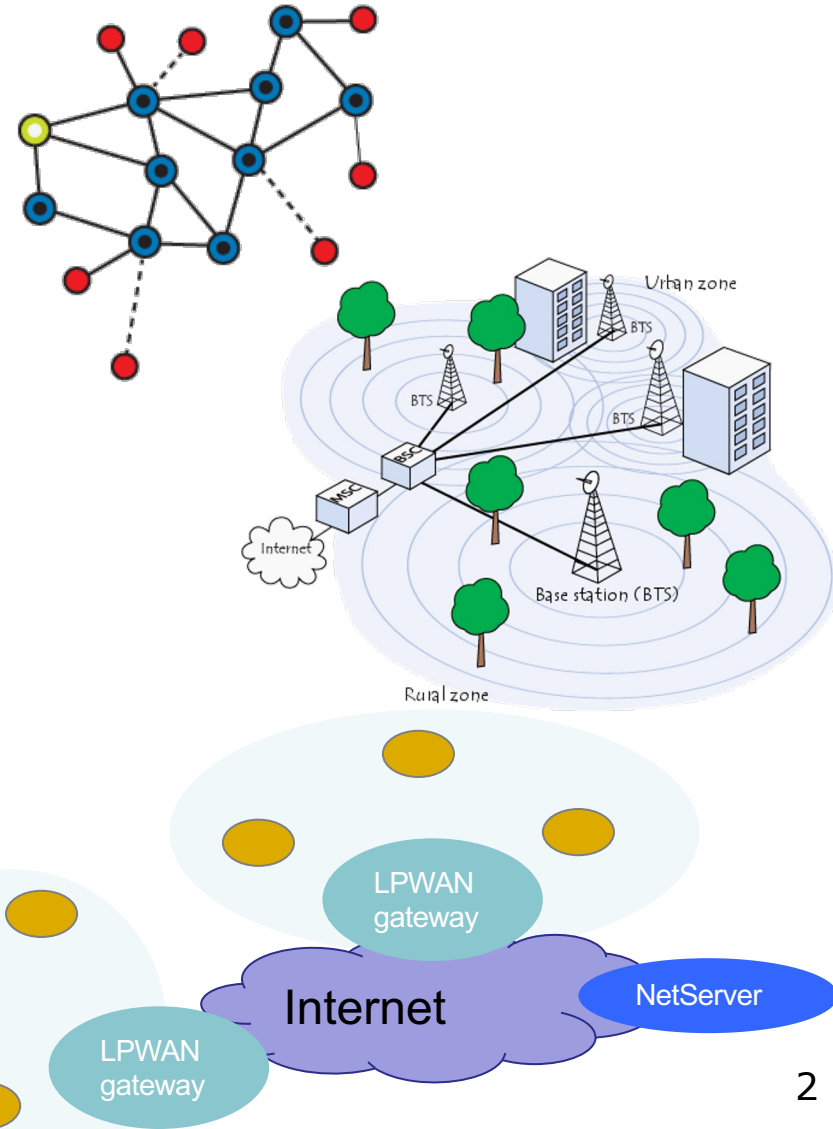
Cellular

- GSM
- LTE-A
- 5G

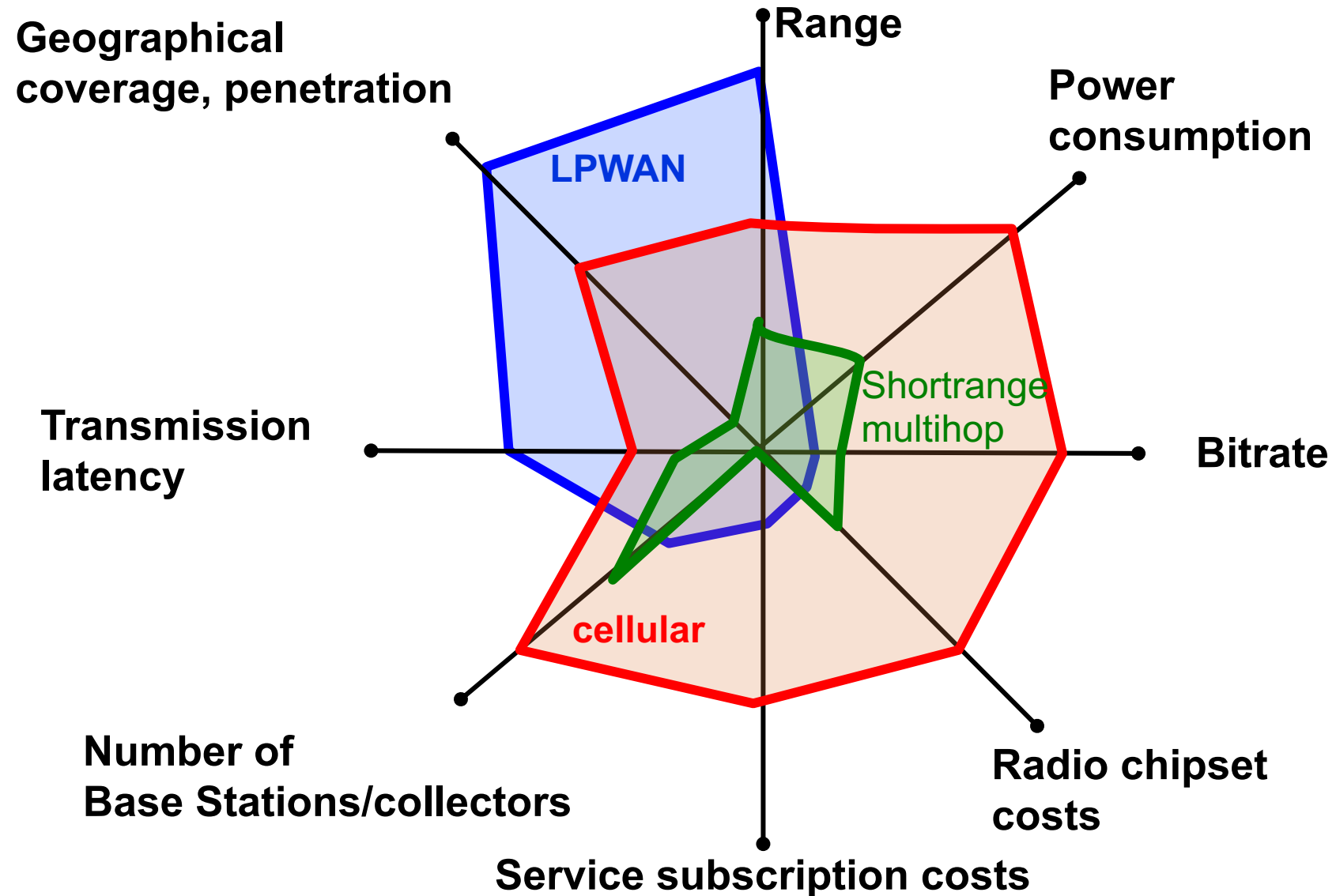


Low Power Wide Area Networks (LPWAN)

- SIGFOX
- LoRa

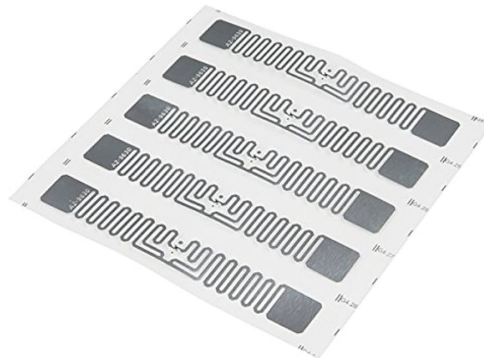


Quick comparison



Radio Frequency IDentification

- Radio Frequency IDentification, uses radio frequency communication to automatically identify, track and manage objects, people or animals
- It uses TAGS that are similar to a bar code or QRCode, but RFID does not rely on LOS
- A typical RFID tag consists of a microchip attached to a radio antenna (RX/TX) mounted on a substrate.
- The chip can store as much as 2 kilobytes
- Passive tags, are powered by the RF field generated by the reader.
- Active tags require their power source

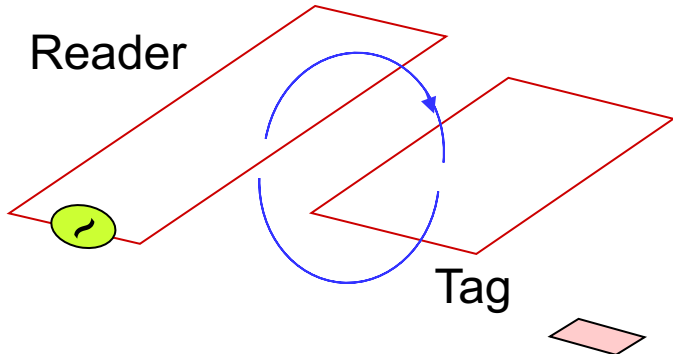


Passive Tags

- Passive RFID tags have no power source and no transmitter
- They are cheaper than active tags (20 cents to 50 cents) and require no maintenance
- Much shorter read range than active tags (few cms).
- They operate at:
 - Low frequency (LF): 124 kHz, 125 kHz or 135 kHz
 - High frequency (HF): 13.56 Mhz
 - Ultra-high frequency (UHF): 860 MHz to 960 MHz
 - Some systems also use 2.45 Ghz

Inductive Coupling

- Interaction between Reader and Tag is due to inductive coupling
 - production of electric current by passing a wire through a magnetic field
 - Devices have coils built into them: the coil in the reader antenna and the coil in the tag antenna form an electromagnetic field
 - The tag draws power from the field, uses the power to run the circuitry on the chip and then changes the electric load on the antenna.
 - The reader antenna senses the change in the magnetic field and converts these changes into the ones and zeros that computers understand



Field generated by Reader (**Cause**)

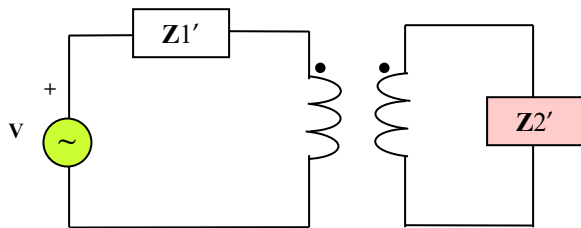
- Based on Biot and Savart's (Ampere's) Law

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad \mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{2\pi} \frac{\hat{\mathbf{I}} \times \hat{\mathbf{r}}}{|\mathbf{r}|}$$

Induced EMF in Tag (**Effect**)

- Faraday's Law

$$\frac{d\Phi_B}{dt} = \frac{d}{dt} \int_{\Sigma(t)} \mathbf{B}(t) \cdot d\mathbf{A} \quad \nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$



Circuit representation

Active Tags

- They require an internal power source and are woken up when they receive a signal from a reader --> conserve battery life.
- An active tag working as a beacon emits a signal with its unique identifier at pre-set intervals --> triangulation in location systems
- They generally cost from \$10 to \$50 (memory, battery life, sensors)
- They usually operate at 455 MHz, 2.45 GHz, or 5.8 GHz, read range of 20-100m



Readers

- They retrieve the data stored on an RFID tag.
- Three main functions: energizing, demodulating, decoding
 - They need to provide energy to the tag
 - The information sent by the tag must be demodulated (like an AM radio).
 - The encoded information is decoded by the reader's on-board microcontroller



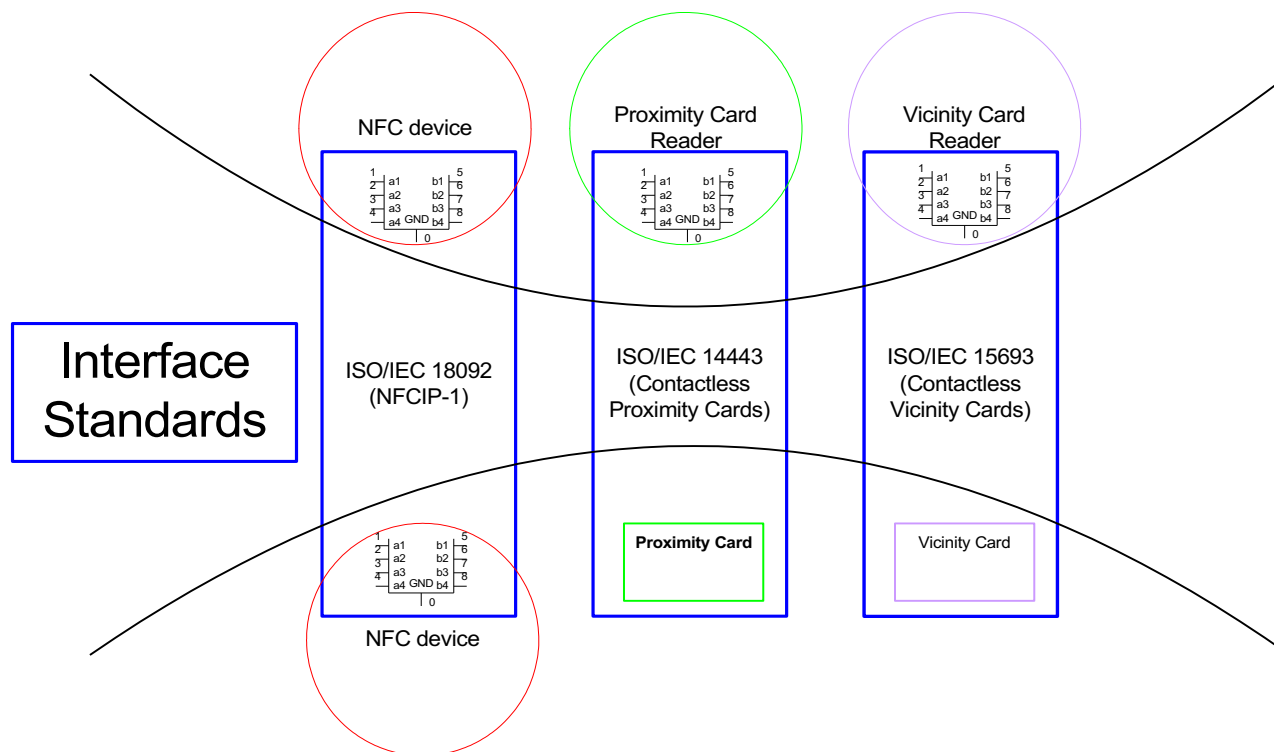
An example: Near Field Communication

Wireless short range communication technology

- NFC is designed for short distance wireless communication
- Allows intuitive initialization of wireless networks
- NFC is complementary to Bluetooth and 802.11 with their long distance capabilities
- NFC also works in dirty environment
- NFC does not require line of sight
- Easy and simple connection method
- Provides communication method to non-self powered devices
- NFC was approved as an ISO/IEC standard and later as an ECMA standard

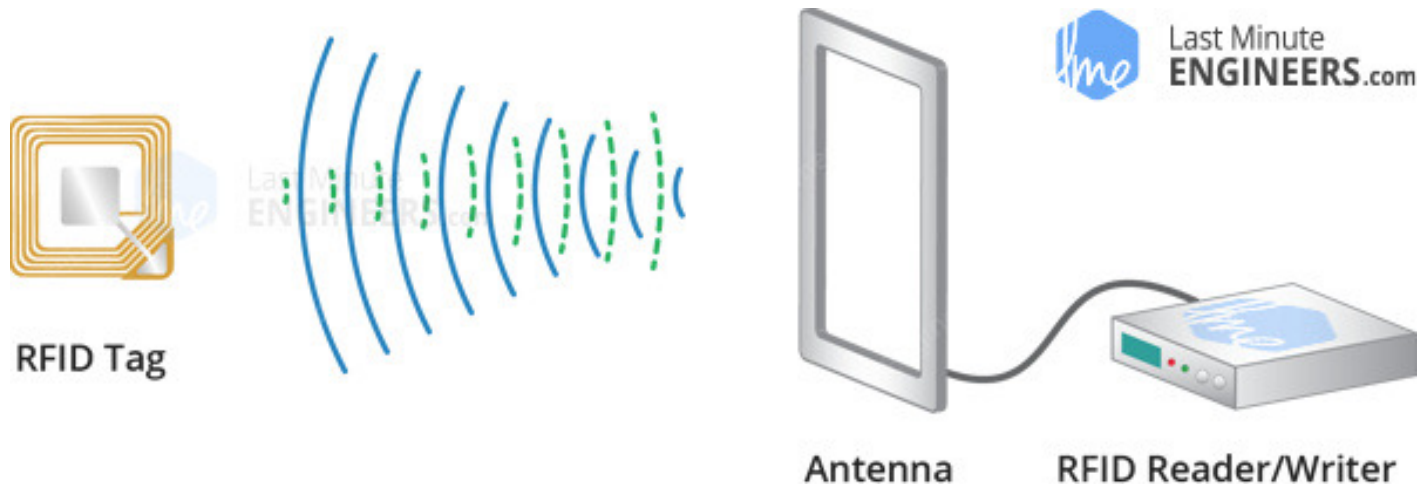
NFC Interface and protocols

- Gateway between
 - ECMA-340
 - ISO/IEC 14443 PCD mode
 - ISO/IEC 15693 VCD mode

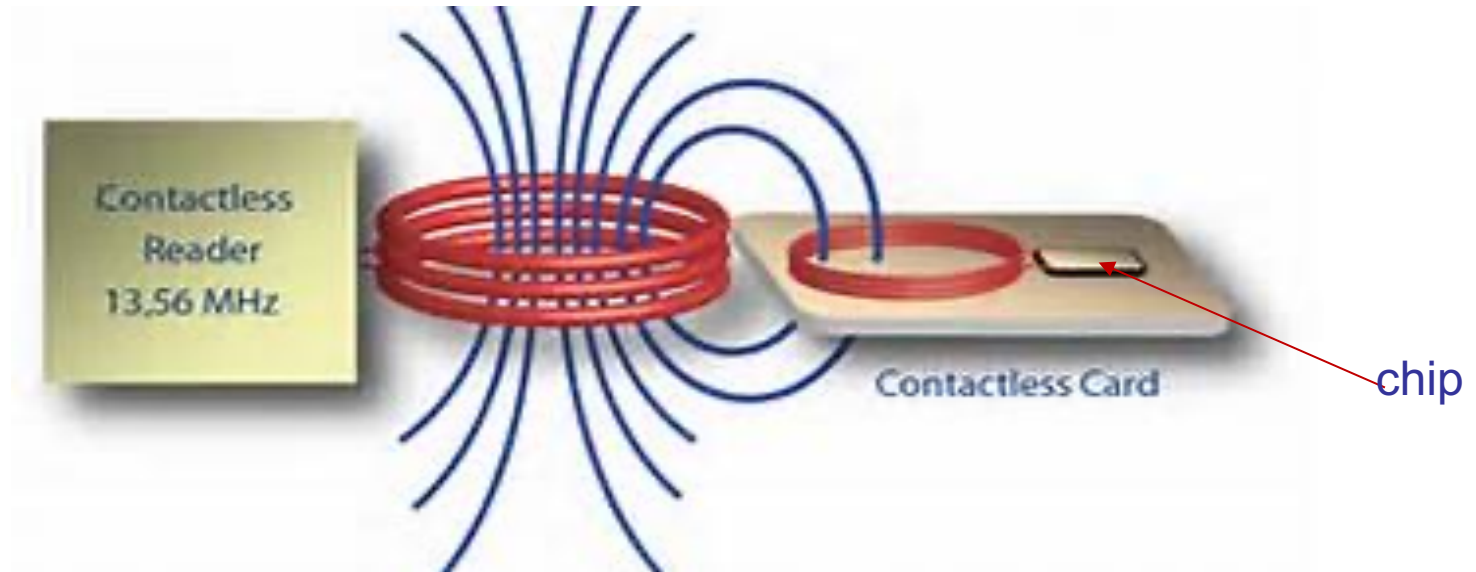


Near Field Communication

- Based on an inductive RF link on 13.56 MHz
- Active and passive NFC mode at 3 different transfer speeds (106, 212, 424 kbps)
- Operating distance typical up to 10 cm
- Compatible with today's field proven contactless RFID technology
- 3 Operating modes
 - Read-write
 - Peer to peer
 - Card emulation



NFC: functional principles



- Energy from Reader activates the chip inside the NFC Tag (tens of μW to few mW) according to inductive coupling
 - Tag and Reader are a few **centimeters** apart
- Chip generates talk-back signals once powered up
- Tag communicates above signals back to Reader

Peer to peer mode



- Peer-to-Peer mode is used to establish a two-way communication channel between a pair of NFC-enabled devices.
- Each NFC-enabled device serves as an endpoint, meaning the two systems can initiate a communication as equals, or peers.
- This mode uses either a passive or active communication scheme.

Read-Write mode



- In Read/Write mode, the system performs the functions of a contactless reader/writer.
- The system's NFC Integrated Circuit interacts with an NFC-enabled device – such as a contactless smartcard, an NFC tag, or an NFC-enabled smartphone and either reads data in from the device or writes data out to it.
- This mode is used to get information or initiate an action.

Card Emulation mode



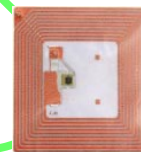
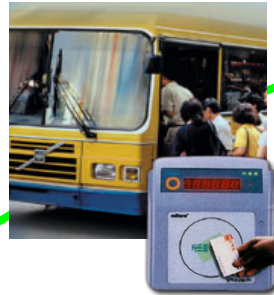
Card Emulation

- Card Emulation Mode lets the system behave as an ISO/IEC 14443-compliant contactless smartcard.
- This means that the NFC-enabled device can be used in the existing contactless card infrastructure, for things like ticketing, access control, transit, tollgates, and contactless payments.

Near Field Communication Applications

Smart Key

- for mobile local payment

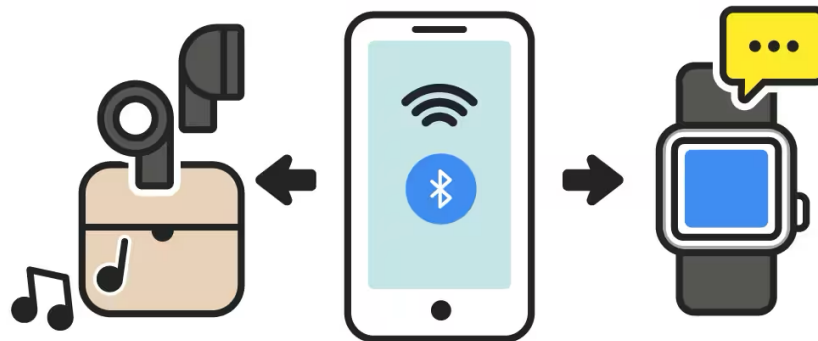


**Low cost solution to
distribute info / services**
-e.g. by passive loops
embedded in paper media

NFC enables a unique blend of valued applications

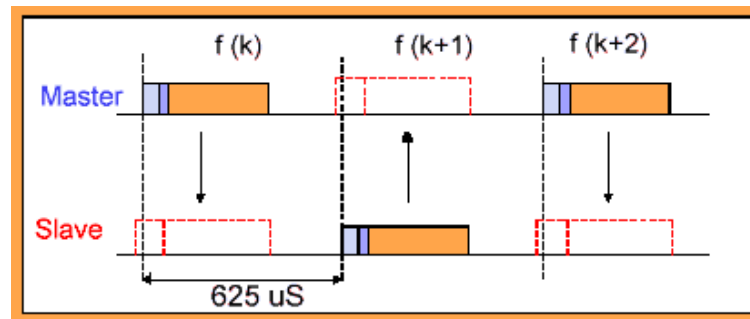
IEEE 802.15.1 (Bluetooth)

- A **cable-replacement** technology that can be used to connect almost any device to any other device
- Radio interface enabling electronic devices to communicate wirelessly via short range (10 meters) ad-hoc radio connections
- a standard for a **small , cheap radio chip to be plugged into computers, printers, mobile phones, etc**



What is Bluetooth?

- Uses **FHSS** with a radio range of **2.45 GHz** and **PSK** modulation
 - 1600 hops per second through 79 1MHz channels
 - Spreads Bluetooth traffic over the entire ISM band
 - All slaves follow the master for frequency hop sequence



- Theoretical maximum bandwidth is 1 Mb/s
- Several Bluetooth devices can form an ad hoc network called a “piconet”
 - In a piconet one device acts as a master (sets frequency hopping behavior) and the others as slaves

Bluetooth Architecture

– Piconet

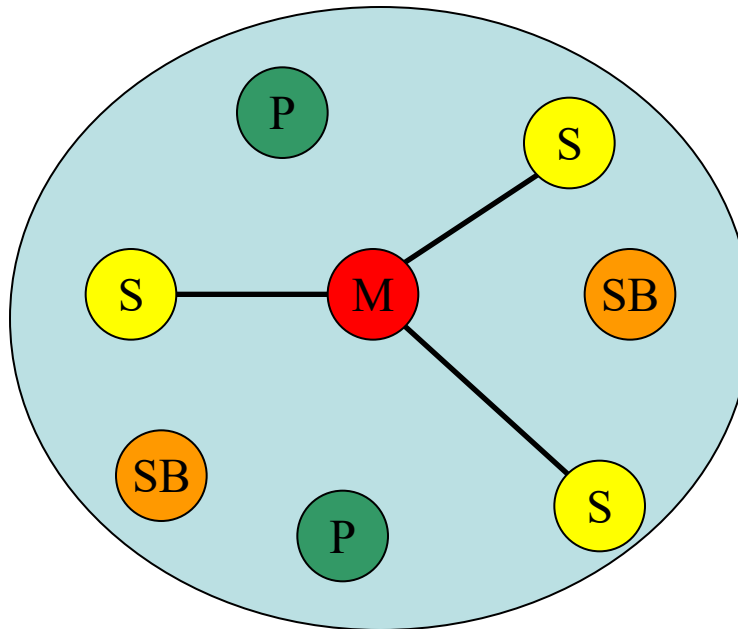
- Each piconet has one master and up to 7 simultaneous slaves
 - Master : device that initiates a data exchange.
 - Slave : device that responds to the master
- Piconet Addressing
 - Active Member Address (AMA, 3-bits)
 - Parked Member Address (PMA, 8-bits)

– Scatternet

- Linking of multiple piconets through master or slave devices
- Bluetooth devices have point-to-multipoint capability to engage in Scatternet communication.

Piconet

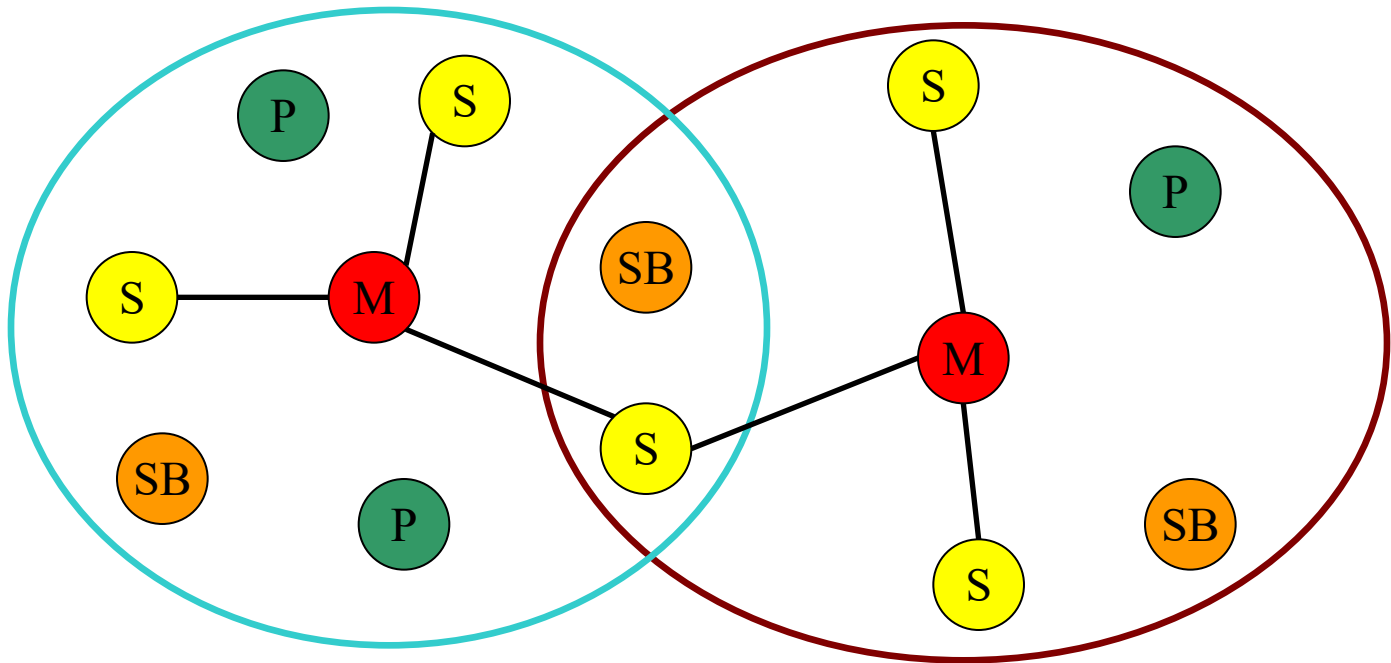
- All devices in a piconet hop together
 - Master gives slaves its clock and device ID
 - Master can connect to 7 simultaneous or 200+ inactive (parked) slaves
- Non-piconet devices are in standby
- Each piconet has maximum capacity (1 Mbps)
- Unique hopping pattern/ID



M=Master P=Parked
S=Slave SB=Standby

Scatternet

- Piconets can coexist in time and space
- Devices can be slave in one piconet and master of another



Physical links

- Between master and slave(s), different types of links can be established. Two link types :
 - **Synchronous Connection Oriented (SCO)**
 - Support symmetrical, circuit-switched, point-to-point connections
 - Typically used for voice traffic.
 - Data rate is 64 kbit/s
 - Retransmission is not necessary, since data is real-time
 - Up to 3 SCO links per piconet
 - **Asynchronous Connection-Less (ACL)**
 - Support symmetrical and asymmetrical, packet-switched, point-to-multipoint connections.
 - Typically used for data transmission .
 - Up to 433.9 kbit/s in symmetric or 723.2/57.6 kbit/s in asymmetric
 - One ACL link between a master and a slave

ACL Links vs. SCO Links

	Intended Traffic Type	Retransmission	Max # links between master and slave	Supported during hold mode	Switched connection type
ACL	Data	Yes	1	No	Packet
SCO	Time bounded info (Audio or Video)	No	3	Yes	Circuit

Bluetooth States

- **Standby**

- State in which Bluetooth device is inactive, radio not switched on, enable low power operation.

- **Page**

- Master enters page state and starts transmitting paging messages to Slave using earlier gained access code and timing information.

- **Page Scan**

- Device periodically enters page state to allow paging devices to establish connections.

- **Inquiry**

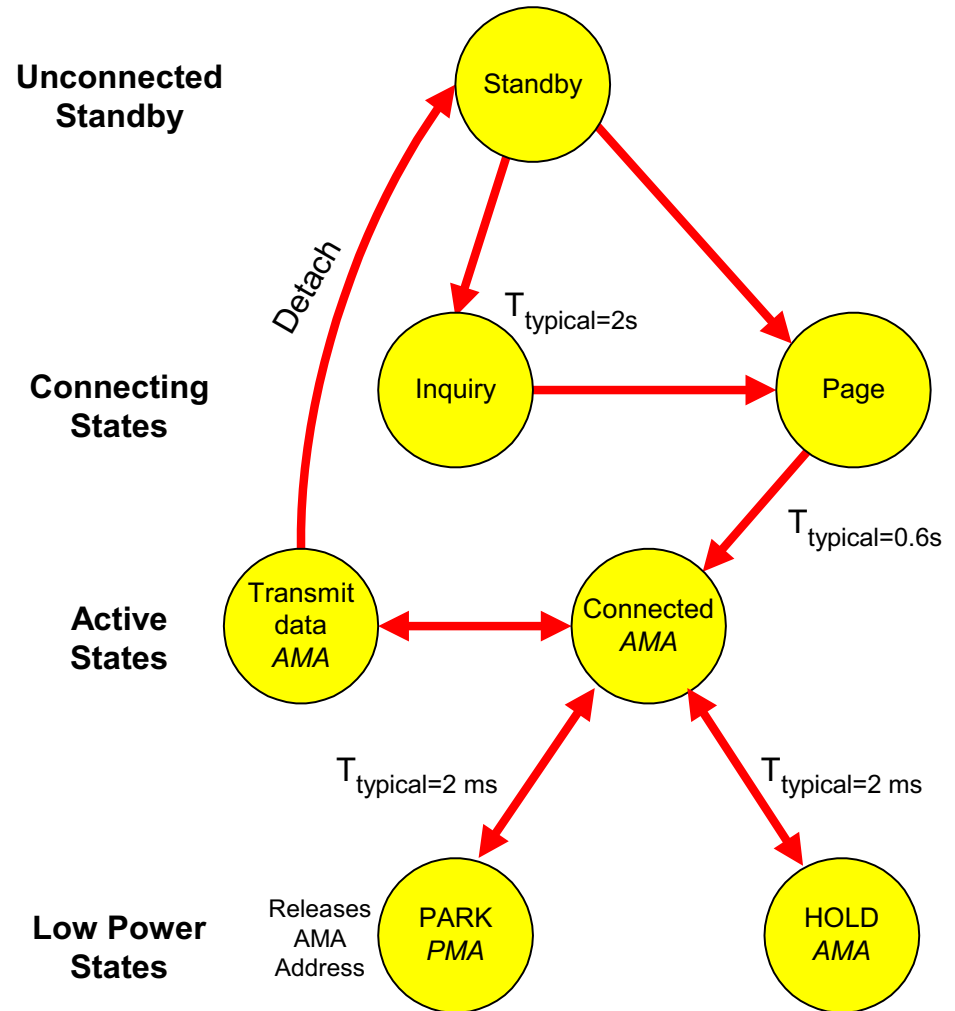
- State in which device tries to discover all Bluetooth enabled devices in the close vicinity.

- **Inquiry scan**

- Most devices periodically enter the inquiry scan state to make themselves available to inquiring devices.

Bluetooth States

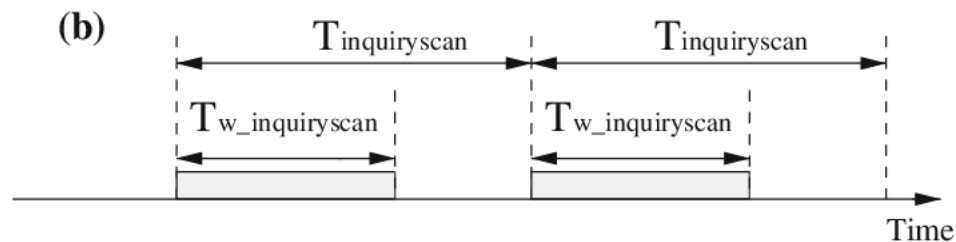
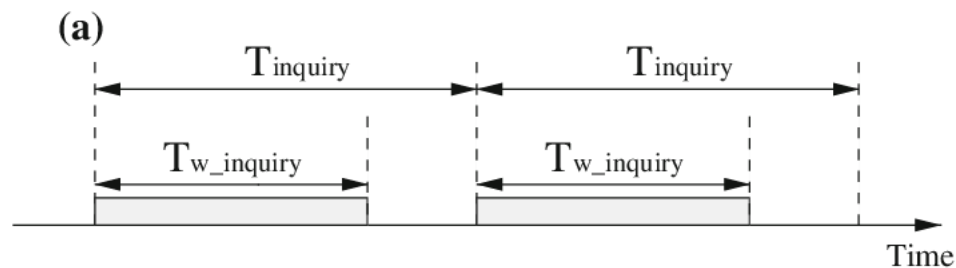
- **Transmit**
 - Sending data
- **Connected**
 - Actively on a piconet (master or slave)
- **Park/Hold**
 - Low Power connected states



Bluetooth: Hello, Anyone Around?

– Inquiry Procedure

- Sends an inquire request for nearby devices (within 10 m)
- Discoverable devices issue an inquiry response
- Uses 32 inquire channels to send out inquiry messages
- inquiry is broken up into 2 hop trains (16 freqs/channels)
- Receivers perform an **inquire scan** long and frequent enough for an inquiring device to send the inquire on 16 frequencies



Bluetooth: Hello, Anyone Around?

– Inquiry Scan

- A device periodically listens for inquiry packets at a single frequency – chosen out of 16 frequencies
- Stays in the state long enough for a inquiring device to cover 16 frequencies
- Will re-enter inquiry scan state even after responding to inquire

– Inquiry Response

- When radio receives inquire, it will wait between 0 and .32 seconds before responding with an FHS packet containing:
 - Device ID
 - Clock
- After inquiring radio is done with inquiring procedure, it knows all of the radios (that are discoverable) within range

Paging: Will you connect to me?

- Very similar to inquire
- Still have not synchronized clocks or frequencies
- Establishes actual Piconet connection with a device that it knows about
- Connection process involves a 6 steps of communication between the the master and the slave

Step	Message	Direction	Hopping Pattern	Pattern Source and Clock
1	Slave ID	Master to Slave	Page	Slave
2	Slave ID	Slave to Master	Page Response	Slave
3	FHS	Master to Slave	Page	Slave
4	Slave ID	Slave to Master	Page Response	Slave
5	1st Master Packet	Master to Slave	Channel	Master
6	1st Slave Packet	Slave to Master	Channel	Master

Paging Steps

- Step1:
 - Device broadcasts a page message out to the device that it wants to set up a connection with
 - similar to inquire messages (2 frequency trains of 16 frequencies)
 - Once the device receives a page response, it will stop paging and move on to step 2
- Step 2:
 - In the page response, an acknowledgement is sent back to the master containing the slave ID
- Step 3:
 - In the response, the frequency hopping generator is stopped and the master issues an FHS packet to the slave

Paging Steps

– Step4:

- The slave issues a final response transmission that is aligned to the slave's native clock
- Using the data from FHS packet, the slave calculates the master's frequency hopping pattern and synchronizes

– Step 5:

- When the master receives the packet, it jumps back to its frequency hopping pattern and assigns the slave an Active Member Address (AMA) for the piconet
- Master sends out a poll packet to ensure that the slave is on its frequency hopping pattern

– Step 6:

- Once the slave receives the poll packet, the slave replies with a packet to ensure that it is on the right channel
- At the conclusion of step 6, a new synchronized connection is established between master and slave

Service Discovery Protocol (SDP)

Idea:

- Traditional LANs: Find a connection to a printer (or other resource) and keep that connection for a long time
- Bluetooth: Walk into an area, find a printer (or other resource), use it, then walk away forgetting any details of the connection

Client-Server mode:

- SDP Server is any Bluetooth device that offers services to other Bluetooth device (ex. Bluetooth-enabled printer, etc.)
 - Each SDP Server maintains its own database (SDP Database) that contains information about the services that it offers
- SDP Client is any device using services offered by an SDP Server

SDP query:

- The client queries an SDP server to find out services available
- When the SDP server gets a query, it looks up the service that the client is requesting and returns information to the client on how to connect to the service

What is Bluetooth Low Energy?

- Traditional Bluetooth is connection oriented
- When a device is connected, a link is maintained, even if there is no data flowing.
- Low-energy modes (park, sniff) allow devices to sleep, reducing power consumption to give months of battery life
- Bluetooth low energy is a NEW, open, short range radio technology
 - Different to Bluetooth classic
 - Optimized for ultra low power
 - Enable coin cell battery use cases
 - $< 20\text{mA}$ peak current
 - $< 5\text{ uA}$ average current

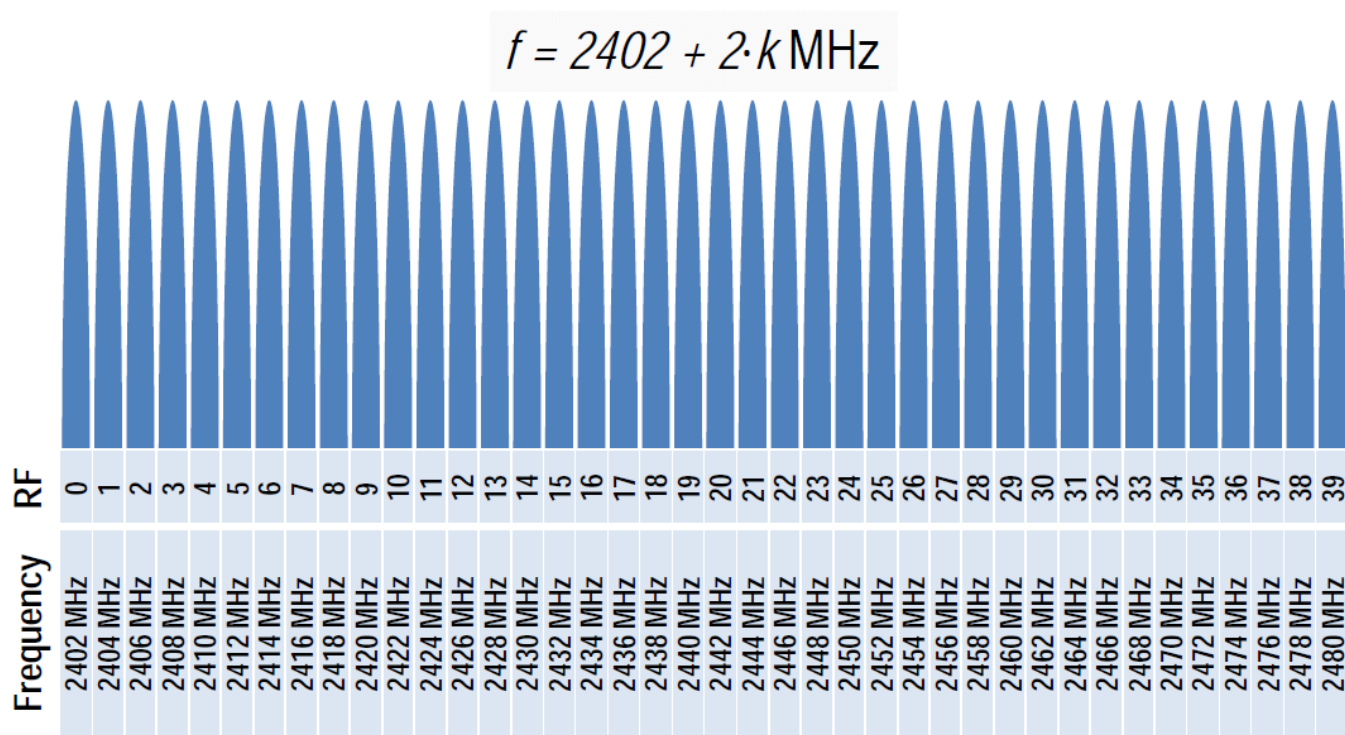


Basic Concepts of Bluetooth 4.0

- Everything is optimized for power consumption
 - Shorter packets to reduce TX peak current and RX time
 - Reduced number of RF channels to improve discovery and connection time
 - Simplified state machine
- Data Throughput
 - For Bluetooth low energy, data throughput is not a meaningful parameter.
 - It does not support streaming.
 - It has a data rate of 1Mbps, but it is not optimized for file transfer.
 - It is designed for sending small chunks of data

Physical Layer

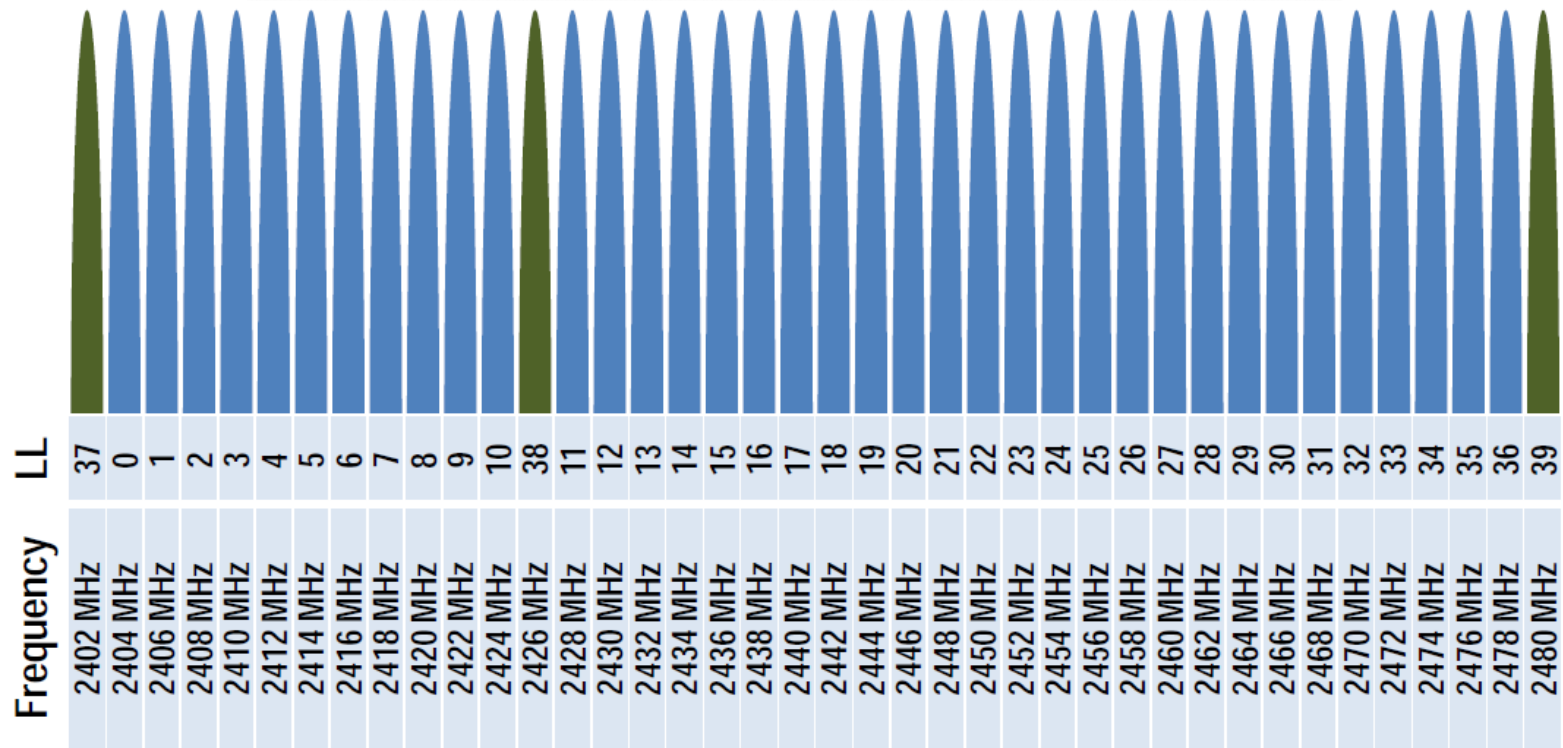
- 2.4 GHz ISM band
- 1Mbps GFSK (FSK with gaussian filtering) modulation
 - Larger modulation index than Bluetooth
- FHSS 40 Channels on 2 MHz spacing



Physical Channels

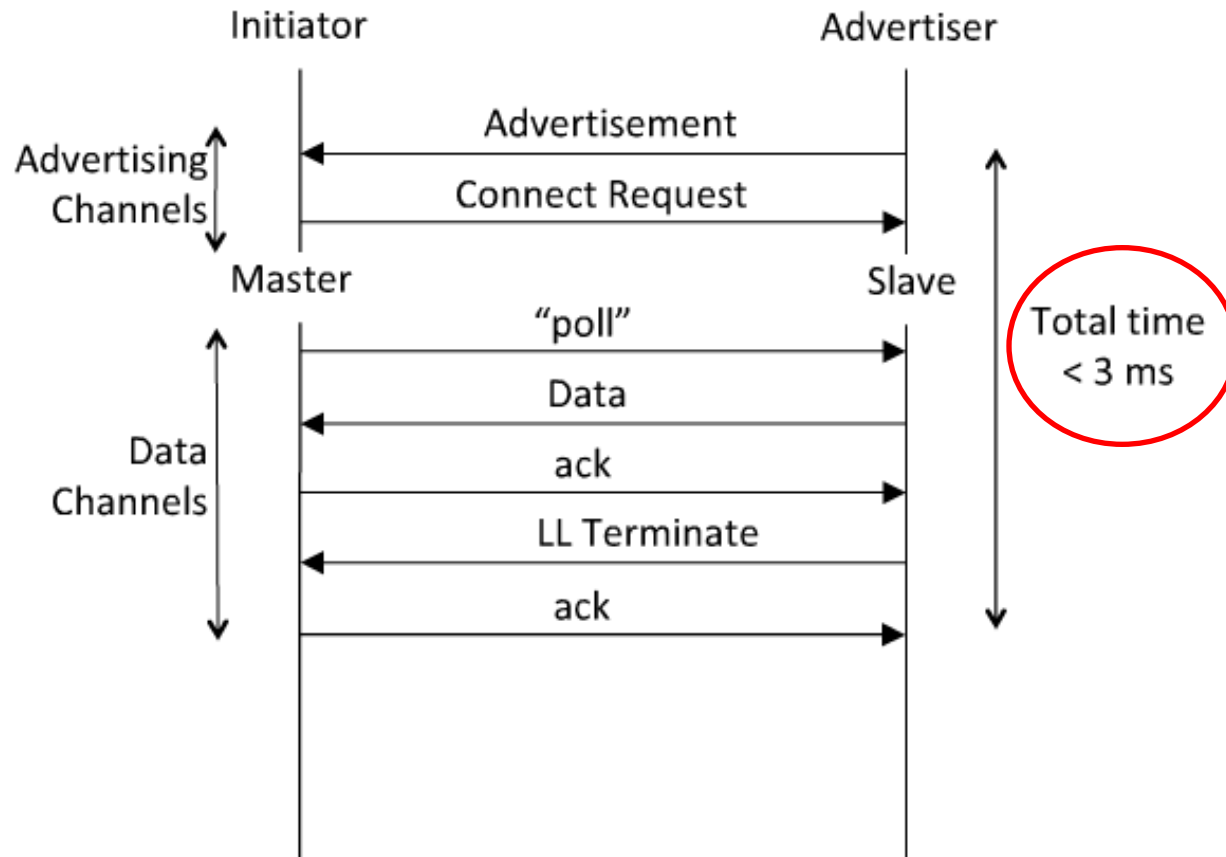
- Two types of channels
- Dedicated channels to perform advertisement

3 Advertising Channels and 37 Data Channels



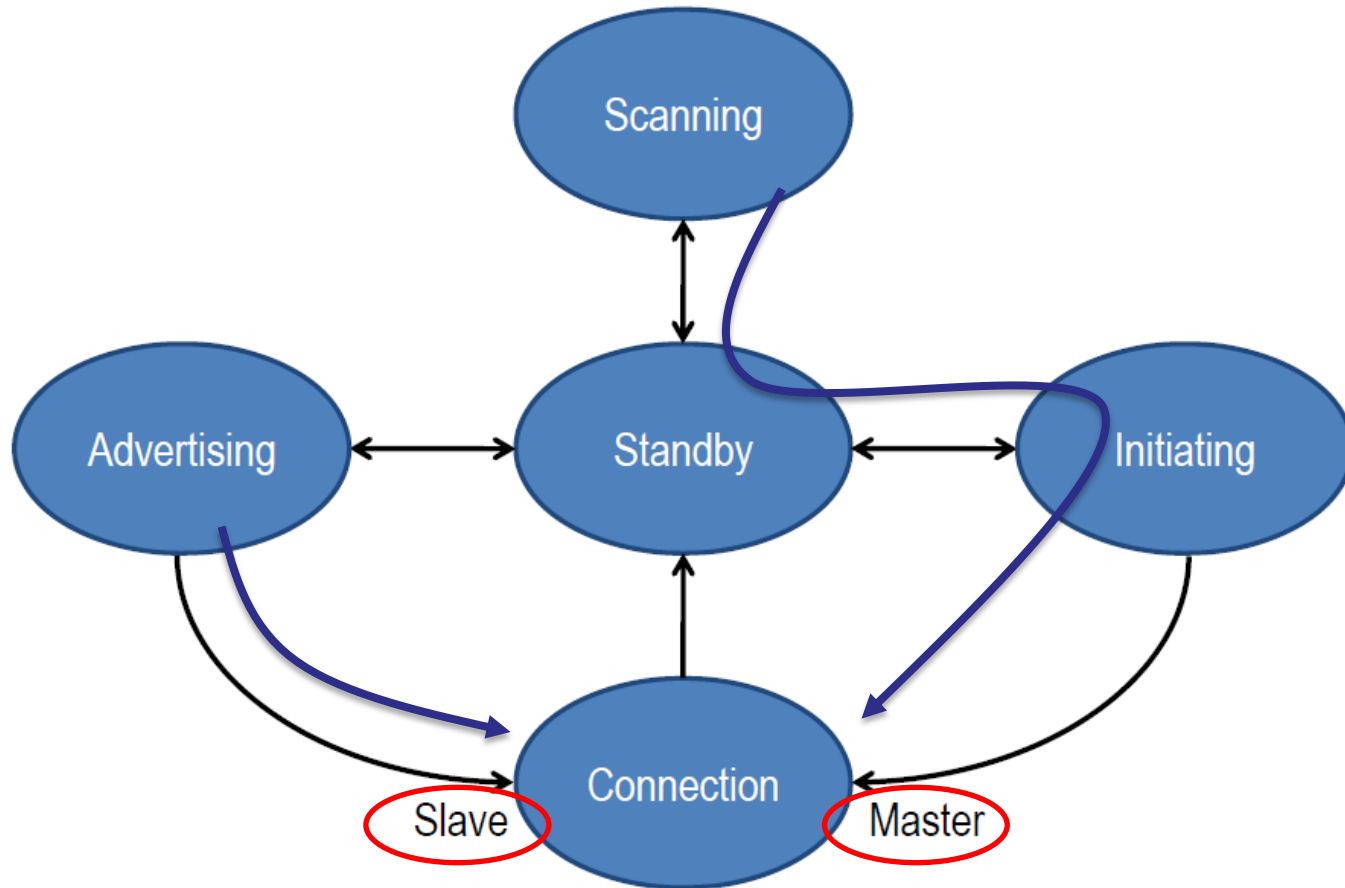
Simplified Connection via Advertisement

- Very low latency connection

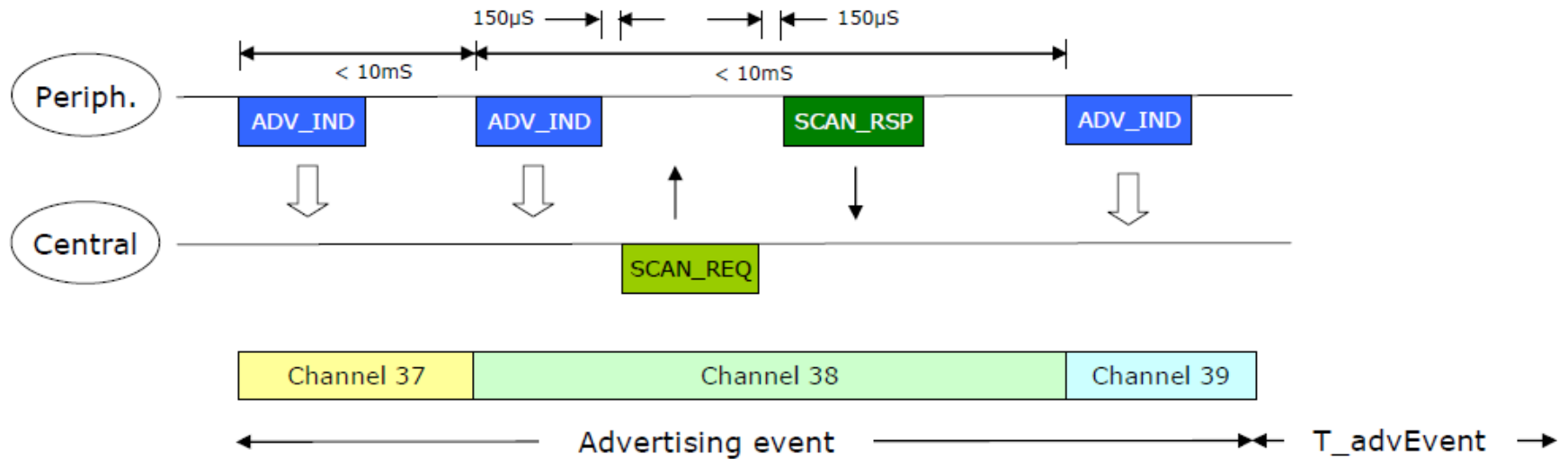


Simplified State Machine

- Reduced number of states

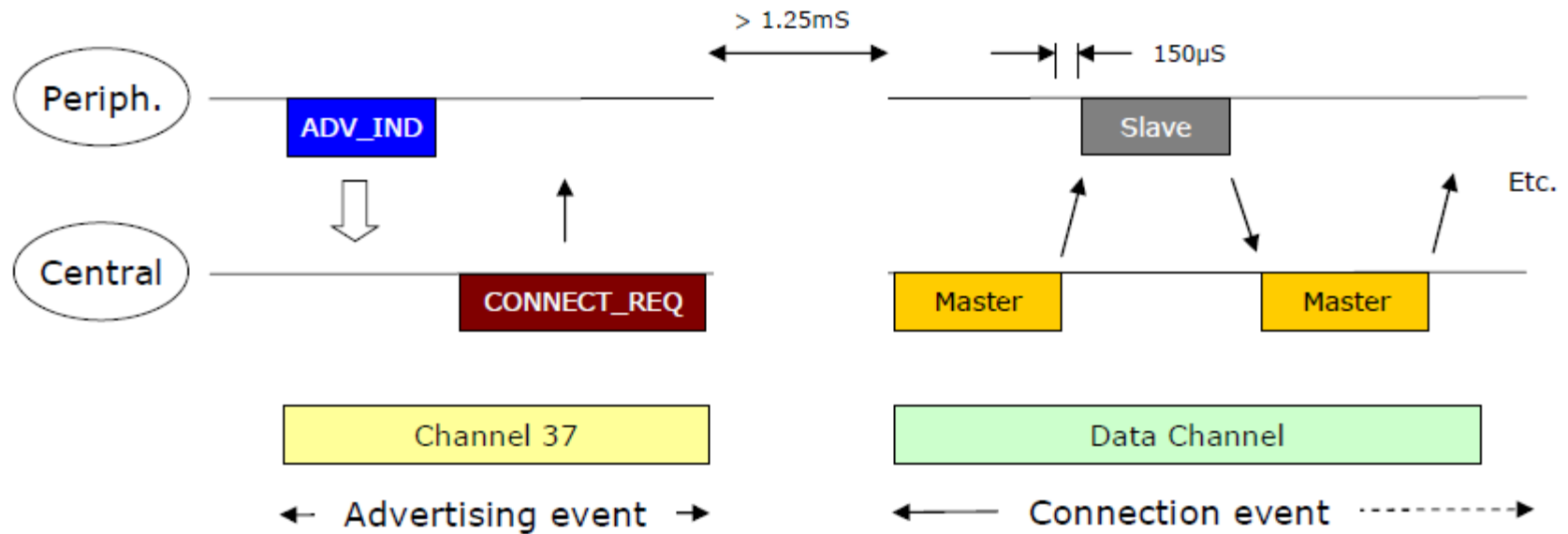


Advertising



- Devices can advertise for a variety of reasons:
 - To broadcast promiscuously
 - To transmit signed data to a previously bonded device
 - To advertise their presence to a device wanting to connect
 - To reconnect asynchronously due to a local event

Data transactions



- Once a connection is made:
 - Master informs slave of hopping sequence and when to wake
 - All subsequent transactions are performed in the 37 data channels
 - Both devices can go into deep sleep between transactions

How low can the energy get?

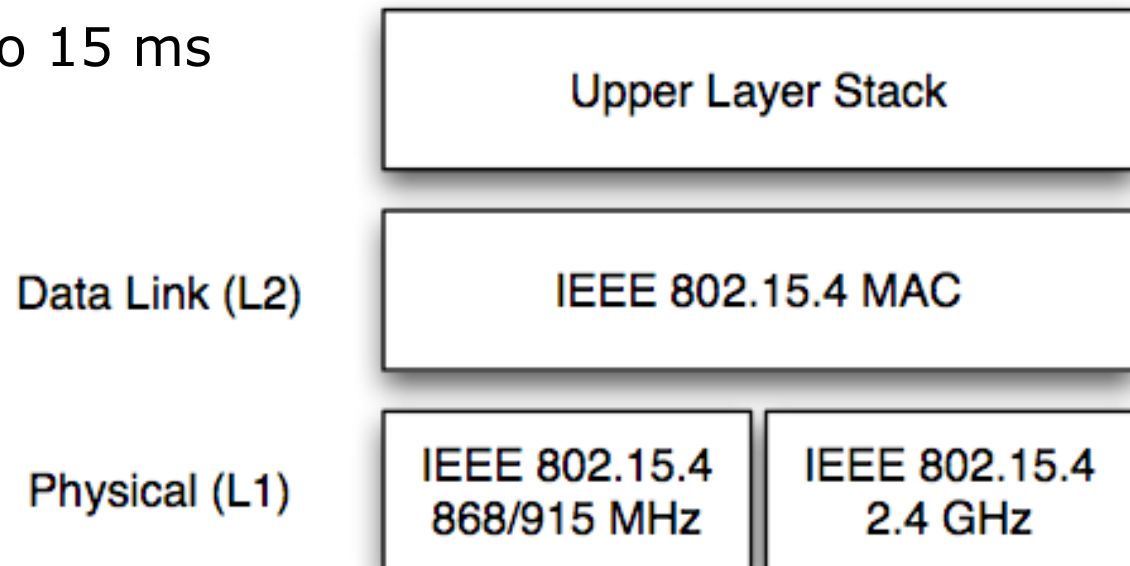
- Calculate energy per transaction
 - Assume an upper bound of 3ms per minimal transaction
 - Estimated TX power is 15mW (mostly TX power amp)
 - For 1.5v battery, this is 10mA.
 - $0.015\text{W} * 0.003 \text{ sec} = 45 \text{ micro Joule}$
- How long could a sensor last on a battery?
 - An example battery: Lenmar WC357, 1.55v, 180mAh, \$2-5
 - $180\text{mAh}/10\text{mA} = 18\text{Hr} = 64,800 \text{ seconds} = 21.6\text{M transactions}$
 - Suppose this sensor sends a report every minute = 1440/day
 - For just the BLE transactions, this is 15,000 days, or > 40 years
 - This far exceeds the life of the battery and/or the product
- This means that battery will cost more than the electronics
 - This sensor could run on scavenged power, e.g. ambient light

Bluetooth Security

- The following are the three basic security services specified in the Bluetooth standard:
 - **Authentication**
 - verifying the identity of communicating devices. User authentication is not provided natively by Bluetooth.
 - **Confidentiality**
 - preventing information compromise caused by eavesdropping by ensuring that only authorized devices can access and view data.
 - **Authorization**
 - allowing the control of resources by ensuring that a device is authorized to use a service before permitting it to do so.

The IEEE 802.15.4 standard

- The most important standard for home networking, industrial control and building automation
- The Zigbee Alliance is an association of companies involved with building higher-layer standards based on IEEE 802.15.4
- Self configuring and self maintaining
- Low cost and power for low data rate wireless connectivity
- Up to 64k nodes
- Range: 10 - 20 m.
- Latency: down to 15 ms



Bluetooth vs IEEE 802.15.4.

Bluetooth based WPAN

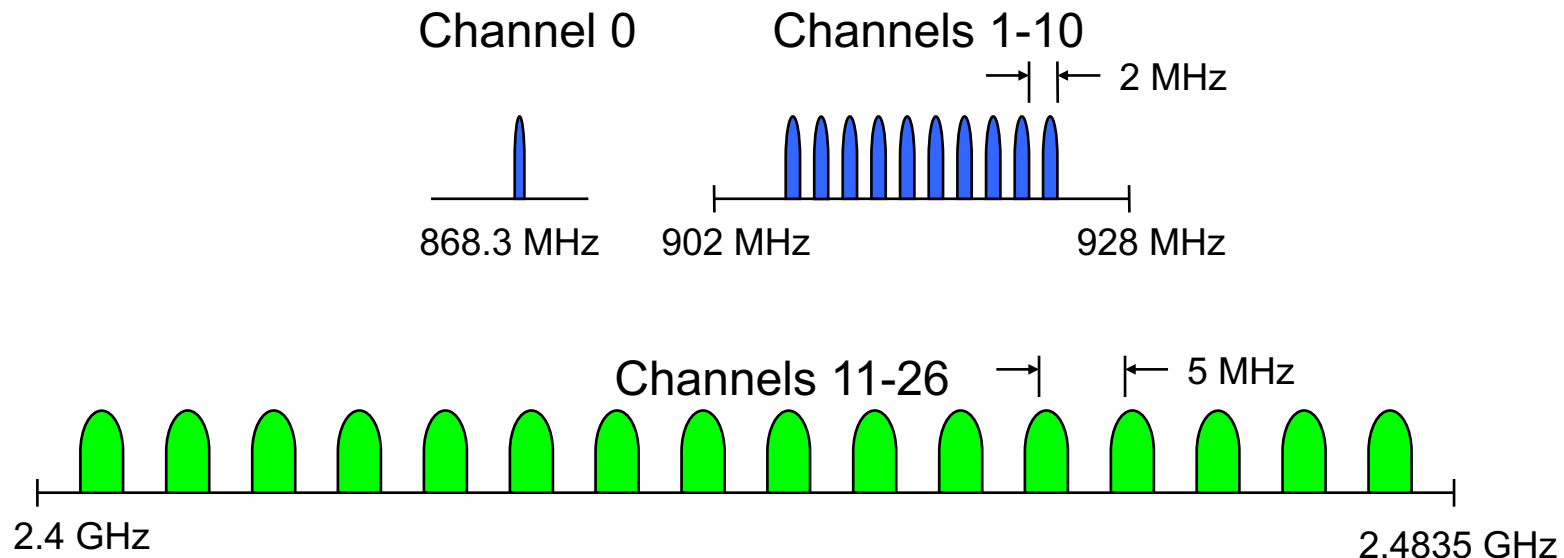
- Few devices
- Data range is 10m
- Data rate is nearly 1Mb/s
- Power consumption is low
- Battery life is low
- Star only

IEEE 802.15.4 LR-WPAN

- Many devices
- Data range is nearly 10m
- Data rate is 20, 40, 250kb/s
- Power consumption is ultra low
- Battery lasts years
- peer to peer and star

802.15.4 PHY - Channelization

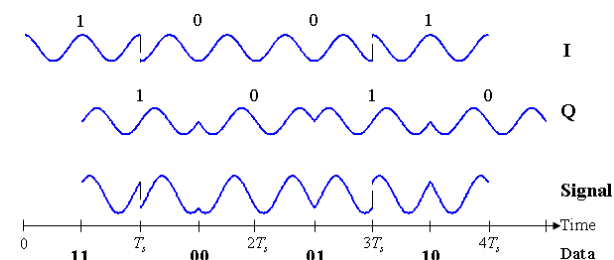
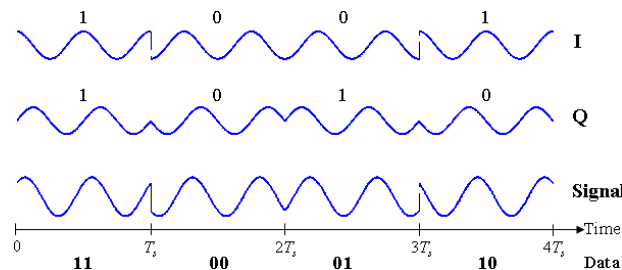
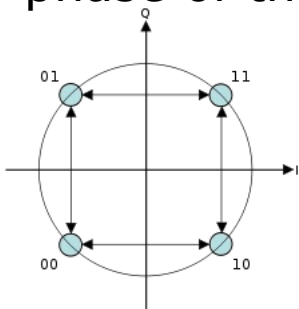
- 27 frequency channels are available across all the 3 bands.
 - 1 channel between 868.0 & 868.6 Mhz (DSSS)
 - 10 channels between 902.8 & 928.0 (DSSS)
 - 2.4 Ghz PHY has 16 channels (DSSS)
- implement dynamic channel selection to avoid interference.
- receiver energy detection, link quality indication, and channel switching enable channel assessment.
- used to establish initial operating channel and change channels



802.15.4 - Modulation

- Different kind of modulation are used depending on the frequencies used:
 - On 868MHz/915MHz
 - ASK, BPSK, O-QPSK
 - On 2,4 GHz
 - O-QPSK

OQPSK (Offset Quadrature Phase-Shift Keying) is a variant of QPSK where the phase or timing of either the in-phase (I) or Quadrature (Q) component is shifted relative to each other by a one bit-period or half a symbol-period. This limits the phase of the total signal to no more than 90° at a time respect to a traditional QPSK scheme, allowing the phase of the signal to shift by as much as 180° at a time.



Non-Beacon vs Beacon Modes

- Non-Beacon Mode

- A simple, traditional multiple access system used in simple peer and near-peer networks
- Think of it like a two-way radio network, where each client is autonomous and can initiate a conversation at will, but could interfere with others unintentionally
- However, the recipient may not hear the call or the channel might already be in use

- Beacon Mode

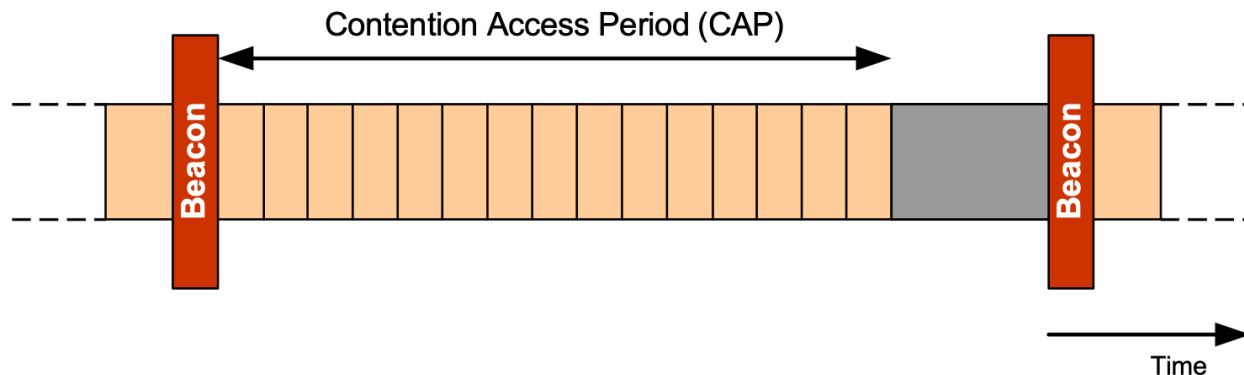
- A very powerful mechanism for controlling power consumption in extended networks like cluster tree or mesh
- Allows all clients in a local piece of the network the ability to know when to communicate with each other
- Here, the two-way radio network has a central dispatcher who manages the channel and arranges the calls

MAC Options – Non-Beacon vs Beacon

- Two channel access mechanisms:
 - **Non-beacon network**
 - Standard ALOHA CSMA-CA communications
 - A device checks if another device is transmitting in the same channel if so backs off for certain time
 - Positive acknowledgement for successfully received packets
 - **Beacon-enabled network**
 - a slotted carrier sense multiple access with collision avoidance (CSMA-CA) mechanism is used
 - very powerful for controlling power consumption in extended networks like cluster tree or mesh
 - a central dispatcher is needed who manages the channel
 - a Superframe structure is needed
 - For dedicated bandwidth and low latency
 - Set up by network coordinator to transmit beacons at predetermined intervals

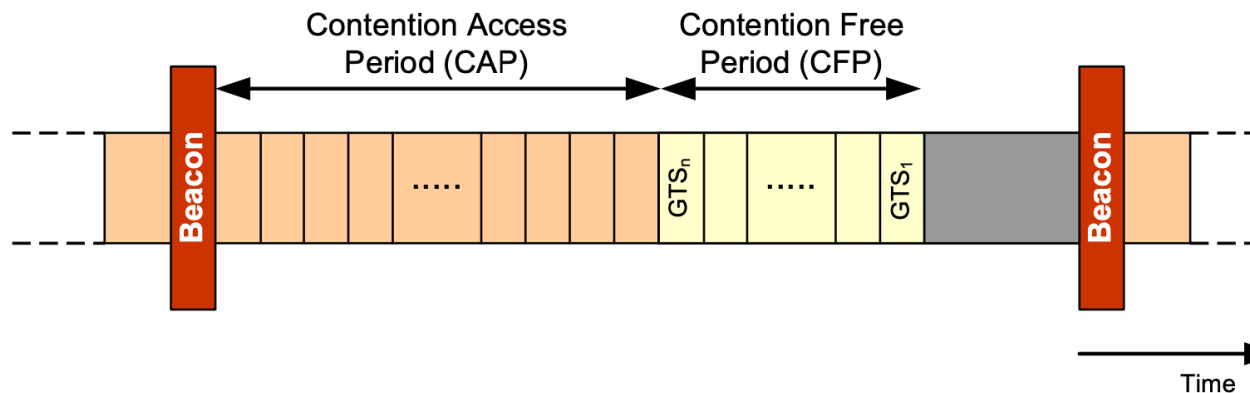
MAC features – Beacon mode

- In Beacon mode, the Coordinator sends out a periodic train of beacon signals containing information that allows network nodes to synchronize their communications.
- A beacon also contains information on the data pending for the different nodes of the network.
- Normally, two successive beacons mark the beginning and end of a superframe.
- A superframe contains 16 timeslots that can be used by nodes to communicate over the network (there may also be a dead period at the end of the superframe).
- The total time interval of these timeslots is called the Contention Access Period (CAP), during which nodes can attempt to communicate using slotted CSMA/CA



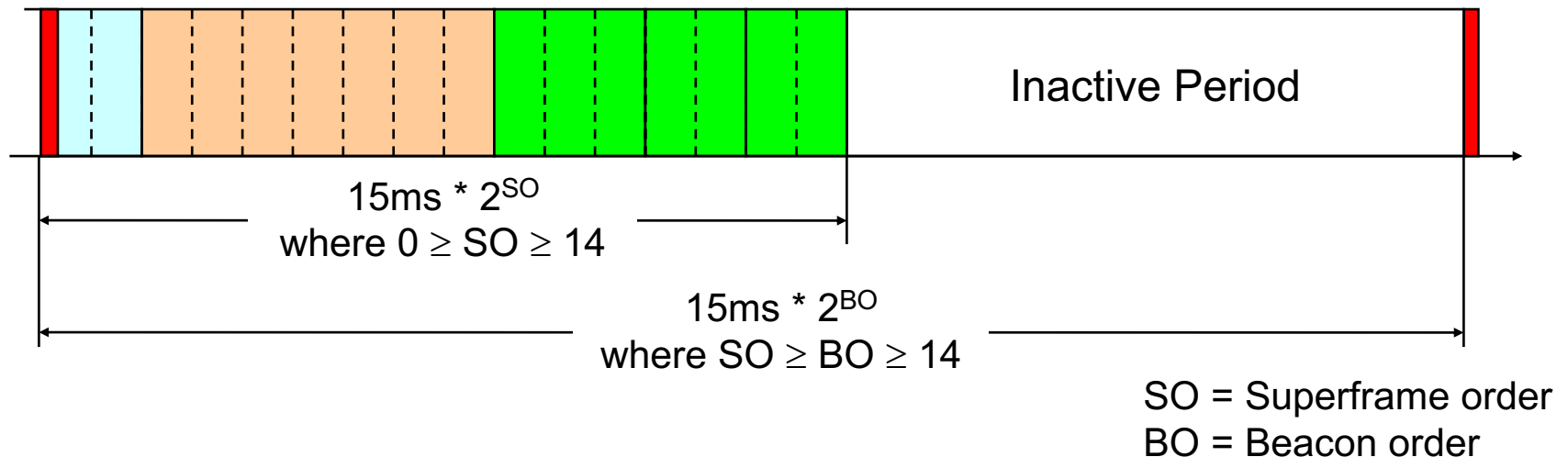
Superframe

- A superframe is divided into two parts
 - **Inactive**: all station sleep and **Active**: divided into 16 slots
 - 16 slots can further divided into two parts
 - Contention access period
 - Contention free period
- There are consecutive timeslots called Guaranteed Timeslots (GTSs) - in fact, one GTS can be multiple timeslots.
 - They are located after the CAP and the total time interval of all GTSs (for all nodes) is called the Contention Free Period (CFP).
 - Communication in the CFP does not require use of CSMA/CA. Use of GTSs reduces the CAP, and the superframe then consists of a CAP followed by a CFP (and possibly a dead period).



Superframe

- Certain applications require dedicated bandwidth to achieve low latency for this it can operate in optional superframe mode
- PAN coordinator, transmits superframe beacons in predetermined intervals
- The channel access in the time slots is contention-based but PAN coordinator may assign time slots to a single device requiring dedicated bandwidth or low-latency transmissions. These assigned time slots are called guaranteed time slots (GTS) and together form a contention-free period.

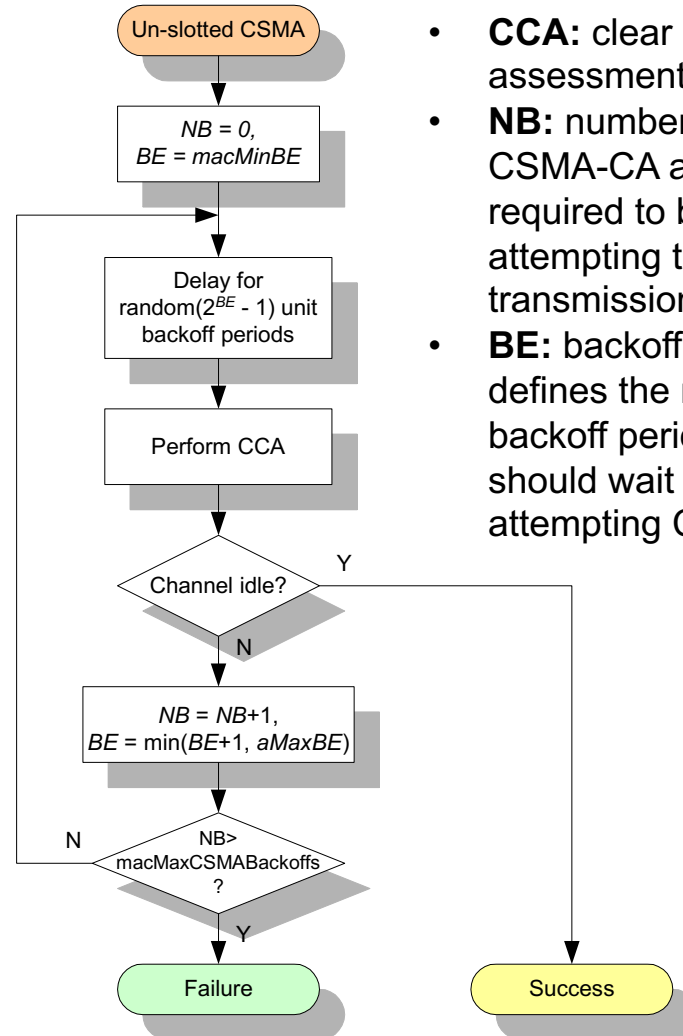


GTS Concepts

- A **guaranteed time slot (GTS)** allows a device to operate on the channel within a portion of the superframe
- A GTS shall only be allocated by the PAN coordinator
- The PAN coordinator can allocated up to **7 GTSs** at the same time
- The PAN coordinator decides whether to allocate GTS based on:
 - Requirements of the GTS request
 - The current available capacity in the superframe

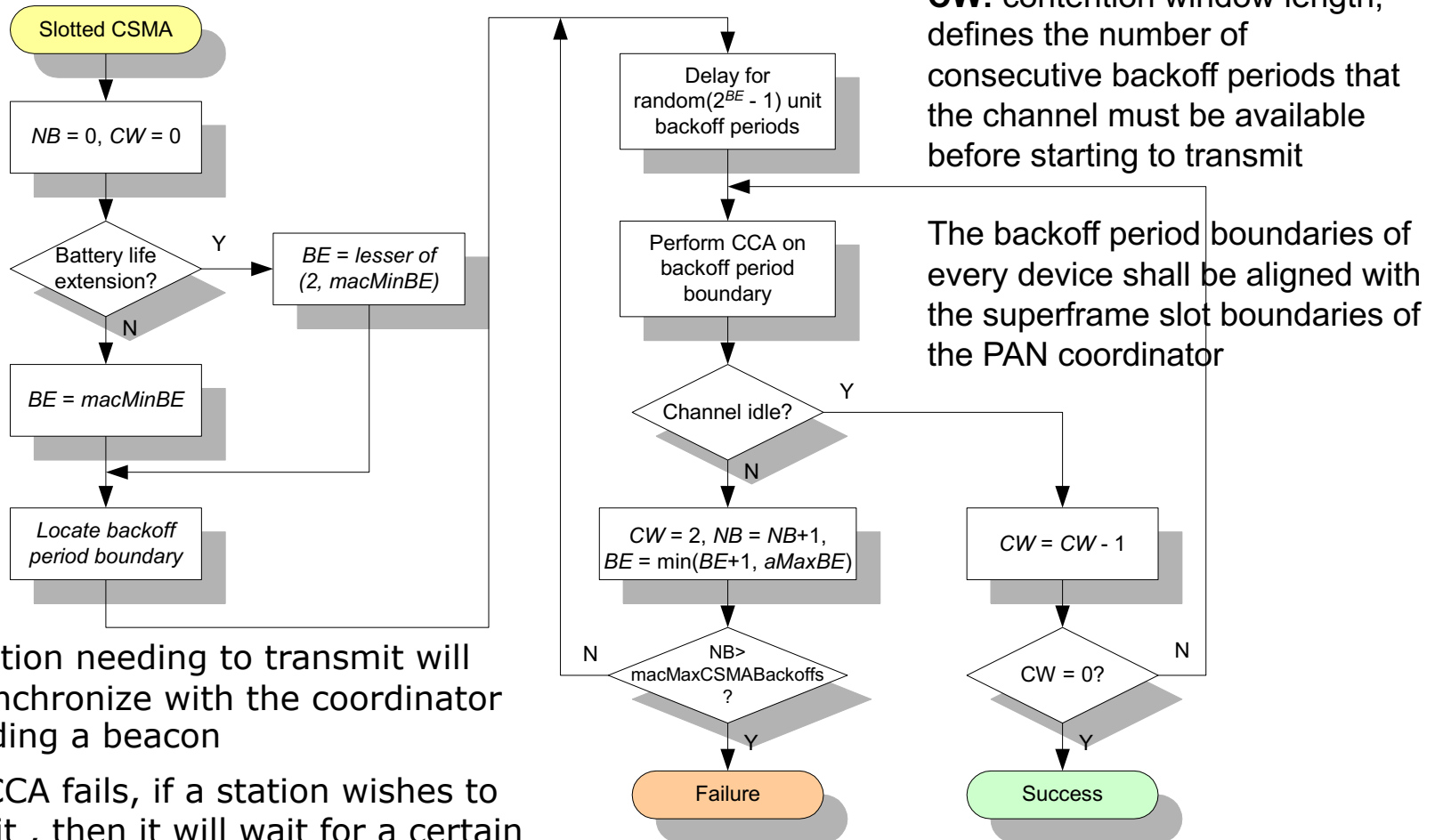
Un-slotted CSMA Procedure - non-beacon nets

- Backoff exponent and number of attempts are initialized
- Clear Channel Assessment is performed after waiting a random number of delay units selected in the backoff window
- At each unsuccessful trial the backoff window size is doubled
- Only a limited number of attempts is allowed (macMaxCSMABackoffs)



- **CCA:** clear channel assessment
- **NB:** number of times the CSMA-CA algorithm was required to backoff while attempting the current transmission
- **BE:** backoff exponent, defines the number of backoff periods a node should wait before attempting CCA

Slotted CSMA Procedure – beacon-enabled nets



- The station needing to transmit will first synchronize with the coordinator by sending a beacon
- While CCA fails, if a station wishes to transmit, then it will wait for a certain number of backoff periods
- The CW will decrement once only at the boundaries of each backoff period till it becomes 0, as long as the CCA succeeds; Once the CW becomes 0, the medium is allocated to the node and the data transmission takes place and an acknowledgement will be sent

IEEE 802.15.4 Device Classes

- Full function device (FFD)
 - Any topology
 - PAN coordinator capable
 - Talks to any other device
 - Implements complete protocol set
- Reduced function device (RFD)
 - Limited to star topology or end-device in a peer-to-peer network.
 - Cannot become a PAN coordinator
 - Very simple implementation
 - Reduced protocol set

IEEE 802.15.4 Definitions

– Network Device:

- An RFD or FFD implementation containing an IEEE 802.15.4 medium access control and physical interface to the wireless medium.

– Coordinator:

- An FFD with network device functionality that provides coordination and other services to the network.

– PAN Coordinator:

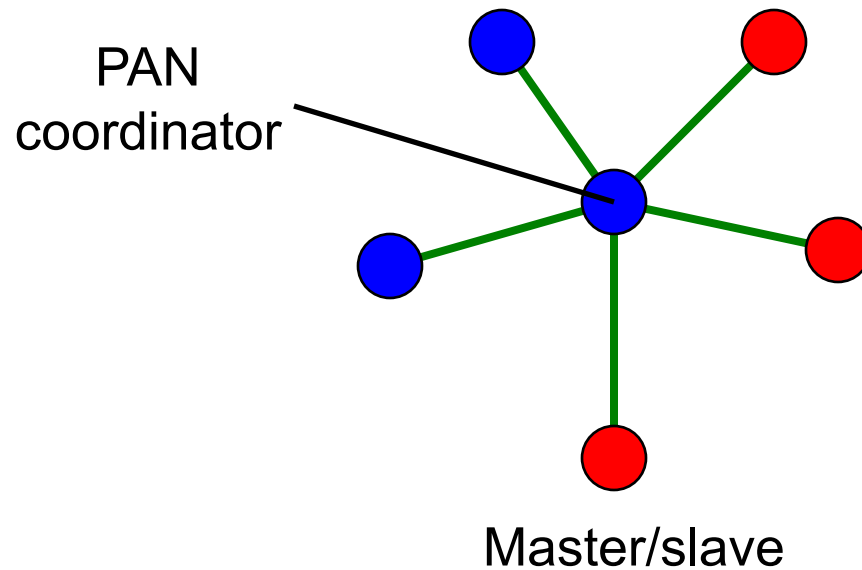
- A coordinator that is the principal controller of the PAN. A network has exactly one PAN coordinator.
- Maintains overall network knowledge

Topologies

Networks can be built as peer-to-peer or stars

- A star topology can provide complete home coverage
 - every network needs at least one FFD to work as the coordinator
- Peer-to-peer networks can form arbitrary connections
 - their extension is only limited by the distance between each pair of nodes
 - they serve as ad hoc networks capable of performing self-organization
 - routing is not directly supported but multihop communication is supported

Star Topology

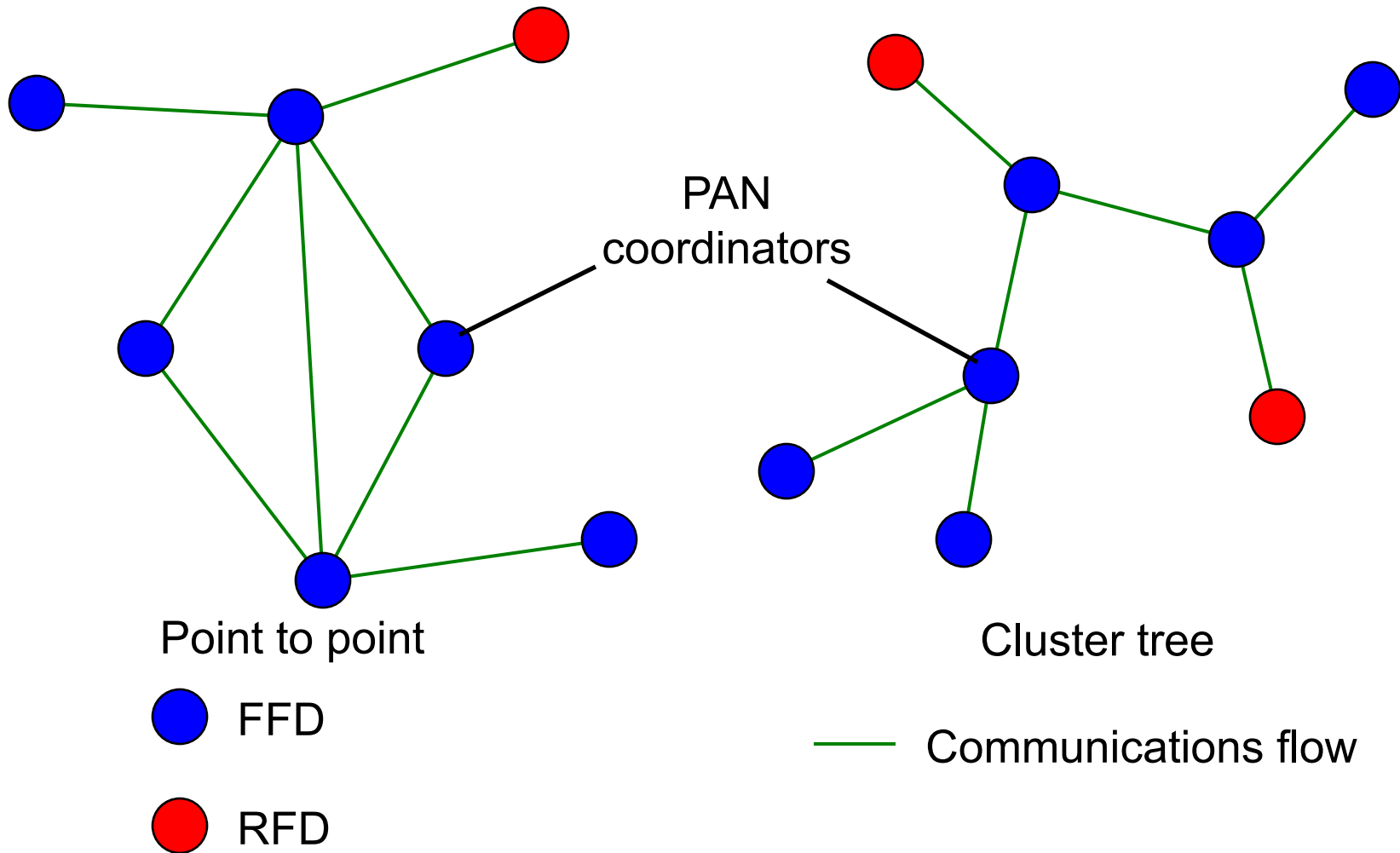


● FFD

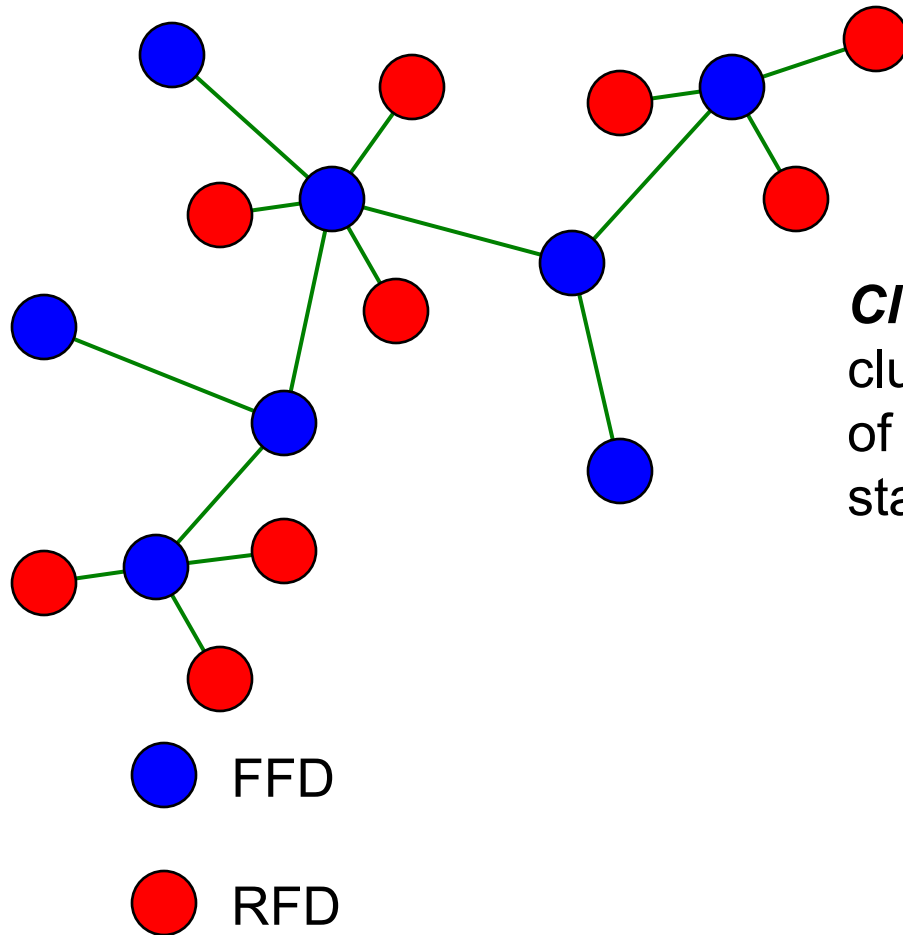
● RFD

— Communications flow

Peer-Peer Topology



Combined Topology

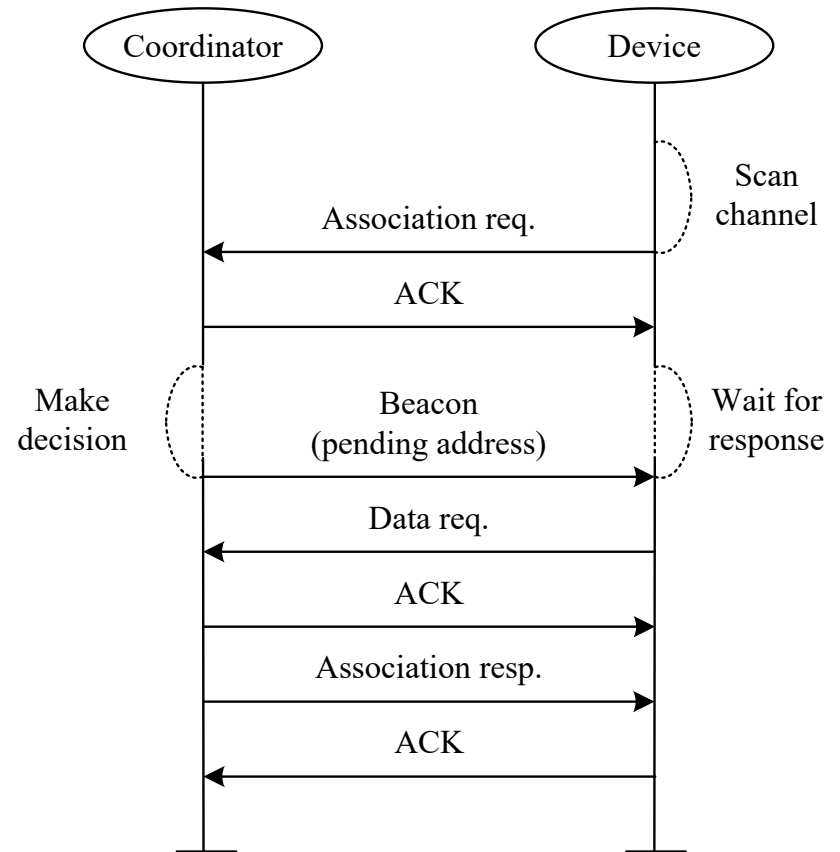


Clustered stars - for example, cluster nodes exist between rooms of a hotel and each room has a star network for control.

— Communications flow

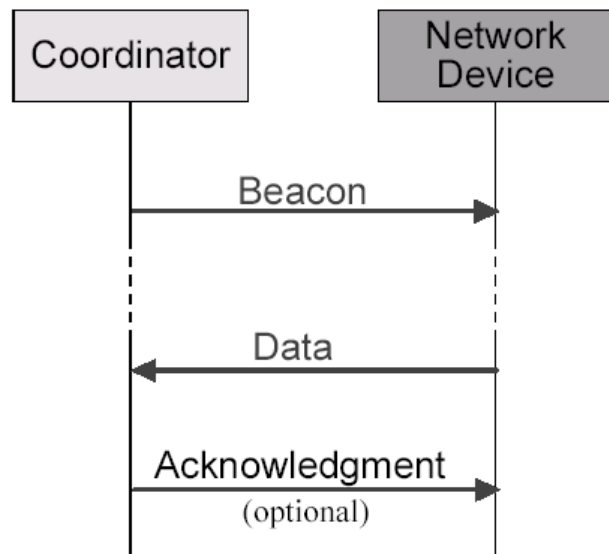
Association Procedures

- A device becomes a joins a PAN by associating with its coordinator
- A coordinator responds to association requests by appending devices' long addresses in beacon frames
- Devices need to send a **data request** to the coordinator to acquire the association result
- After associating to a coordinator, a device will be assigned a **16-bit short address**.

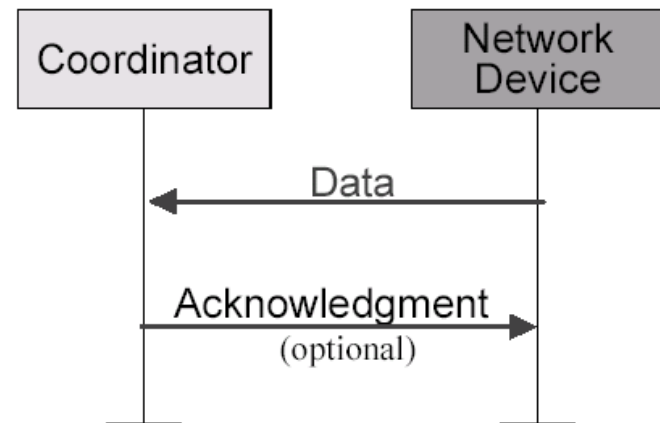


Data Transfer in 802.15.4

- Data transferred from device to coordinator
 - In a **beacon-enabled network**, a device finds the beacon to synchronize to the superframe structure. Then it uses **slotted CSMA/CA** to transmit its data.
 - In a **non-beacon-enabled network**, device simply transmits its data using **unslotted CSMA/CA**



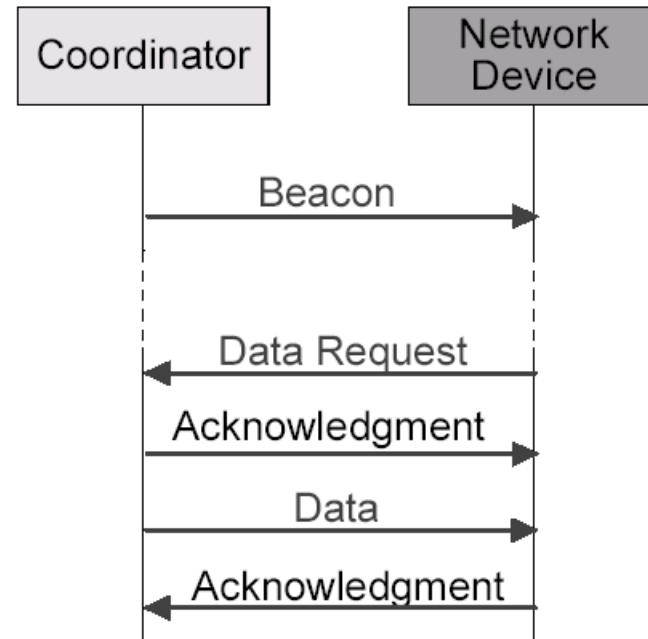
Communication to a coordinator
In a **beacon-enabled** network



Communication to a coordinator
In a **non beacon-enabled** network

Data Transfer in 802.15.4

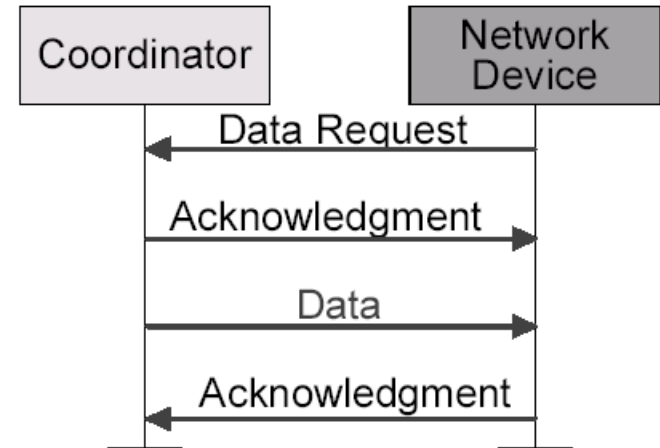
- Data transferred from coordinator to device in a **beacon-enabled** network:
 - The coordinator indicates in the beacon that some data is pending.
 - A device periodically listens to the beacon and transmits a Data Request command using **slotted** CSMA/CA.
 - Then ACK, Data, and ACK follow ...



Communication from a coordinator
In a **beacon-enabled** network

Data Transfer in 802.15.4

- Data transferred from coordinator to device in a **non-beacon-enable** network:
 - The device transmits a Data Request using **unslotted** CSMA/CA.
 - If the coordinator has its pending data, an ACK is replied.
 - Then the coordinator transmits Data using **unslotted** CSMA/CA.
 - If there is no pending data, a data frame with zero length payload is transmitted.



Communication from a coordinator in a **non beacon-enabled** network

The frame structure

- 4 frame types (data, acknowledgment, beacon and MAC command)
- Frame control field indicates the type of frame transmitted, the format of the address field, and controls the acknowledgment.
- Multiple address types:
 - 64 bit physical address
 - short 16 bit network assigned address
- Payload field variable with size of mac frame ≤ 127 bytes

Octets:2	1	0/2	0/2/8	0/2	0/2/8	variable	2
Frame control	Sequence number	Destination PAN identifier	Destination address	Source PAN identifier	Source address	Frame payload	Frame check sequence
		Addressing fields					
MAC header						MAC payload	MAC footer

Bits: 0-2	3	4	5	6	7-9	10-11	12-13	14-15
Frame type	Security enabled	Frame pending	Ack. Req.	Intra PAN	Reserved	Dest. addressing mode	Reserved	Source addressing mode

Frame control field

Beacon Frame Format

Octets:2	1	4 or 10	2	variable	variable	variable	2
Frame control	Beacon sequence number	Source address information	Superframe specification	GTS fields	Pending address fields	Beacon payload	Frame check sequence
MAC header			MAC payload				MAC footer

Bits: 0-3	4-7	8-11	12	13	14	15
Beacon order	Superframe order	Final CAP slot	Battery life extension	Reserved	PAN coordinator	Association permit

- Transmitted by network coordinator nodes
- Beacons add a new level of functionality to a network
- Client devices can wake up only when a beacon is to be broadcast, listen for their address, and if not heard, return to sleep
- Beacons are important for mesh and cluster tree networks to keep all of the nodes synchronized without requiring nodes to consume precious battery energy listening for long periods of time

MAC Command Frame

Octets:2	1	4 to 20	1	variable	2
Frame control	Data sequence number	Address information	Command type	Command payload	Frame check sequence
MAC header			MAC payload		MAC footer

- Mechanism for remote control/configuration of client nodes
- Allows a centralized network manager to configure individual clients no matter how large the network
- Command Frame Types
 - Association request
 - Association response
 - Disassociation notification
 - Data request
 - PAN ID conflict notification
 - Orphan Notification
 - Beacon request
 - Coordinator realignment
 - GTS request

Data Frame Format

Octets:2	1	4 to 20	variable	2
Frame control	Data sequence number	Address information	Data payload	Frame check sequence
MAC header			MAC Payload	MAC footer

- Provides up to 104 byte data payload capacity
- Data sequence numbering to ensure that all packets are tracked
- Robust frame structure improves reception in difficult conditions
- Frame Check Sequence (FCS) ensures that packets received are without error

ACK Frame Format

Octets:2	1	2
Frame control	Data sequence number	Frame check sequence
MAC header		MAC footer

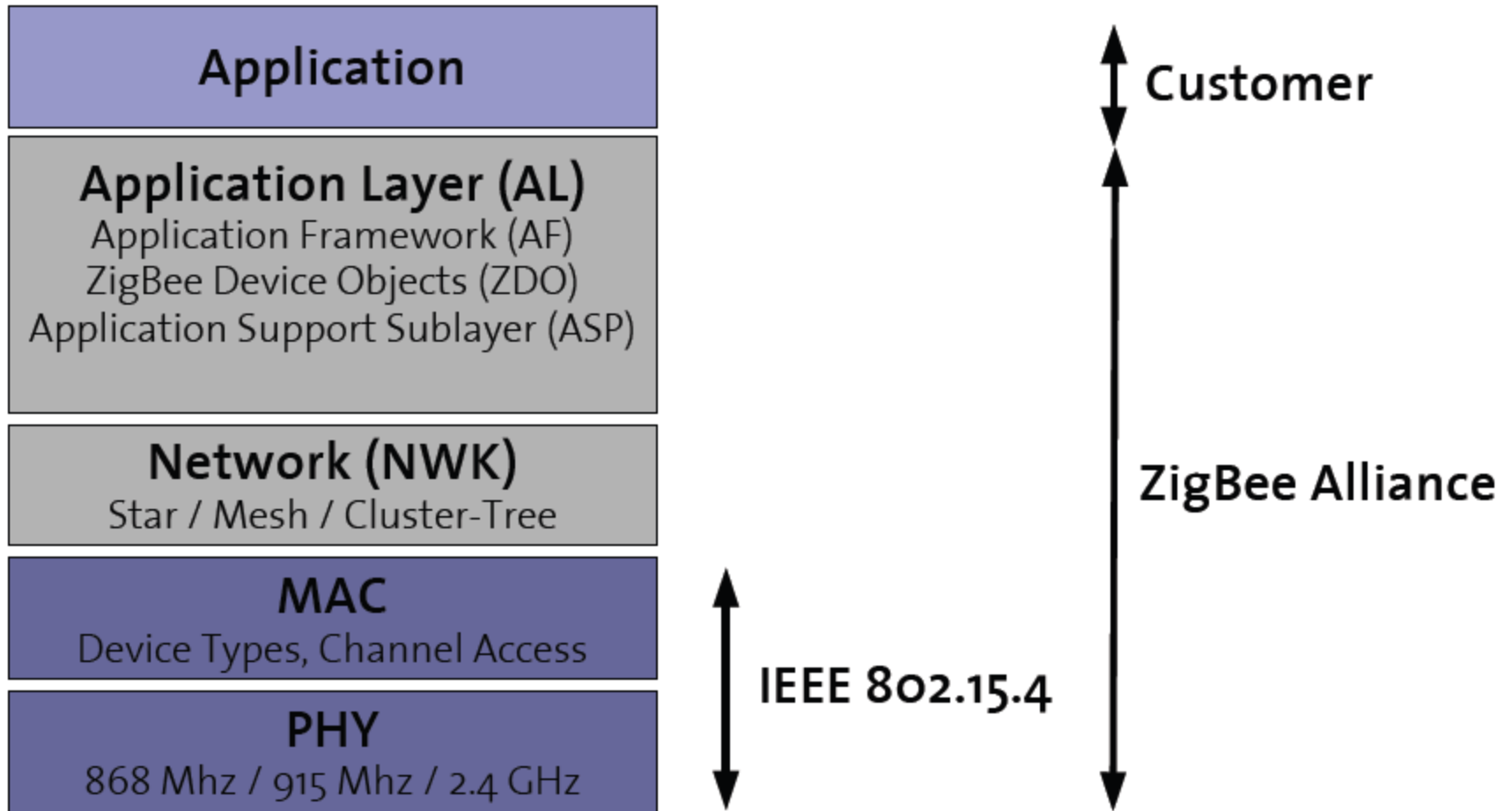
- Provides active feedback from receiver to sender that packet was received without error
- Short packet that takes advantage of standards-specified “quiet time” immediately after data packet transmission

802.15.4 and the ZigBee Alliance

- IEEE 802.15.4
 - Composed of many of the individuals and companies that make up the ZigBee Alliance
 - Developed the basic PHY and MAC standard with the requirement that 802.15.4 be simple and manageable and that high-level functionality (networking, security key management, applications) be considered
- The ZigBee Alliance is
 - A consortium of end users and solution providers, primarily responsible for the development of the 802.15.4 standard
 - Developing applications and network capability utilizing the 802.15.4 packet delivery mechanism
 - Addresses application and interoperability needs of a substantial part of the market



ZigBee Protocol Stack



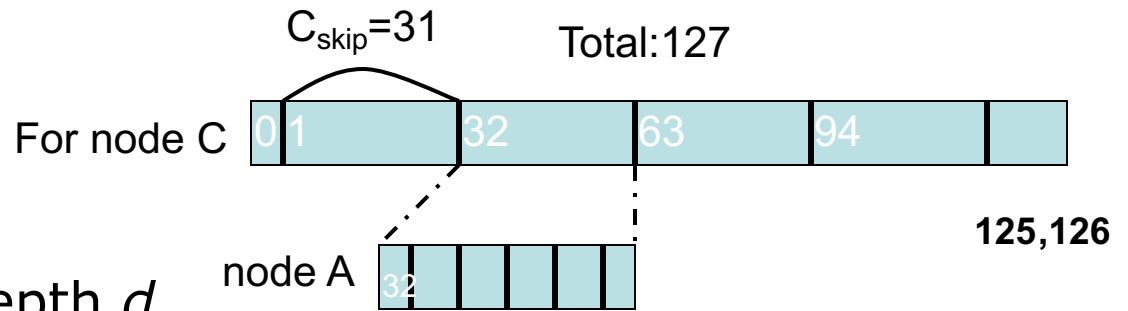
ZigBee Network Layer Overview

- Three kinds of devices in the network layer
 - **ZigBee coordinator**: responsible for initializing, maintaining, and controlling the network
 - **ZigBee router**: form the network backbone
 - **ZigBee end device**: must be connected to router/coordinator
- In a tree network, the coordinator and routers can announce **beacons**.
- In a mesh network, there is **no regular beacon**.
 - Devices in a mesh network can only communicate with each other in a **peer-to-peer** manner

Address Assignment

- In ZigBee, network addresses are assigned to devices by a distributed address assignment scheme
- ZigBee coordinator determines three network parameters
 - the maximum number of children (C_m) of a ZigBee router
 - the maximum number of child routers (R_m) of a parent node
 - the depth of the network (L_m)
- A parent device utilizes C_m , R_m , and L_m to compute a parameter called C_{skip}
 - which is used to compute the size of its children's address pools

$$C_{skip}(d) = \begin{cases} 1 + C_m \cdot (L_m - d - 1), & \text{if } R_m = 1 \quad \dots\dots\dots(a) \\ \frac{1 + C_m - R_m - C_m \cdot R_m^{L_m - d - 1}}{1 - R_m}, & \text{Otherwise} \quad \dots\dots\dots(b) \end{cases}$$



- If a parent node at depth d has an address A_{parent}
 - the n^{th} child router is assigned to address:

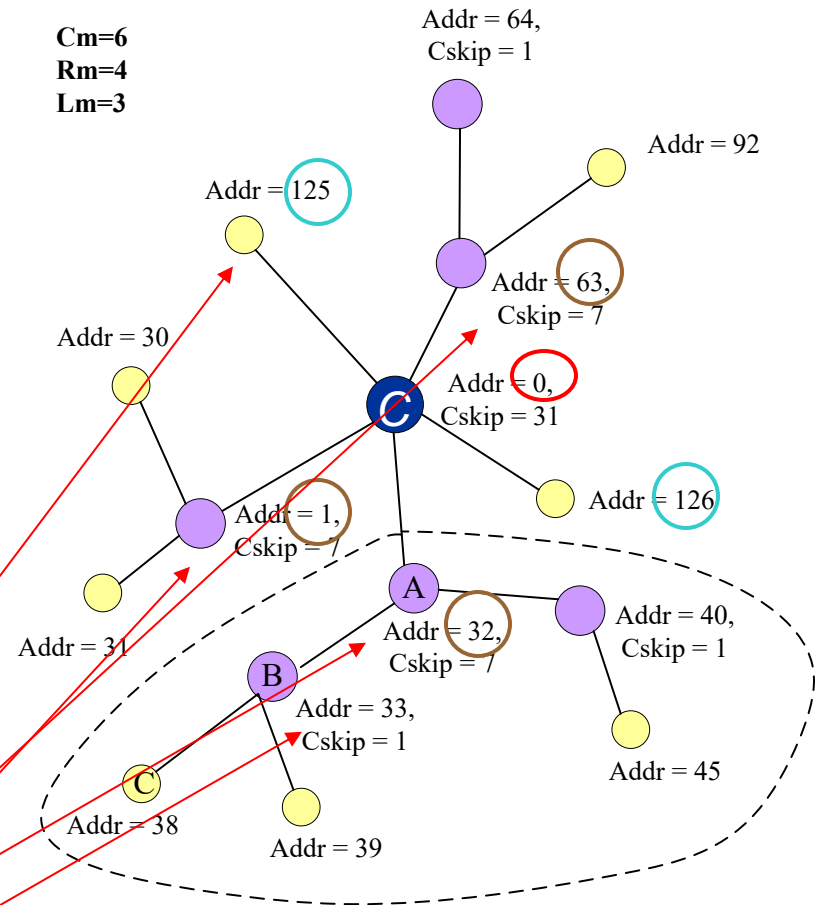
$$A_{parent} + (n-1) \times C_{skip}(d) + 1$$

- n^{th} child end device is assigned to address

$$A_{parent} + R_m \times C_{skip}(d) + n$$

Example: (Addr=0, Cskip=31)

- 1st child non router of C $0+4 \times 31+1=1$
- 1st child router of C $0+(1-1) \times 31+1=1$
- 3rd child router of C $0+(3-1) \times 31+1=63$
- 2nd child router of C $0+(2-1) \times 31+1=32$ (A)
- 1st child router of A $32+(1-1) \times 7+1=33$ (B)



Introduction to Z-Wave

- Zensys a Danish-American company founded in 1999 invented the Z-wave technology.
- They are basically providers of Integrated Single chip Solutions. While trying to embed intelligence and RF communication into their products they stumbled upon the idea to come up with a new technology combining the pros of the existing technologies.



Z-Wave

- Low power radio protocol for Home automation (lighting, heating, ...) applications
 - Low-throughput: 9 and 40 kbps
 - Battery-operated or electrically powered
 - Frequency range: 868 MHz in Europe, 908 MHz in the US
 - Coverage Range: about 50 m (more outdoor, less indoor)
 - Mesh architecture possible to increase the coverage
 - Access method type CSMA / CA
 - Z-Wave Alliance: more than 100 manufacturers in



Z-Wave PHY/MAC

Public PHY/MAC specification ITU G.9959

<https://www.itu.int/rec/T-REC-G.9959-201501-I/en>

Z-Wave Frequency Allocation ZAD12837-10

<https://zwavealliance.org/wp-content/uploads/2018/03/ZAD12837-10.pdf>

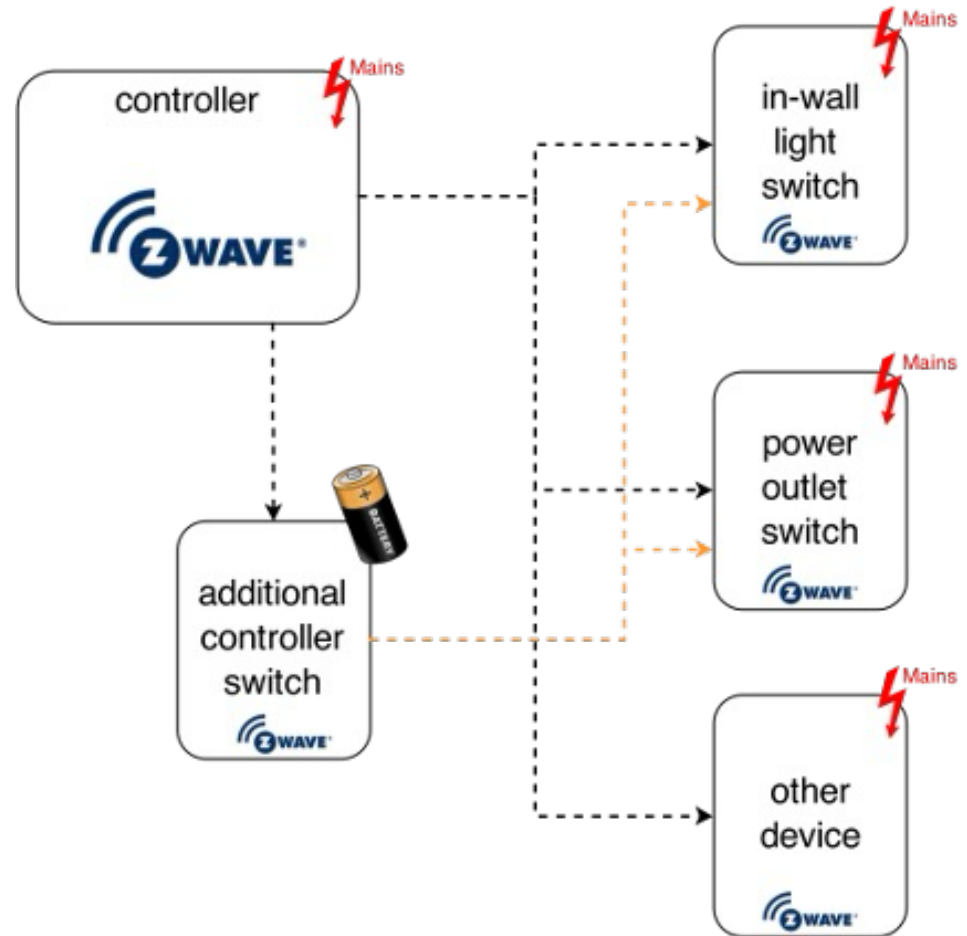
Sub GHz, license free ISM bands, region/Country specific

Data rate \ region frequency & channel(s)	US / Canada / South America	EU / UAE / China / India	AU / NZ / Brazil	Japan / Taiwan / Korea / Singapore
9.6kbps FSK Modulation	908 MHz Ch 1	868 MHz Ch 1	921 MHz Ch 1	NA
40kbps FSK Modulation	908 MHz Ch 1	868 MHz Ch 1	921 MHz Ch 1	NA
100kbps GFSK Modulation	916 MHz Ch 2	869 MHz Ch 2	919 MHz Ch 2	919-926 MHz Ch 1 - 3

We have different frequency allocation for different regions.

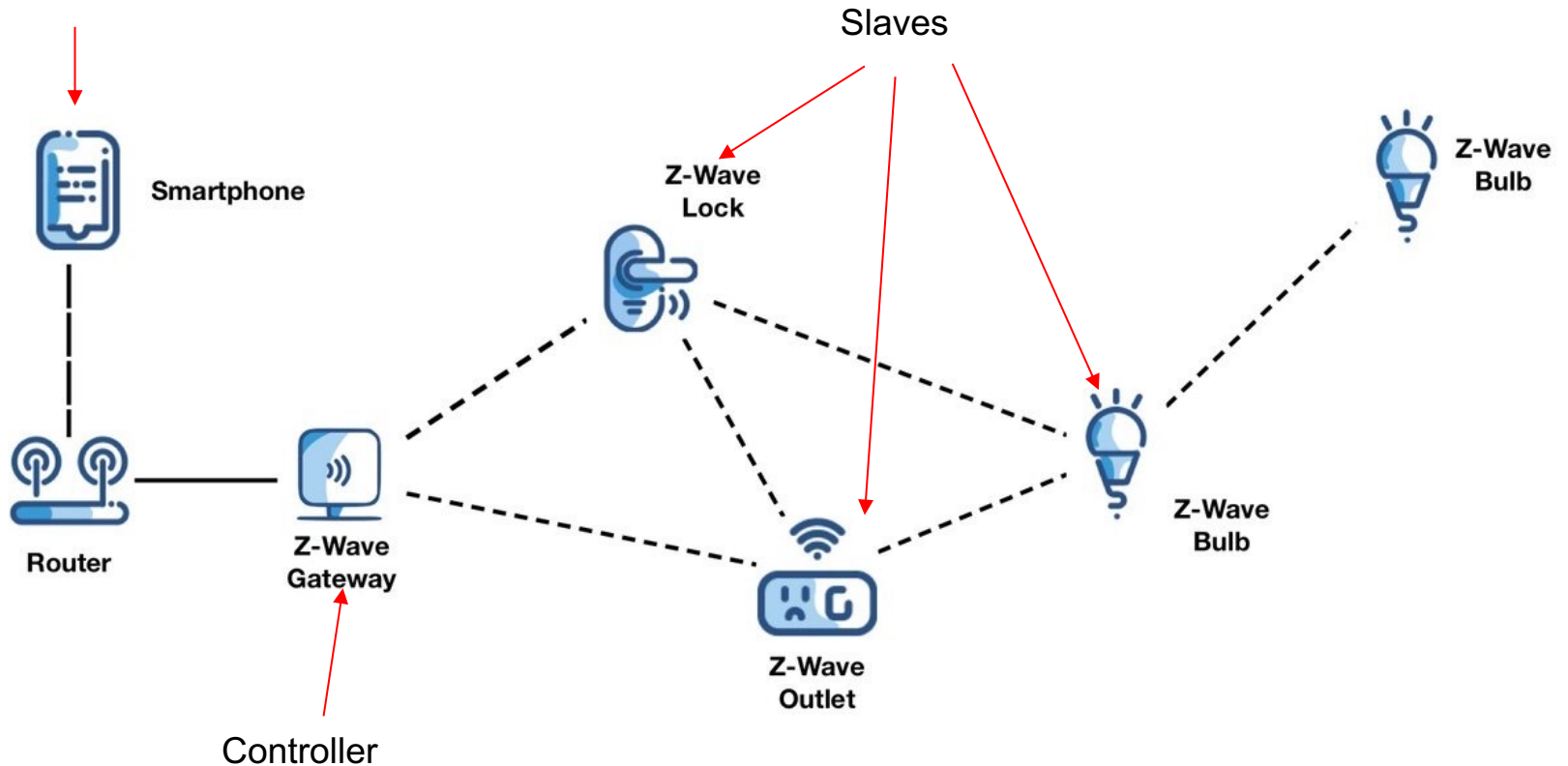
Home Area Network (HAN)

- HAN allows devices to communicate together and make it possible to monitor the whole automated system remotely and anywhere inside the house.
- These technologies provide an auto-balancing and self-monitoring power smart grid capabilities.
- A controller manages house automation and warns house owners of any problems



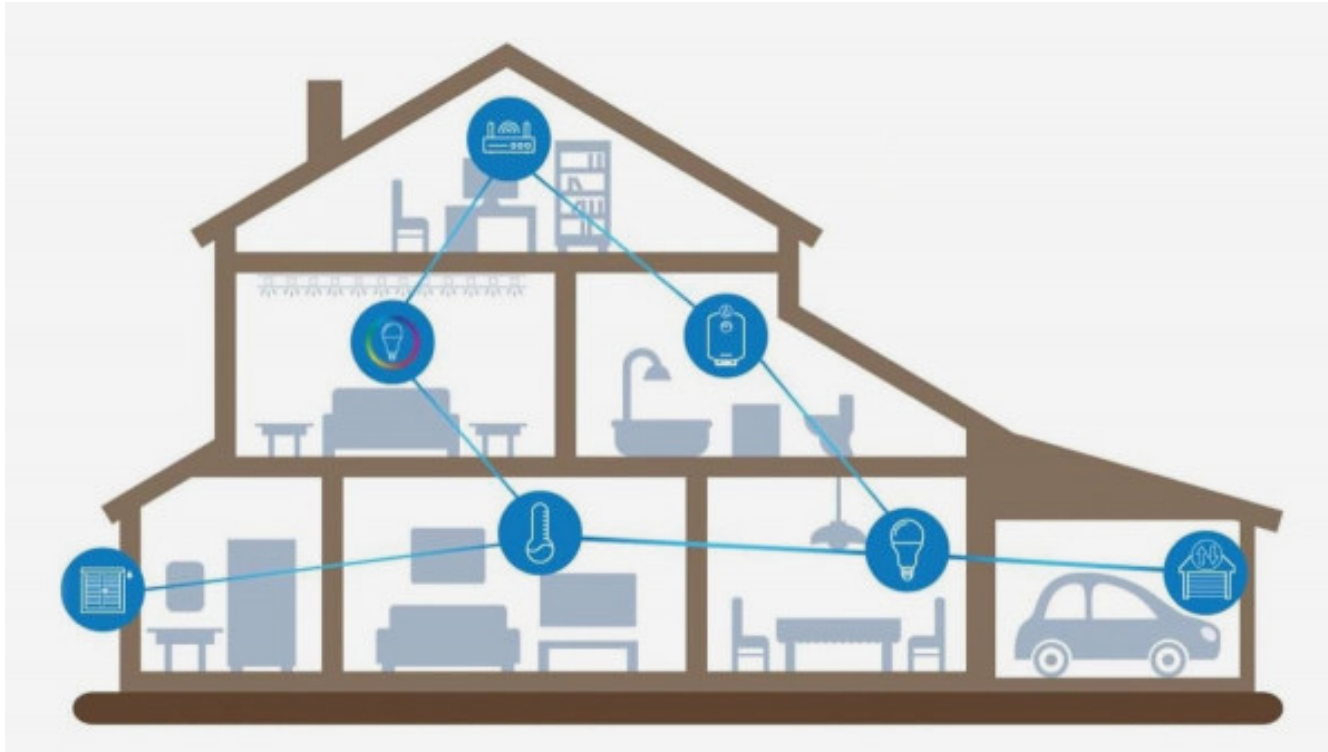
Topology

Control interface



A Z-wave network is structured according to a mesh topology

Typical Z-Wave Network



Note: Z-Wave networks devices manufactured by different companies are compatible with one another.

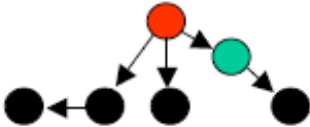
Controllers

1. PortableController
2. Static Controller
3. Installer Controller
4. Bridge Controller

Slaves

1. Slave
2. Routing Slave
3. Enhanced Slave

Z-Wave Node Types

Archetypes	Illustration	Main Network System Functions
Controllers 		• Have a complete topology map and can calculate routes • Can address all nodes in the network • Can act as repeater in the network • Can be portable or static in the network • Can have the added functionalities of Bridge and/or SUC + SIS
Routing Slaves 		• Have a partial topology map and cannot calculate routes • Can address a subset of the network • Can act as a repeater in the network
Slaves 		• Have no topology map and cannot calculate routes • Can only respond to requests from Controllers or Routing Slaves

Network Node Type Controllers

There are two types of controllers:

- Portable Controllers – Controllers that can be moved around in a Network.
- Static Controllers – Controllers that cannot be moved around in a Network.
- Bridge – This makes it possible for the Z-Wave to control up to 128 devices.

A “Virtual Node” is created by a Bridge that act as proxies for the non Z-Wave devices present on the other network which thereby can be controlled by the Z-Wave Network.

NOTE: A non Z-Wave Network can be imagined as a Bluetooth Network.



A Portable
Controller



Static Controller



Bridge

Network Node Types – Routing Slaves

- Routing Slave is an AC powered device present in a fixed location.
 - It can receive commands and replies from/to all nodes.
 - It can send unsolicited routed frames to up to 5 other nodes.
- Enhanced Slave is the same as a routing slave but can support an external EEPROM.



Routing
Slave

Network Node Types - Slaves

- Slave – Must be in a fixed position and should be listening every time.
 - It can receive commands and send reply using the same route.



Slave

Typical Node type controllers



A Portable
Controller



Static Controller



Bridge

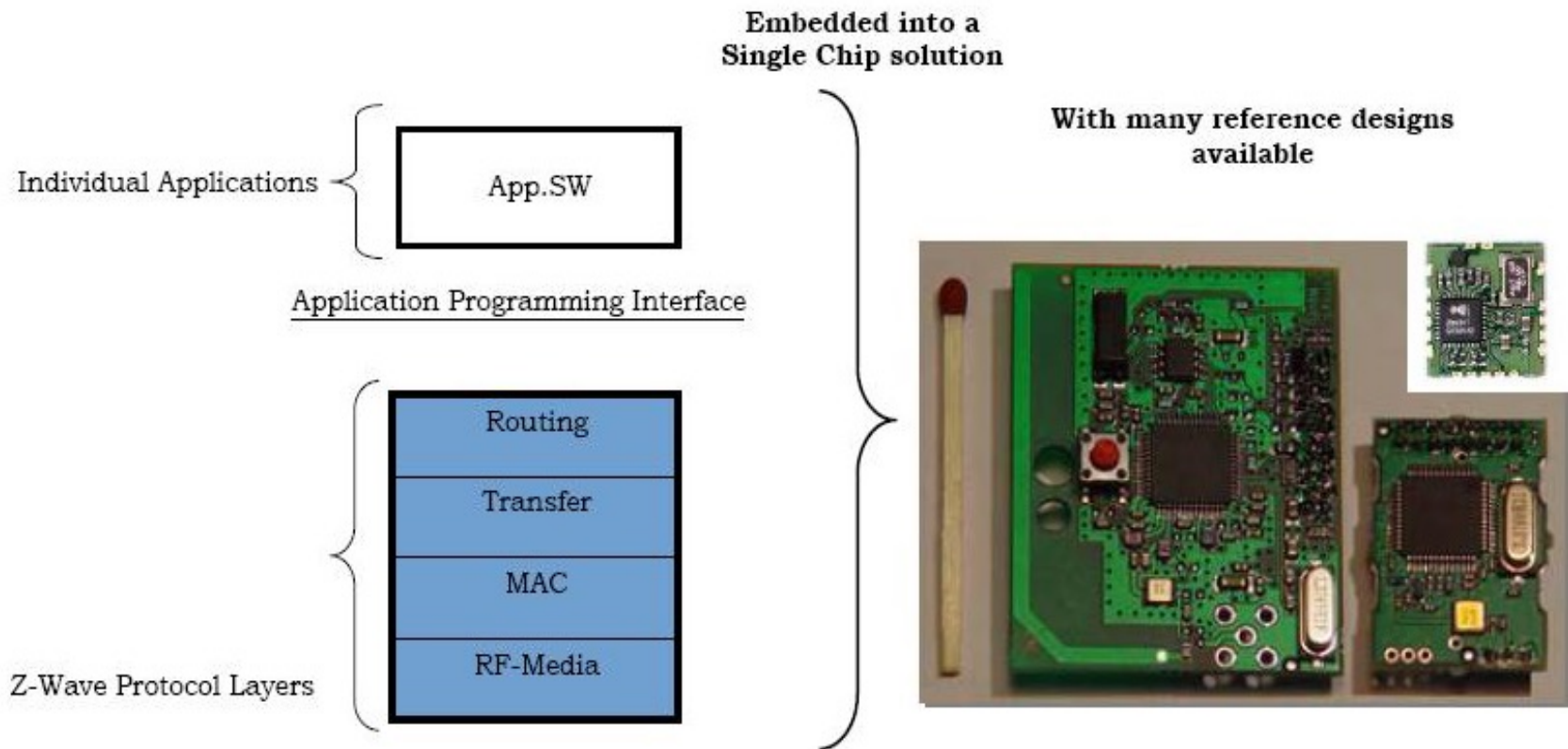


Routing
Slave



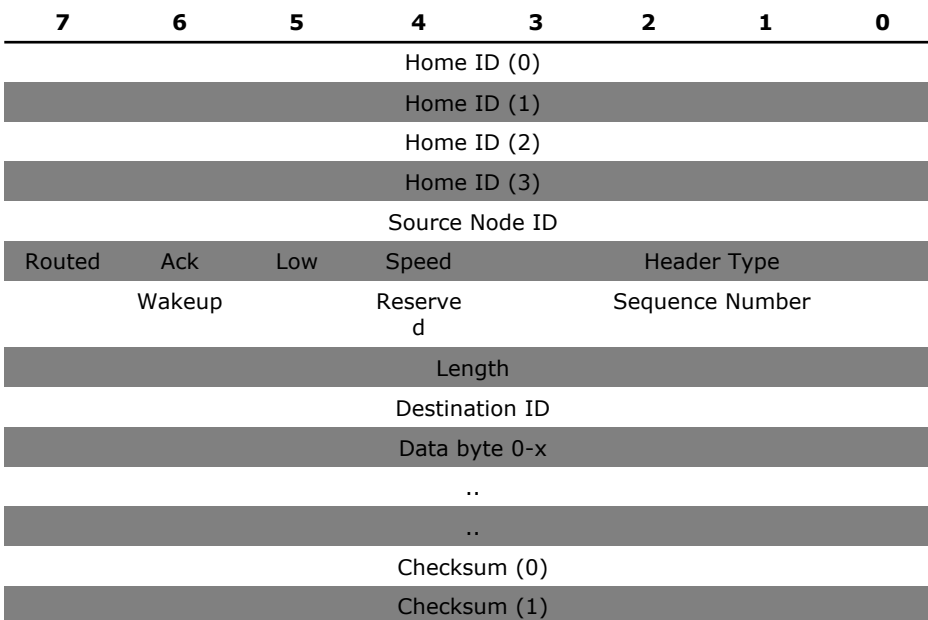
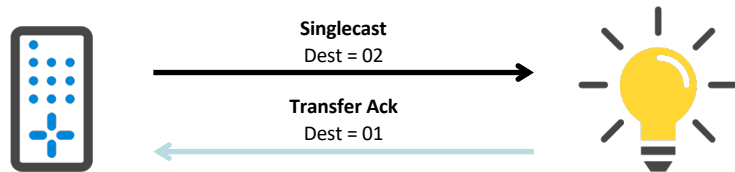
Slave

Z-Wave Protocol Layers



Z-Wave PHY/MAC - ADDRESSING

- Every device controlled by a microcontroller is assigned a common Home ID
- Once the device is added to the network the device looks for other devices and adds them to the same network
- When two different networks are present they know each other but they are not interoperable.



- Home ID, 32 bit network identifier
 - Randomly generated in all unincluded nodes
 - Home ID will be the same as Controller Home ID upon inclusion to its network
- Node ID, 8 bit node identifier
 - 0 at startup
 - Changed at inclusion into network
 - Max nodes in a network 2^{32}
- Frame Types
 - Singlecast
 - Multicast
 - Broadcast

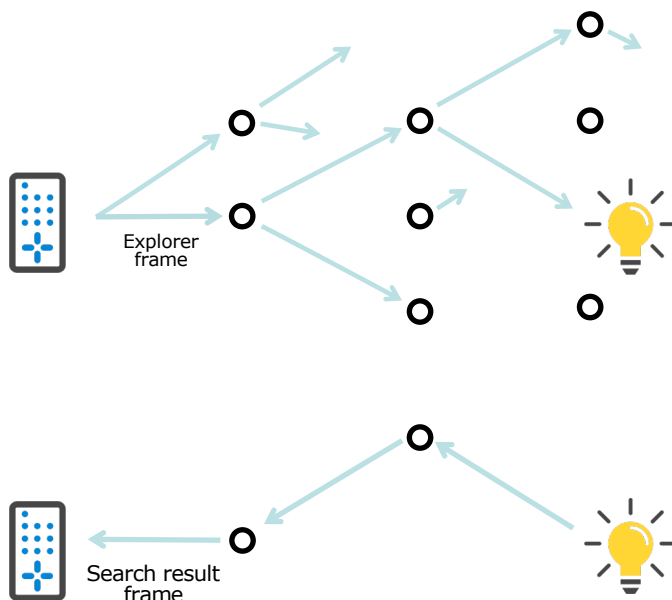
Z-Wave Routing

- Source Routing is applied – 4 hops max.
- The routing table is held by the controller
 - It is built during inclusion through neighbor discovery.
 - Neighbor discovery can be invoked at any time to refresh the routing table hold by the controller
- The Controller caches the last working route (LWR), and uses this by default
- Similarly the slave nodes relies on the response routes, i.e. Respond through the route a request was received through.
 - If a slave sends unsolicited reports, it relies on assigned return routes.



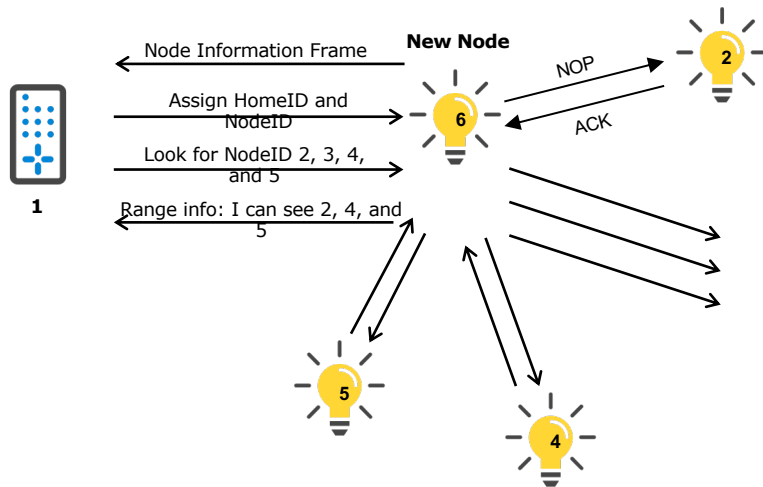
Dynamic route resolution

- Explorer frames are used as a last resort if the destination cannot be reached.
- The network is flooded with, and if the destination is reached, it acks the frame.
- The ack is used to update the “last working route” / response route.



- Explorer Frames
 - It is used to discover/rediscover routes
 - Application command is included in the explorer frame
- Explorer Frame Response
 - Explorer response uses source routing
 - Becomes the LWR when received
 - Implements random back off to avoid collisions

Z-Wave Device Inclusion



- Inclusion
 - Home ID and Node ID are assigned
 - New node is asked to look for neighbors
- Secure Bootstrapping
 - Exchange of keys
- Device probing
 - All CC information are extracted and configured if needed

Z-Wave vs. ZigBee

Physical Layer		Z-Wave	ZigBee
	RF band (MHz)	868/908 (all chips) 2400 (400 series chip)	868/915/2400
	Range (m)	30 (indoors) 100 (outdoors)	10-100
	Bit rate (kb/s)	9.6/40(200 series chip) 200 (only 400 series chip)	20/40/250
	Receiver sensitivity (dBm)	-101 (at 40kb/s)	-85 (2.4GHz band) -92 (868/915 MHz bands)

Z-Wave vs. ZigBee (cont.)

Link Layer		Z-Wave	ZigBee
	MAC mechanism	CSMA/CA	TDMA + CSMA/CA (beacon mode) and CSMA/CA (beaconless mode)
	Message size(Bytes)	64 (max MAC payload in 200 series chip)	127 (max)
	Error control	8-bit check-sum, ACKs (optional)	16-bit CRC, ACKs (optional)

Z-Wave vs. ZigBee (cont.)

Communication modes		Z-Wave	ZigBee
	Unicast	Yes	Yes
	Broadcast	Yes	Yes
	Multicast	Yes	Yes (NWK and APL layers). Not supported by MAC
	Other modes	No	Indirect addressing

Z-Wave vs. ZigBee (cont.)

	Z-Wave	ZigBee
Identifiers	32-bit (home ID), 8-bit (node ID)	16-bit and 64-bit MAC addresses 16-bit NWK identifiers
Device types	Controllers and slaves	Corrdinator, router and end device

Z-Wave vs. ZigBee (cont.)

		Z-Wave	ZigBee
Network Layer	Multihop solution	Source routing	Mesh routing, tree routing and source routing
	Hop limit	4	30/10/5 (Mesh routing/tree routing/source routing)
Application Layer	Command space	32,768	65,536 (clusters)
	Device type space	N/A	65,536

Z-Wave vs. ZigBee (cont.)

	Z-Wave	ZigBee
End-to-end reliability	ACKs	ACKs and control of duplicate packets
Security	128 bit AES encryption (400 series chip)	Integrity, confidentiality, access control, and key management

- Origin:
 - First proposed by SemTech (France)
 - Now being developed by the LoRa Alliance
 - <https://www.lora-alliance.org>
 - Cisco, IBM, SemTech, ...
- Secret sauce:
 - Wideband CHIRP-like PHY layer with adaptive rate
 - Simple but effective LoRA WAN MAC protocol
- Capacity & Coverage
 - From order of 10^2 to 10^4 bit/s over up to 1-3 km in EU

LoraWAN: Low Power Wide Area Network protocol

- LoRa® is the physical layer used to create long range communication links for energy limited End Devices (EDs)
- A single gateway or base station can cover entire cities or hundreds of Km²
- LoRaWAN defines the communication protocol and system architecture
- LoRaWAN EDs are asynchronous and communicate when they have data ready to send according to an Aloha scheme. They also:
 - Provide high sensitivity (EndNodes: Up to -137 dBm, GW: up to -142 dBm)
 - Supports long range communication (3-5km in urban, up to 15 Km in rural areas)
 - Enables strong indoor penetration: With High Spreading Factor, Up to 20dB penetration (deep indoor)
 - Occupies the entire bandwidth of the channel to broadcast a signal, making it robust to channel noise.
 - Resistant to Doppler effect, multi-path and signal weakening.
- LoRAWAN gateways have a very high capacity to receive messages from a very high volume of nodes. Such capacity is achieved:
 - by utilizing adaptive data rate
 - by using a multichannel multi-modem transceiver in the gateway so that simultaneous messages on multiple channels can be received.

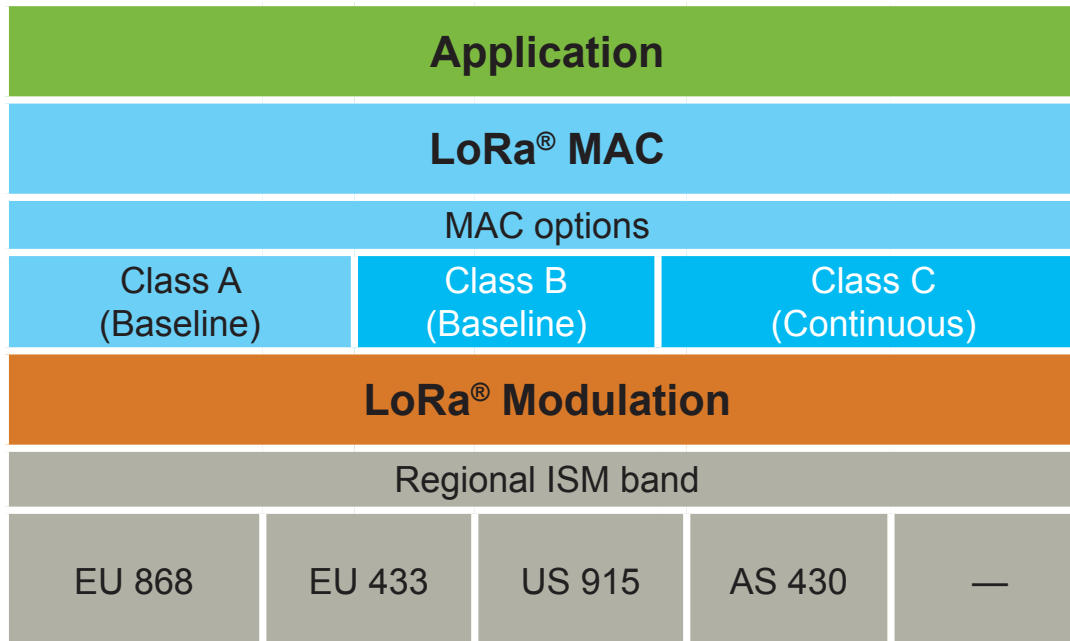
The LoRaWAN Stack

LoRa (from "long range") is a physical proprietary radio communication technique

- It was developed by Cycleo, a company of Grenoble, France, and patented in 2014 (patent 9647718-B2. Cycleo was later acquired by Semtech

LoRaWAN (Wide Area Network) defines the system architecture and is an official standard of the International Telecommunication Union (ITU), ITU-T Y.4480

Together, LoRa and LoRaWAN define a Low Power, Wide Area networking protocol designed to connect battery operated devices to the internet in regional, national or global networks



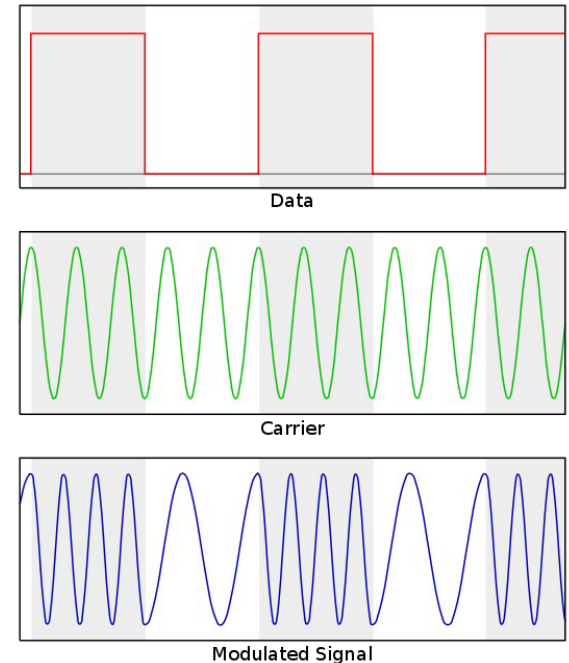
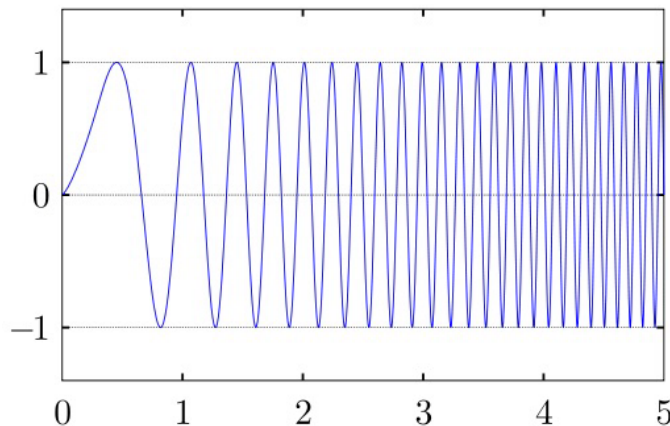
– *Open source*

– *Proprietary
By SemTech™*

– *Region-specific
Carrier frequency*

LoraWAN Modulation

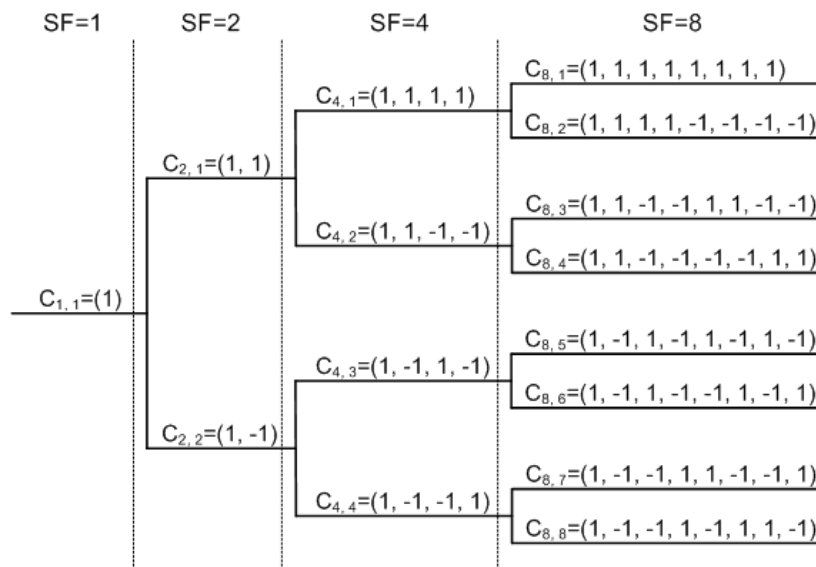
- PHY based on LoRa modulation: a version of Chirp Spread Spectrum (CSS) with a typical channel bandwidth of 125KHz
- The signal is chipped at a higher data rate and modulated onto a chirp signal



- Chirp frequency increases (up-chirp) or decreases (down-chirp) with time to encode information
- the use of broadband chirps should help against heavy multipath fading
- The chipping rate is the number of pulses per second (chips per second) at which the code is transmitted (or received).
- The ratio is known as the spreading factor (SF)

LoraWAN Modulation

- The spreading factor SF is tunable according to the application requirements
 - 7 different spreading factors: bitrates from 0.3 kbit/s to ~11 kbit/s
- Two contrasting objectives:
 - Transmission Rate: “I want to go fast” (low SF)
 - Reliability: “I want to go safe” (high SF)



Transmission rate R_b increases as SF

$$R_b = SF \frac{4}{2^{SF}} \frac{C}{BW} \quad [b/s]$$

The chip rate R_c is associated to the reference bandwidth BW

$$R_c = BW \quad [chips/s]$$

Sensitivity decreases as SF increases

$$P_{min} = -174 + 10 \log_{10} BW + NF + SNR$$

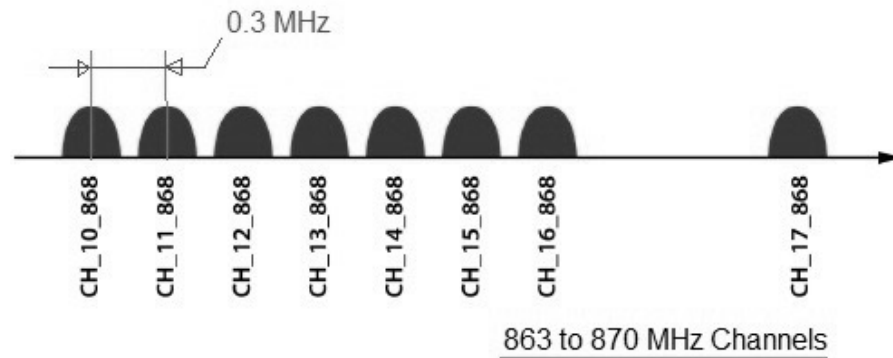
LoraWAN Modulation

- a trade-off can be made between resiliency of a message against interference, throughput and the amount of spectrum used
- The trade-off takes the form of different **modulation types**
- increasing the data rate reduces the spreading factor (Number of chirps per symbol)
- increasing the spreading factor improves sensitivity, decreases throughput and SNR grows but connection is maintained

DataRate	Configuration	Indicative physical bit rate [bit/s]
0	LoRa: SF12 / 125 kHz	250
1	LoRa: SF11 / 125 kHz	440
2	LoRa: SF10 / 125 kHz	980
3	LoRa: SF9 / 125 kHz	1760
4	LoRa: SF8 / 125 kHz	3125
5	LoRa: SF7 / 125 kHz	5470
6	LoRa: SF7 / 250 kHz	11000
7	FSK: 50 kbps	50000
8..15	RFU	

LoraWAN Channeling

- LoRaWAN operates in the sub-gigahertz frequency bands and it's specification varies from region to region because of regulatory requirement.
- LoRaWAN defines ten channels for Europe.
 - Out of which 8 channels are multi data rate from 250 bps to 5.5 kbps.
 - One channel can operate at higher data rate with a speed of 11 kbps..



- The maximum output power allowed by ETSI in Europe is +14dBm, with the exception of the G3 band which allows +27dBm. There are duty cycle restrictions under ETSI but no max transmission or channel dwell time limitations.

LoraWAN devices

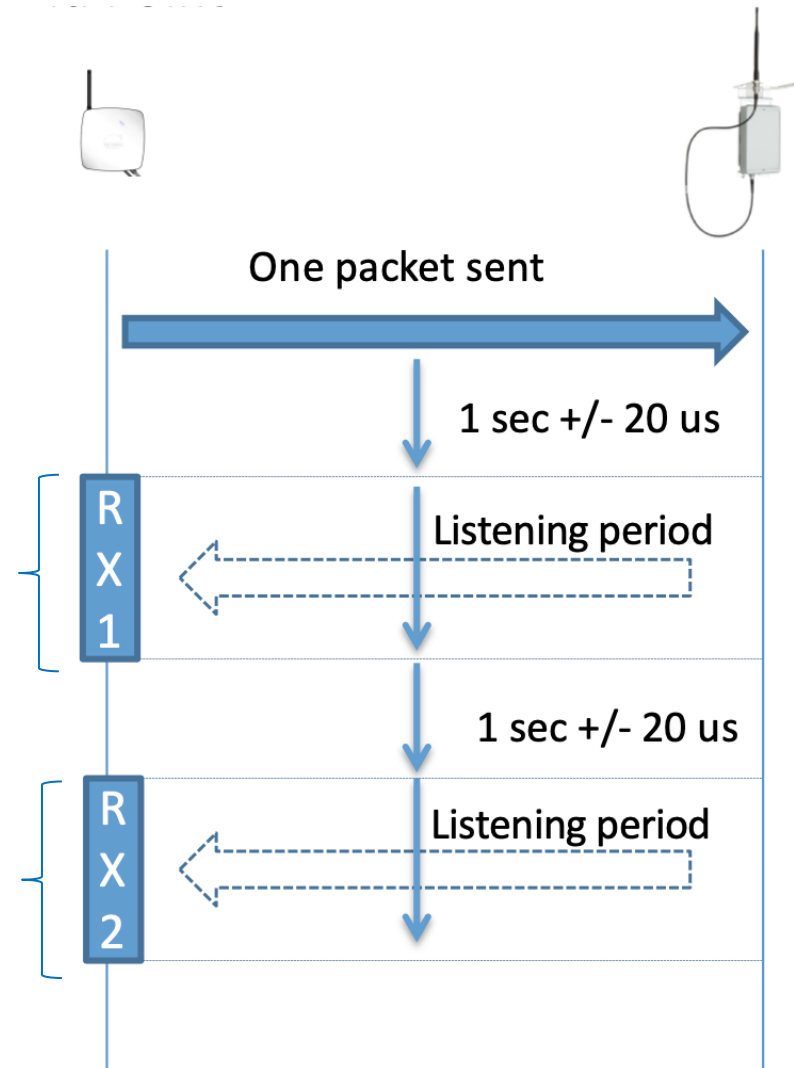
Three classes of devices:

Classes	Description	Intended Use	Consumption	Examples of Services
A (« all »)	Listens only after end device transmission	Modules with no latency constraint	The most economic communication Class energetically.. Supported by all modules. Adapted to battery powered modules	• Fire Detection • Earthquake Early Detection
B (« beacon »)	The module listens at a regularly adjustable frequency	Modules with latency constraints for the reception of messages of a few seconds	Consumption optimized. Adapted to battery powered modules	• Smart metering • Temperature rise
C (« continuous »)	Module always listening	Modules with a strong reception latency constraint (less than one second)	Adapted to modules on the grid or with no power constraints	• Fleet management • Real Time Traffic Management

LoraWAN devices

– Class A (all):

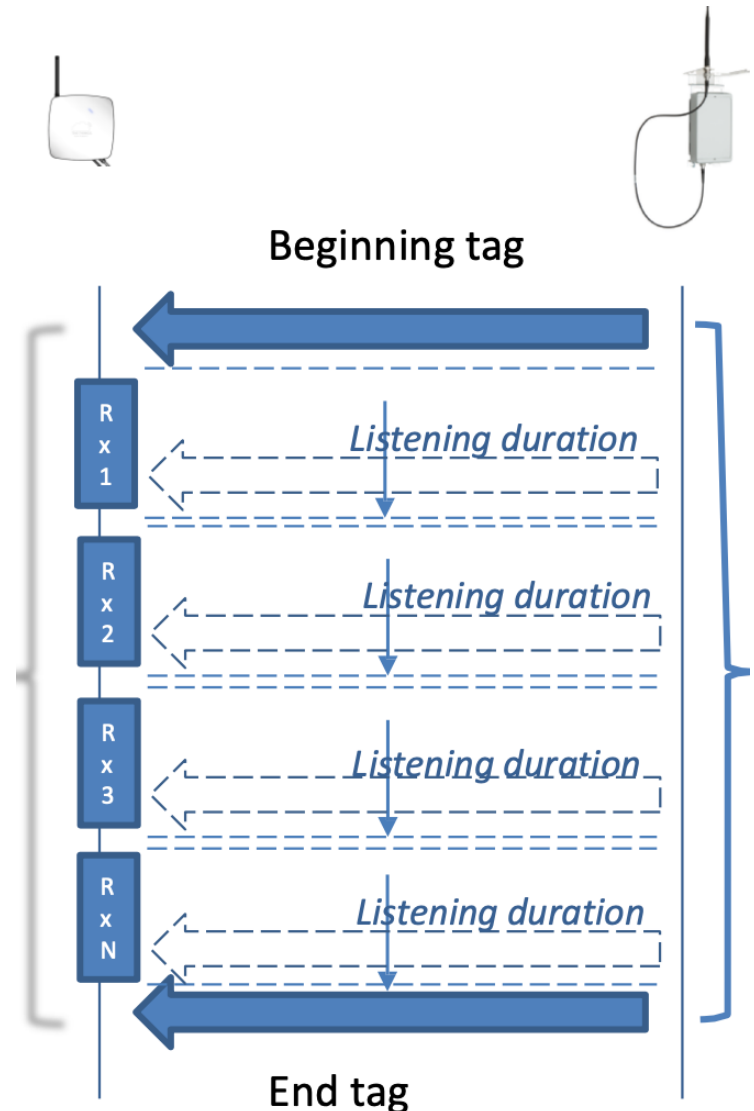
- Mandatory supported by all LoRa devices
- Adaptive modulation
- Very low energy consumption
- Suitable for monitoring services, with mostly uplink traffic
- End-devices start packet exchange with uplink tx followed by two receive windows
- 2 Downlink “slots” following uplink transmissions
 - Uplink messages (from device to server) can be sent at any time
 - The device opens a receive windows after an uplink transmission
 - If the server does not respond in this receive windows the next opportunity will be after the next uplink transmission from the device
- ALOHA-like access in the uplink



LoraWAN devices

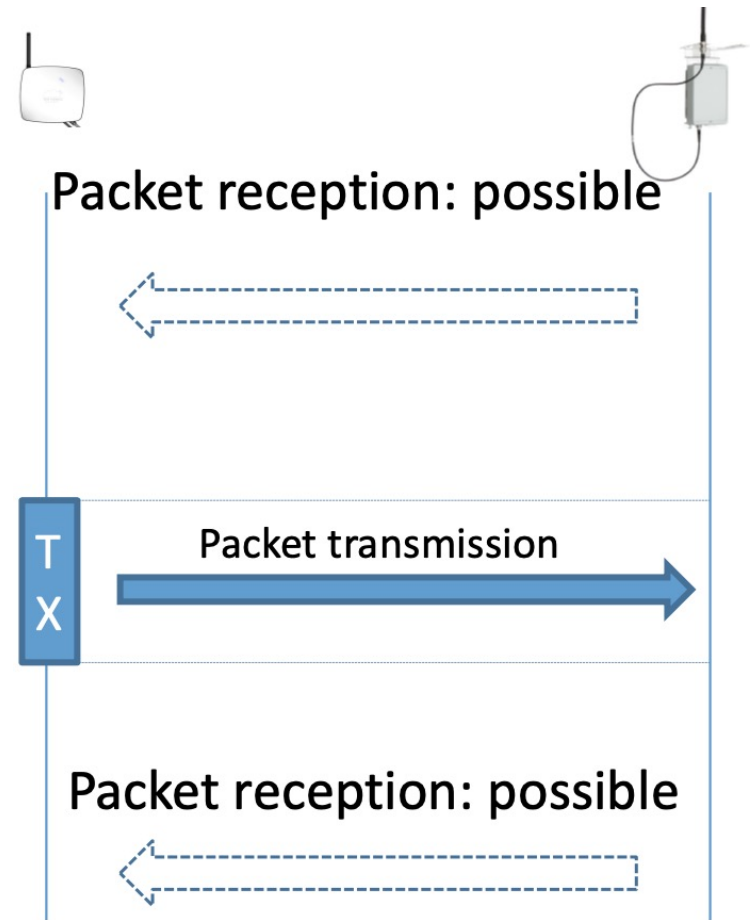
– Class B (Beacon):

- Time synchronization using beacons sent by gateways
 - receive window opens in planned periodic intervals
- Low energy consumption
 - especially for asynchronous bidirectional communications
- Suitable for switches/actuators controlled by a NetServer
 - Command latency in the order of seconds
- “extra” downlink “slots” available in a scheduled fashion
 - Using time-synchronized beacons transmitted by the gateway, the devices periodically open receive windows.

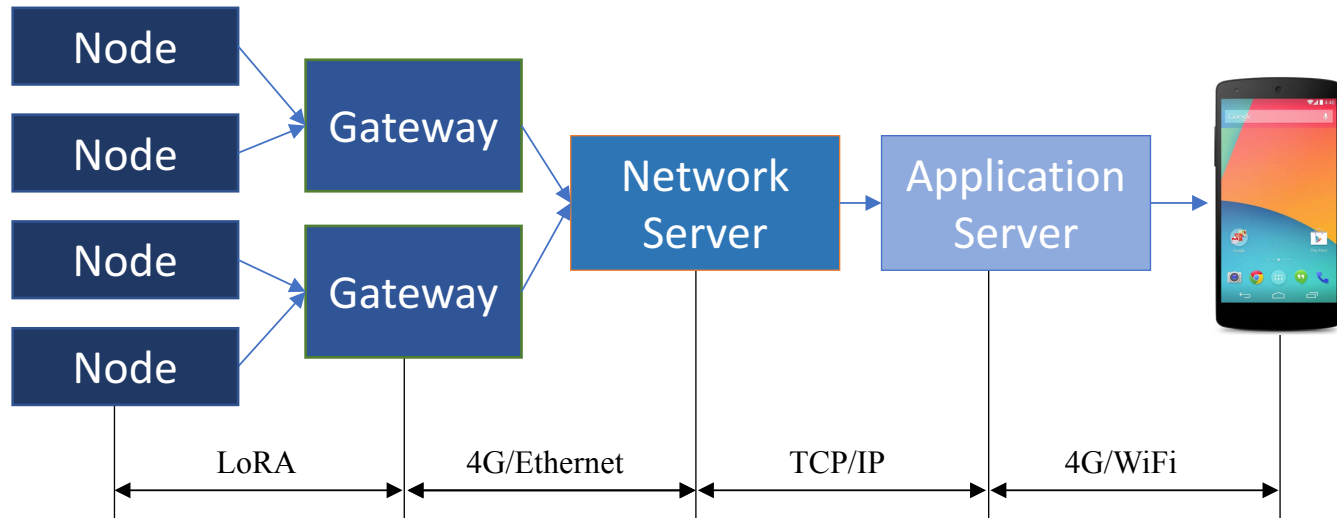


LoraWAN devices

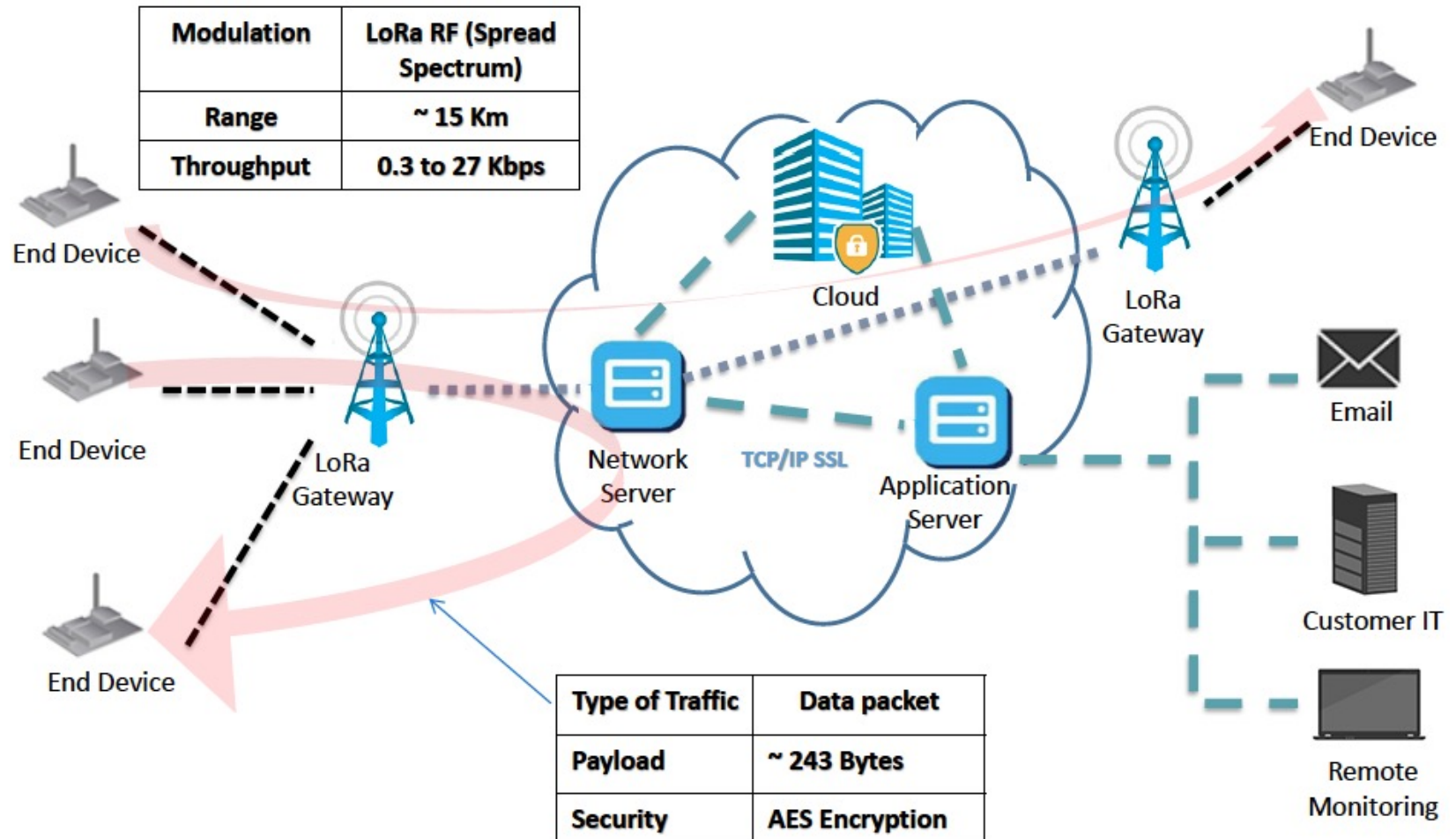
- **Class C (continuously listening):**
 - Almost continuous downlink “slots” (radio always listens)
 - Keep the receive window open unless there is need for transmitting
 - This allows for low-latency communication but is more energy consuming
 - End devices are always ready to receive signals from the gateways (except when transmitting)
 - Medium/High energy consumption at end device
 - Suitable for actuations with strict time constraints
 - Automation, control, etc...



Interactions in LoRaWAN Architecture



LoraWAN Architecture



Lorawan has star topology and consists of BSs relaying data messages between the EDs and an application server

LoraWAN Architecture Components

End-Devices

- A simple ALOHA (random access) scheme is used at the MAC Layer.
- Multiple devices can communicate at the same time on the medium, by using different frequencies or spreading factors (LoRA PHY).
- End-devices are not associated with any specific gateway; data transmitted can be received by multiple gateways.
- Messages sent can be non-confirmable or confirmable (i.e. ACK requested).
- Retransmissions may occur but at the discretion of the end devices.

Gateways

- receive data from multiple clients on the LoRa link, and forward them to a network server via a backhaul (wired) connection.
- In order to make a long range star network viable, it must have a very high capacity

Network Server (back-end)

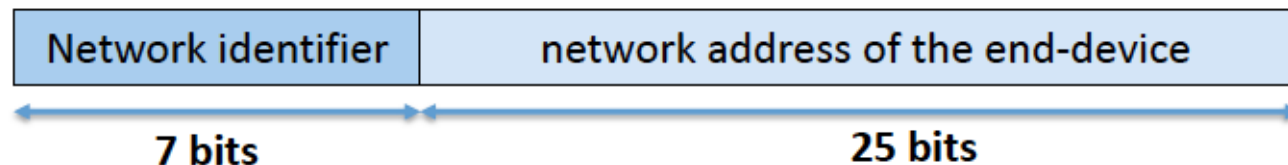
- Performs duplicate detection and filtering.
- Implements the Adaptive Data Rate control of the LoRa devices.
- Decrypts messages and routes them to the application server.

Application Server (back-end)

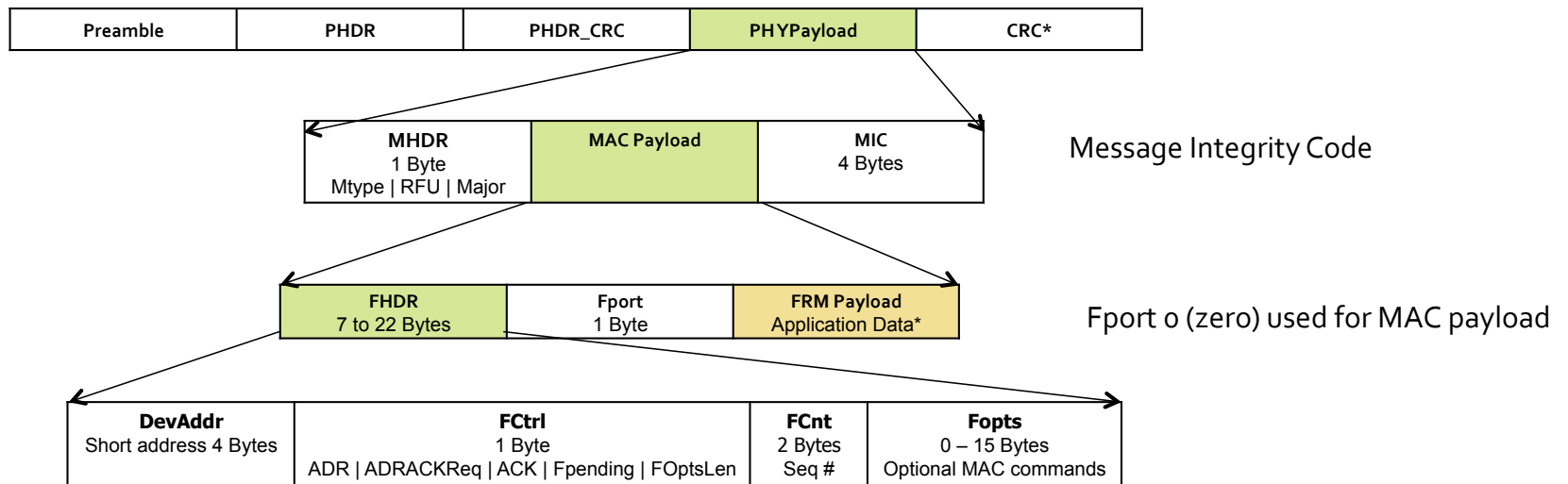
- Performs application-specific data processing

Device identification/addressing

- The DevEUI is an ID in the IEEE EUI64 address space used to identify a device. It is supplied by the device manufacturer.
- (Application or Join Server identifier (AppEUI or JoinEUI): A global application ID in the IEEE EUI64 address space identifying the join server during the over the air activation.
 - This server name is used to find the IP address of the join server via DNS.that uniquely identifies the owner of the end-device.
- The AppKey is an AES-128 root key specific to the end-device
 - Whenever an end-device joins a network via over-the-air activation, the AppKey is used to derive the session keys specific to that end-device to encrypt/verify network communication and application data.
- When a device joins the network, it receives a dynamic (non-unique) 32-bit address End-device address (DevAddr)



LoRaWAN Frame



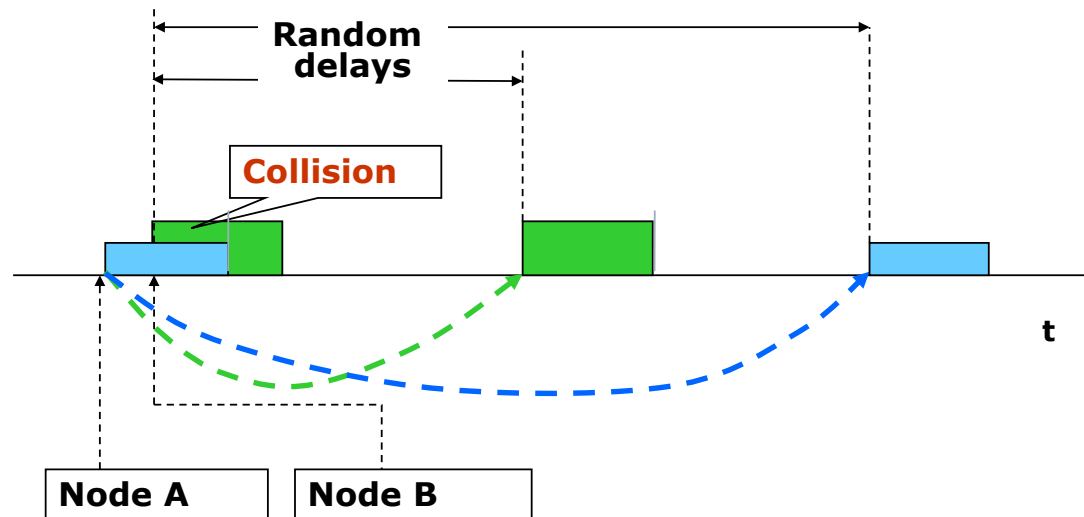
- *Preamble*: used for synchronization and to define the packet modulation scheme
- *Physical HeaDeR (PHDR)*
- *PHDR_CRC, CRC*: fields used for integrity check
- *PHYPayload*: PHY message content
- Mtype: message type
- DevAddr: end device address
- FCtrl: frame control field to manage adaptive data rate, ACK and MAC commands
- FCnt: frame counter to count up each transmitted frame
- Fopts: frame options containing MAC commands

LoRaWAN Messages

- **Uplink / downlink messages**
 - Uplink messages are sent by end-devices to the network server
 - Downlink messages are sent by the network server to the end-devices.
- **Message types : Confirmed / unconfirmed / proprietary**
 - Data messages (both downlink & uplink) can be confirmed or unconfirmed. A confirmed
 - message has to be acknowledged by the receiver.
 - Proprietary messages type can be used to implement non-standard message formats.
- **MAC commands**
 - Network commands added to a message or sent in a separate frame, based on request/answer
 - between end-device and network server

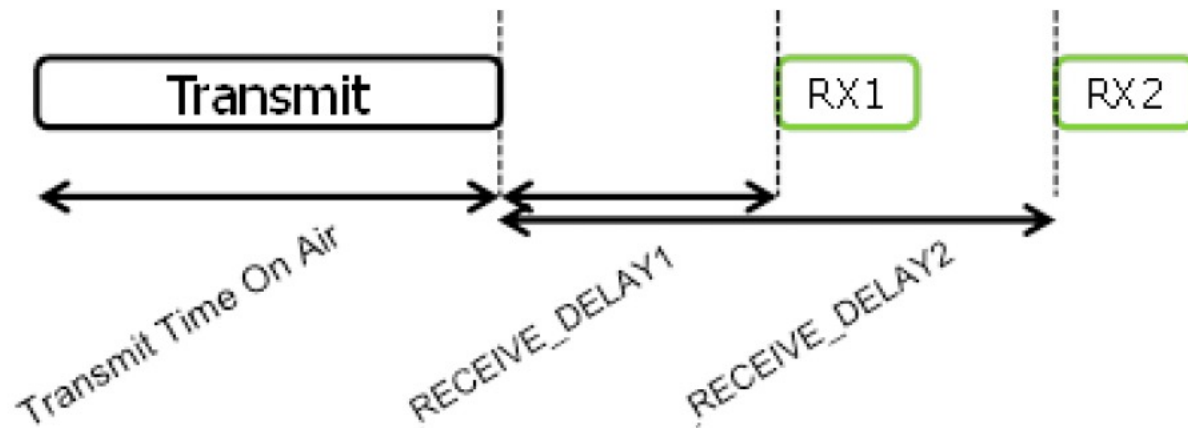
LoRaWAN ALOHA

- No channel feedback required, only the ACK
 - Time is continuous
 - Protocol:
 - The first packet in the transmission queue is
 - transmitted as soon as ready
 - If the ACK does not come, the transmission is re-attempted after a random number of slots X



Receive Window

- Following each uplink transmission, the end-device SHALL open one or two receive windows (RX1 and RX2); if no packet destined for the end-device is received in RX1, it SHALL open RX2.



Security

- The security scheme used by LoRaWAN is based on AES-128, similar to the algorithm used in the 802.15.4 standard.
- Three distinct 128-bit security keys are used:
 - The application key AppKey is only known by the device and by the application.
 - When a device joins the network (this is called a join or activation), an application session key AppSKey and a network session key NwkSKey are generated.
 - The NwkSKey is shared with the network, while the AppSKey is kept private. These session keys will be used for the duration of the session.
 - The NwkSKey is used to validate the integrity of each message by its Message Integrity Code (MIC). This MIC is similar to a checksum, except that it prevents intentional tampering with a message. For this, LoRaWAN uses AES-CMAC
 - The AppSKey is used for encryption of the application payload.



- Origin:
 - Founded in 2009, and growing very fast since then
 - First LPWAN technology proposed in the IoT market
 - \$115 million from investors as Telefonica and Vodafone
- Secret sauce: proprietary network layer protocols
 - UNB modulation, SDR860 band (Europe)
- Coverage:
 - 30-50 km in rural areas and 3-10 km in urban areas

SIGFOX coverage

Source: <http://www.sigfox.com/en/#!/connected-world/sigfox-network-operator>

Current network coverage and network deployment priorities :



Netherlands



Spain



Luxembourg



United kingdom



Denmark



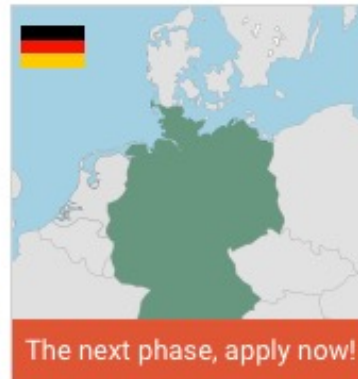
Portugal



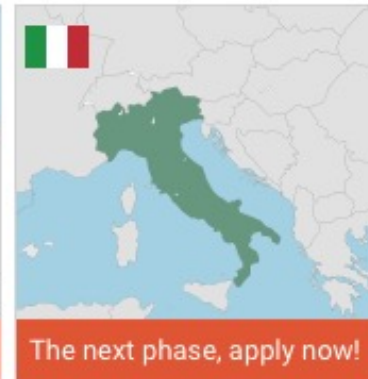
Belgium



France



Germany



Italy

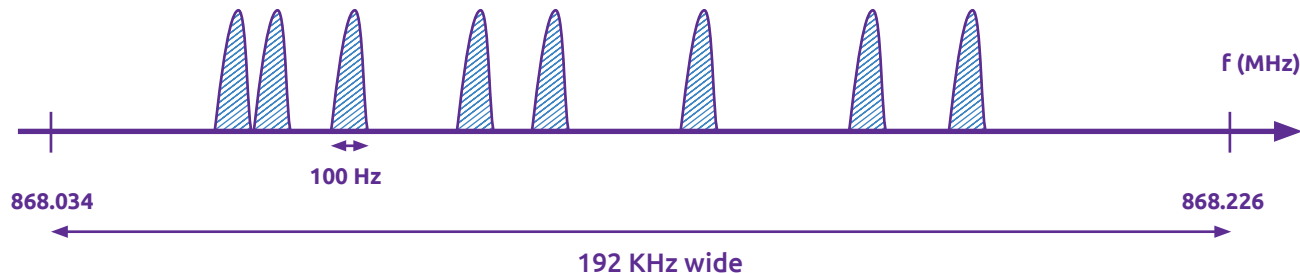
Sigfox

Main features:

- It supports billions of objects broadcasting data, without the need to establish and maintain network connections
- Low throughput (~ 100 bps), Low power, Extended range (up to 50 km)
- 140 messages/day/device
- Subscription-based model with Roaming capability
- Cloud platform with Sigfox-defined API for server access
- The band used depends on the location:
 - in Europe, between 868 and 868.2 MHz;
 - in the rest of the world, between 902 and 928 MHz

Sigfox: spectrum usage

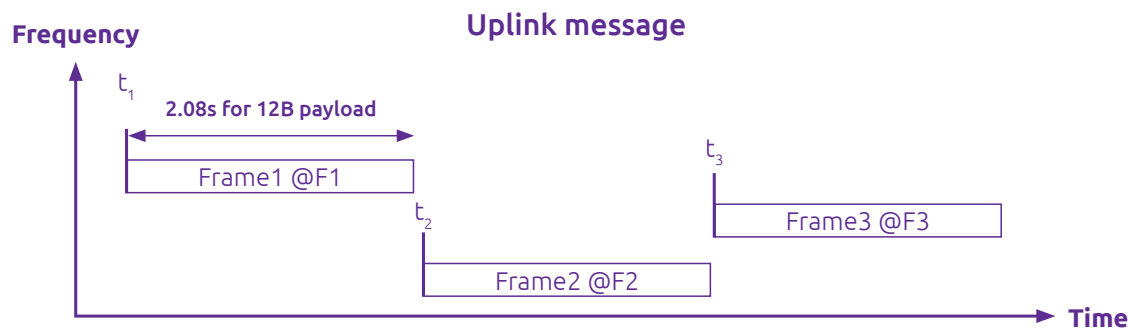
- Rather than spreading the energy across a wide channel, a narrow slice of that energy is confined in these bands
- Ultra-Narrowband wireless modulation takes very narrow parts of spectrum and changes the phase of the carrier radio wave to encode the data
- UNB is extremely robust to interference and jamming
- Sigfox is using 192KHz of the band to exchange messages
- Messages are modulated using ultra-narrowband GFSK (FSK passing pulses through a gaussian filter), each using 100 Hz



- The Sigfox base stations and receivers monitor the full 192 kHz spectrum and look for UNB signals to be demodulated

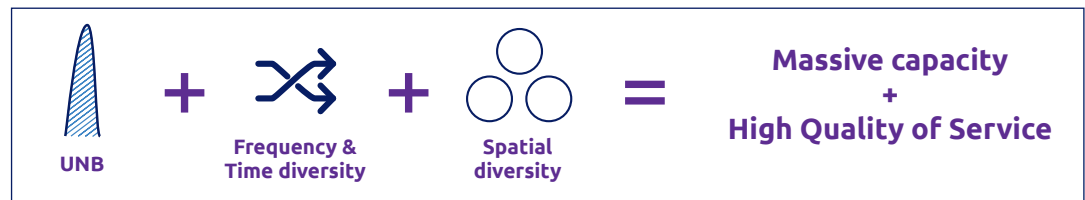
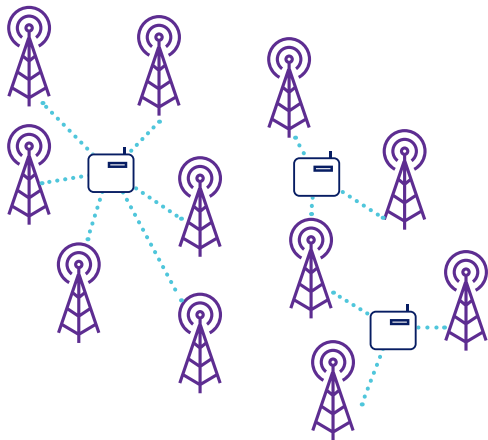
Sigfox: transmission encoding

- Unsynchronized transmission between network and devices
- A device emits an uplink message F1 on a random frequency and then sends 2 replicas (F2 and F3) on different frequencies and time, which is called “time and frequency diversity”
- This type of encoding is called Random Frequency and Time Division Multiple Access (RFTDMA).
 - Each message is 100 Hz wide and transferred with a data rate of 100 or 600 bits per second depending on the region.
 - The message size goes from 0 to 12 bytes
 - a random time delay (500 to 525 ms) is introduced

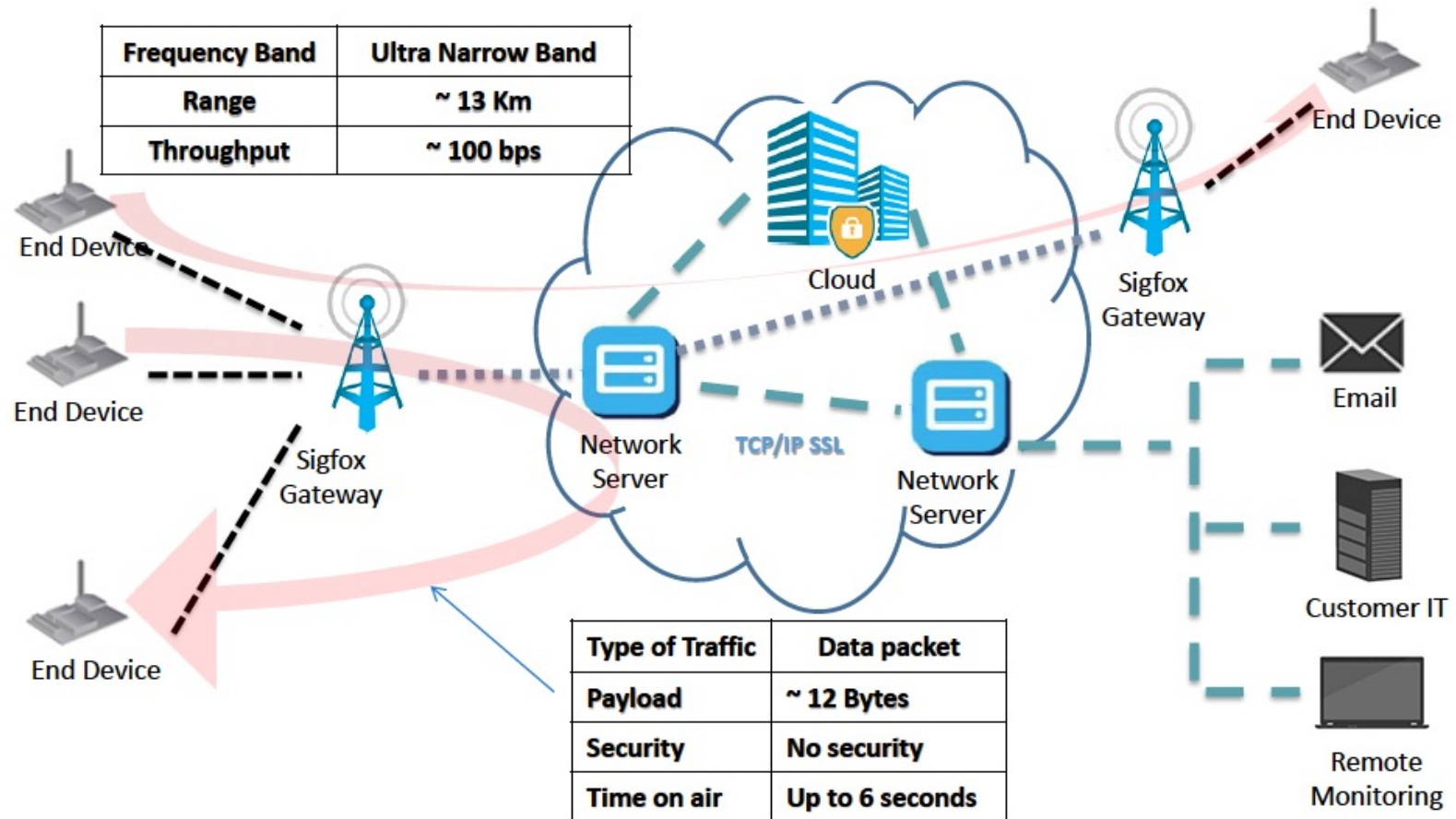


Sigfox: diversity

- An object is not attached to a specific base station unlike cellular protocols.
- The emitted message is received by any base stations that are nearby and on average the number of base stations is 3.
- This is called “spatial diversity”.
- Spatial diversity coupled with the time and frequency diversity of the repetitions are the main factors behind the high quality of service of the Sigfox network.
- These factors improve the capacity of the network enabling Sigfox to scale for the billions of objects.



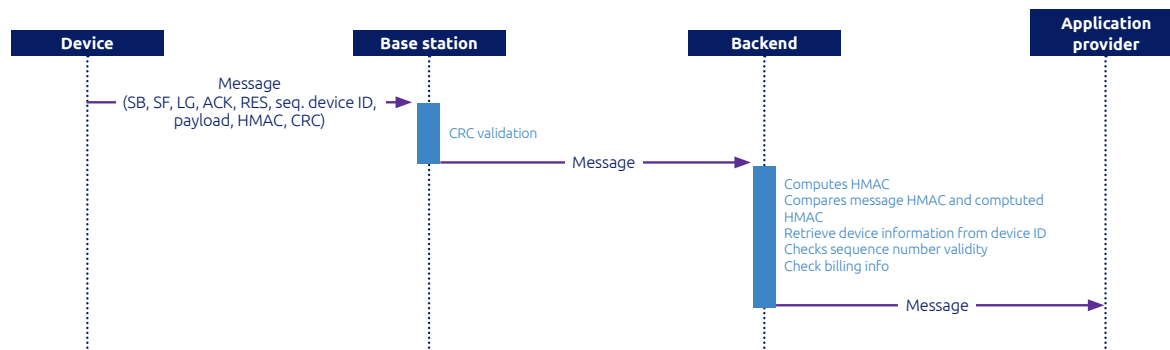
Sigfox Architecture



- The Network consists of bases stations (and associated antennas) in charge of receiving messages from devices and transferring them to Sigfox network servers
- The servers constitute the core network in charge of processing the messages and send them to the customer system available over the cloud

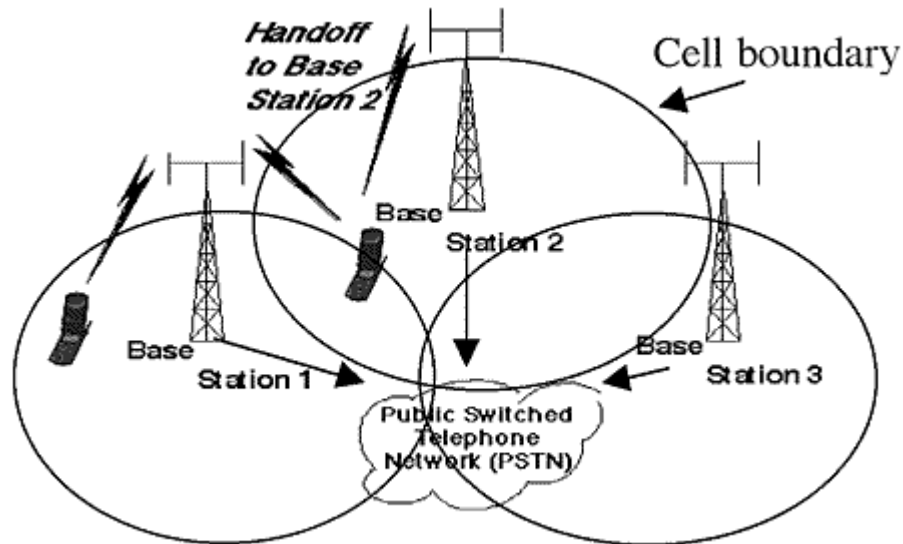
Sigfox: security

- sequence number is used for anti-replay (associated with MAC)
- Each device has a unique symmetrical authentication key
- Each message to be sent or received by the device contains a cryptographic token based on this authentication key.
- Verification of the token ensures:
 - authentication of the sender
 - integrity of the message.
- authentication of communications between the Core Network and application servers relies on VPN or HTTPS
- By default, data is conveyed over the air without any encryption
- Simple encryption is implemented deriving the key from the device key



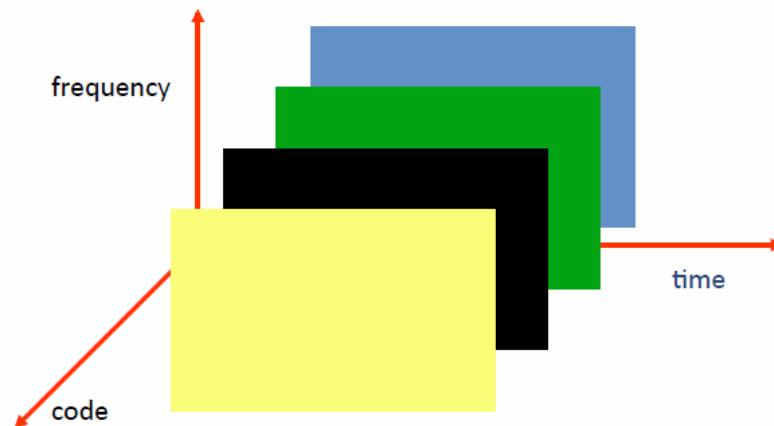
Cellular Networks

- Base stations transmit to and receive from mobile stations at the assigned spectrum
 - Multiple base stations use the same spectrum (spectral reuse)
- The service area of each base station is called a cell
- Each mobile terminal is typically served by the 'closest' base stations
 - Handoff when terminals move



The Multiple Access Problem

- The base stations need to serve many mobile terminals at the same time (both downlink and uplink)
- All mobiles in the cell need to transmit to the base station
- Interference among different senders and receivers
- We need multiple orthogonal access schemes:
 - Frequency Division Multiple Access (FDMA)
 - Time Division Multiple Access (TDMA)
 - Code Division Multiple Access (CDMA)



Cellular Networks Generations

	1G	2G	3G	4G	5G
3GPP releases			4 - 7	8-9, 10-14	15, 16
era	1980s	1990s	2000s	2010s	2020s
services	analog voice	digital voice messages	WB voice packet data	voice, video always on Internet	everything
devices					
data rate	0	100 kbps (GSM/GPRS)	10 Mbps (HSPA)	100+ Mbps (LTE/LTE-A)	10 Gbps (NR)
delay		500 ms	100 ms	10s ms	5 ms

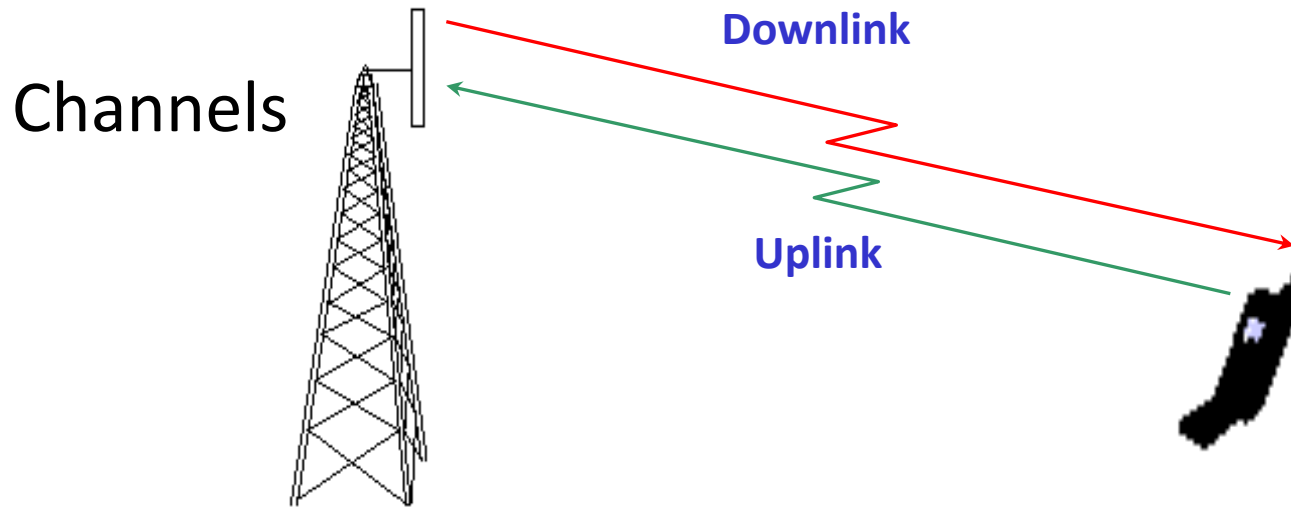
GSM

Abbreviation for Global System for Mobile Communications

Services

- Voice, 3.1 kHz
- Short Message Service (SMS)
 - 160 chars. to be sent between handsets and other stations
- General Packet Radio Service (GPRS)
 - GSM upgrade that provides IP-based packet data transmission up to 114 kbps
 - Users can “simultaneously” make calls and send data
 - GPRS provides “always on” Internet access and the Multimedia Messaging Service (MMS) whereby users can send rich text, audio, video messages to each other
 - Performance degrades as number of users increase
 - GPRS is an example of 2.5G telephony – 2G service similar to 3G

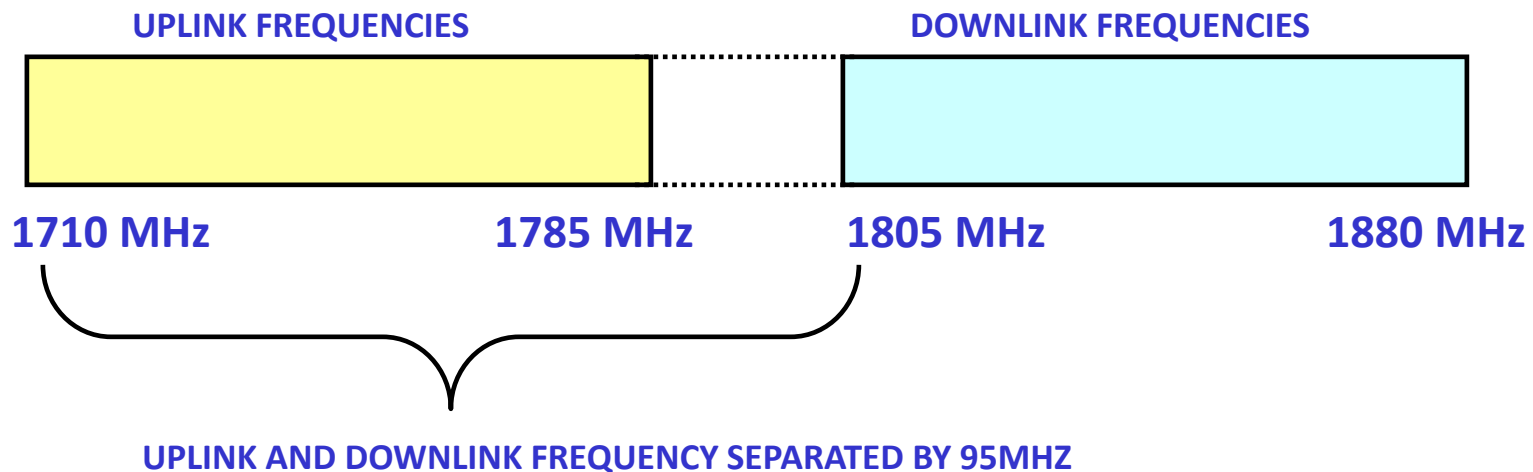
GSM Channels



- Physical Channel: Each timeslot on a carrier is referred to as a physical channel
- Logical Channel: Variety of information is transmitted between the MS and BTS. Different types of logical channels:
 - Traffic channel
 - Control Channel

GSM Frequencies

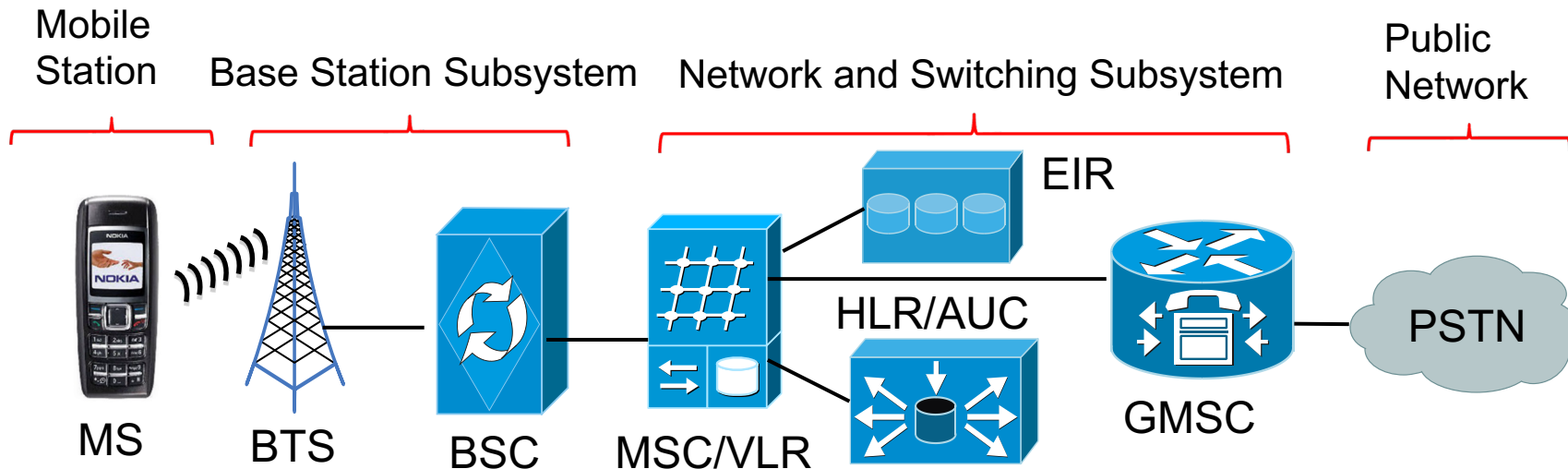
- Originally designed on 900MHz range, now also available on 800MHz, 1800MHz and 1900 MHz ranges.
- Separate Uplink and Downlink frequencies
 - One example channel on the 1800 MHz frequency band, where RF carriers are space every 200 MHz



GSM Architecture

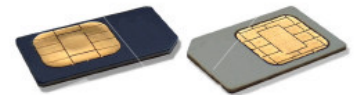
Consists of 3 subsystems:

- Mobile Station
- Base Station Subsystem
- Network/Switching Subsystem



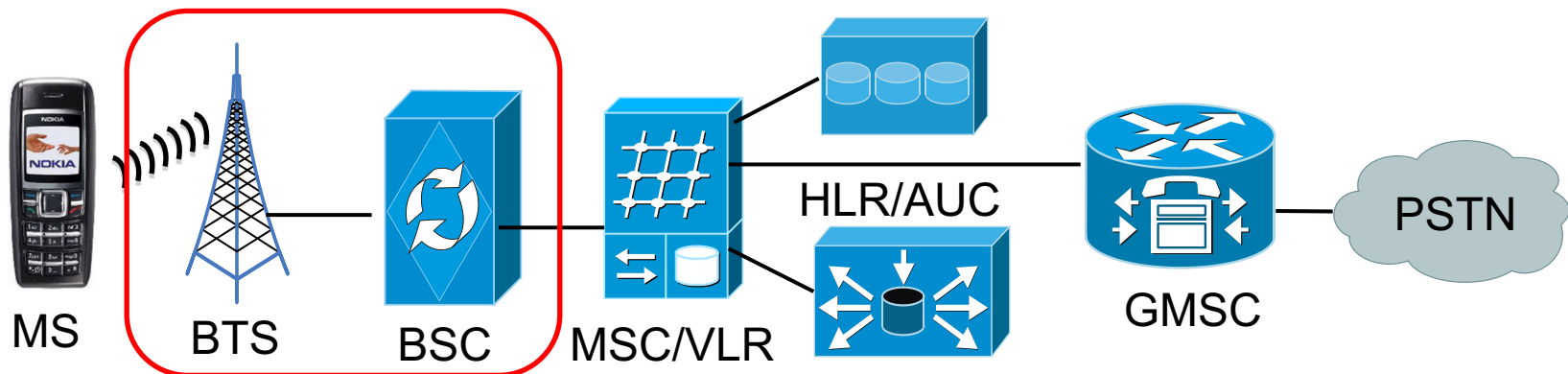
Mobile Station (MS)

- Mobile Equipment
 - Radio equipment
 - User interface
 - Processing capability and memory required for various tasks
 - Call signalling
 - Encryption
 - SMS
 - Equipment IMEI number
- Subscriber identity module
 - A small smart card
 - Encryption codes needed to identify the subscriber
 - Subscriber IMSI number
 - Subscriber's own information (telephone directory)
 - Third party applications (banking etc.)
 - Can also be used in other systems besides GSM, e.g., some WLAN access points accept SIM based user authentication



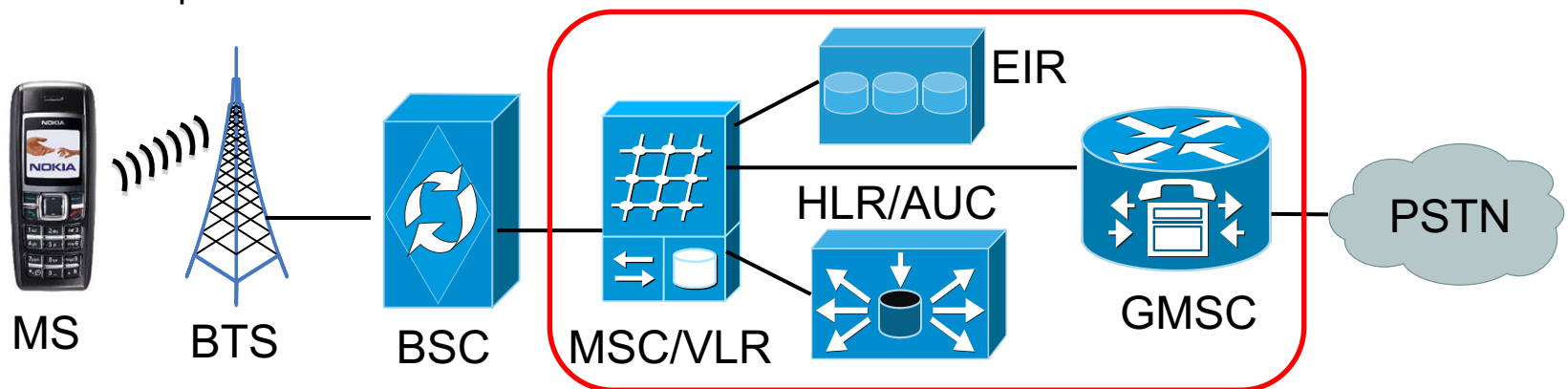
Base Station Subsystem

- Transcoding Rate and Adaptation Unit (TRAU)
 - Performs coding between the 64kbps PCM coding used in the backbone network and the 13 kbps coding used for the Mobile Station (MS)
- Base Transceiver System (BTS)
 - Controls several transmitters
 - Each transmitter has 8 time slots, some used for signaling, on a specific frequency
- Base Station Controller (BSC)
 - Controls the channel (time slot) allocation implemented by the BTSes
 - Manages the handovers within BSS area
 - Knows which mobile stations are within the cell and informs the MSC/VLR about this



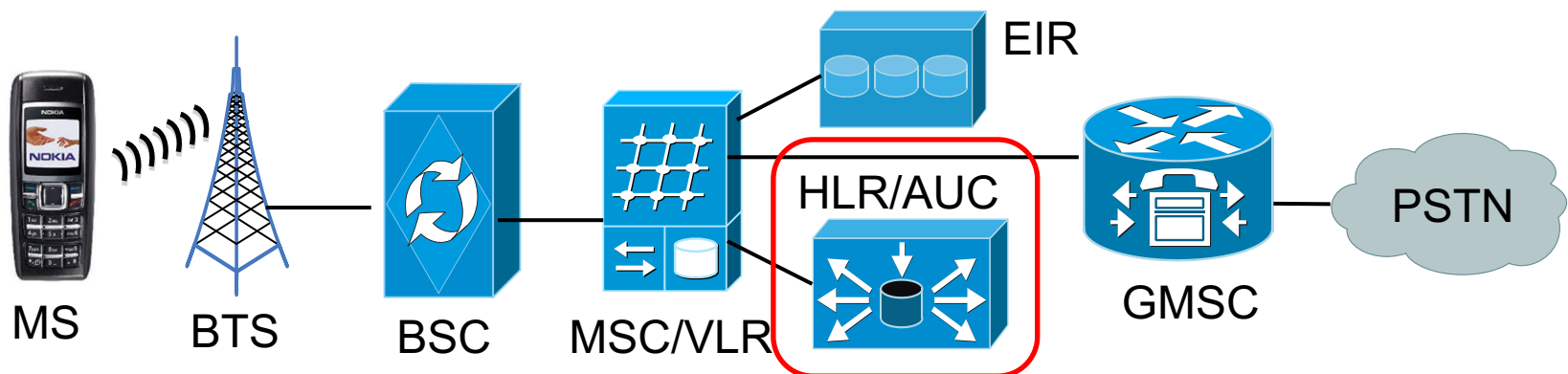
Network and Switching Subsystem

- The backbone of a GSM network is a telephone network with additional cellular network capabilities
- Mobile Switching Center (MSC)
 - An typical telephony exchange (ISDN exchange) supporting mobile communications
 - Visitor Location Register (VLR)
 - A database, part of the MSC containing the location of the active Mobile Stations
- Gateway Mobile Switching Center (GMSC)
 - Links the system to PSTN and other operators
- Home Location Register (HLR)
 - Contain subscriber information, including authentication information in Authentication Center (AuC)
- Equipment Identity Register (EIR)
 - International Mobile Station Equipment Identity (IMEI) codes for e.g., blacklisting stolen phones



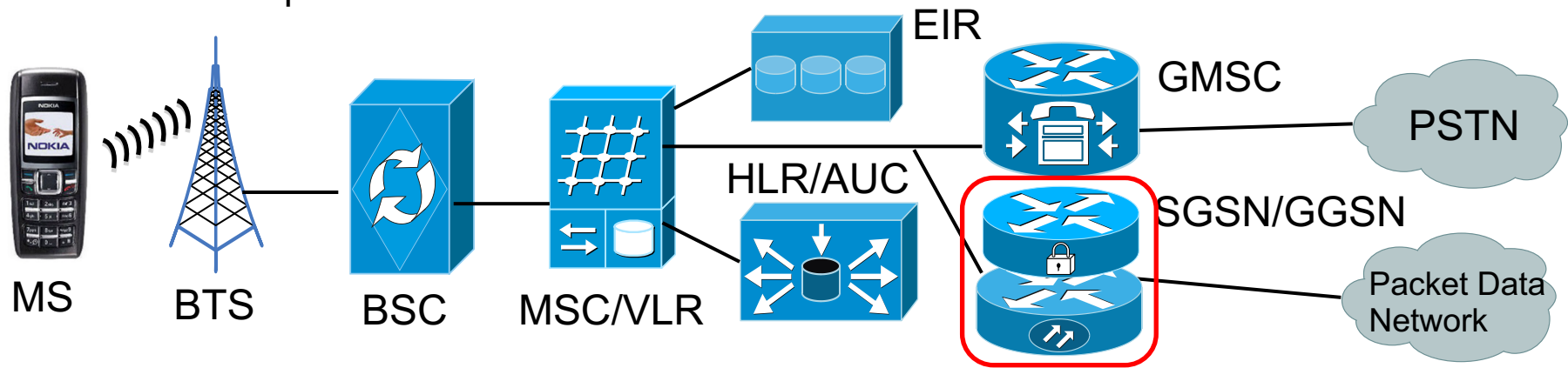
Home Location Register

- One database per operator
- Contains all the permanent subscriber information
 - MSISDN (Mobile Subscriber ISDN number) is the telephone number of the subscriber
 - International Mobile Subscriber Identity (IMSI) is a 15 digit code used to identify the subscriber
 - It incorporates a country code and operator code
 - IMSI code is used to link the MSISDN number to the subscriber's SIM (Subscriber Identity Module)
 - Charging information
 - Services available to the customer
- Also the subscriber's present Location Area Code, which refers to the MSC, which can connect to the MS.



GPRS Service Nodes

- Serving GPRS Support Node (SGSN) is responsible for:
 - Admission control
 - Routing, mobility management, location management, authentication, charging
 - Encapsulation, Receiving and delivering data packets
 - Address translation and mapping
- Gateway GPRS Support Node (GGSN)
 - acts as interface between GPRS backbone and external Packet Data Networks (PDN) or other Public Mobile Land Networks
 - converts GPRS packets coming from SGSN into the appropriate packet data protocol (PDP) format (e.g. IP)
 - converts the PDP addresses of incoming data packets to the GSM address of the destination user, and sends the readdressed packets to the responsible SGSN

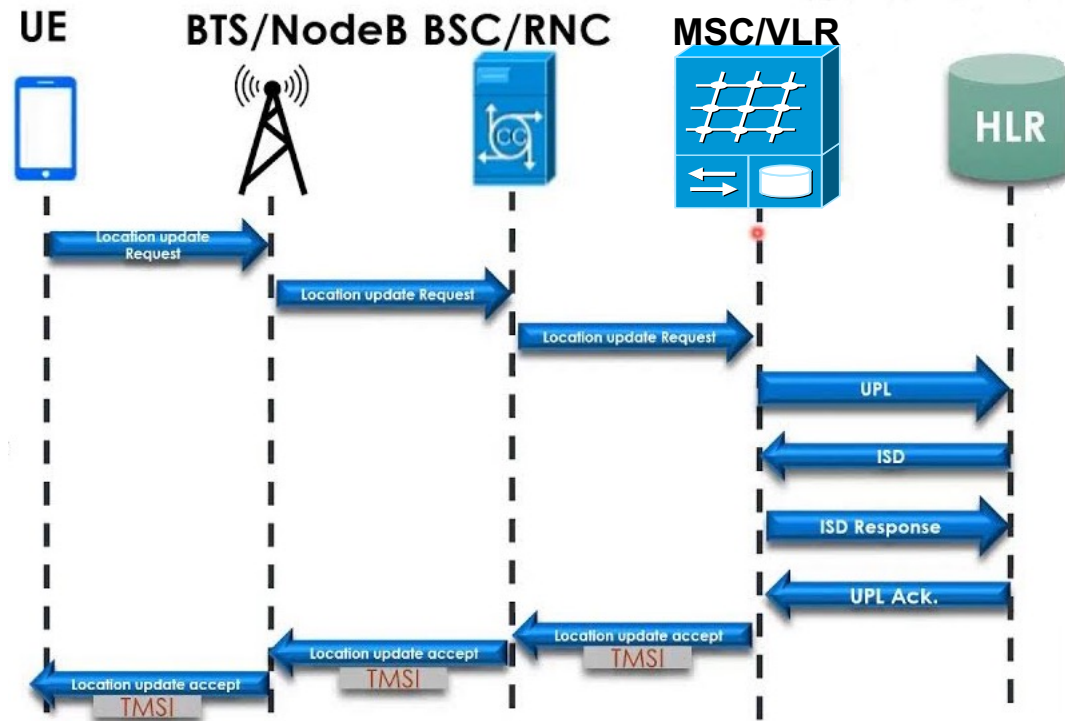


Other Systems

- Operations Support System
 - The management network for the whole GSM network
 - Usually vendor dependent
 - Very loosely specified in the GSM standards
- Value added services
 - Voice mail
 - Call forwarding
 - Group calls
- Short Message Service Center
 - Stores and forwards the SMS messages
 - Like an E-mail server
 - Required to operate the SMS services

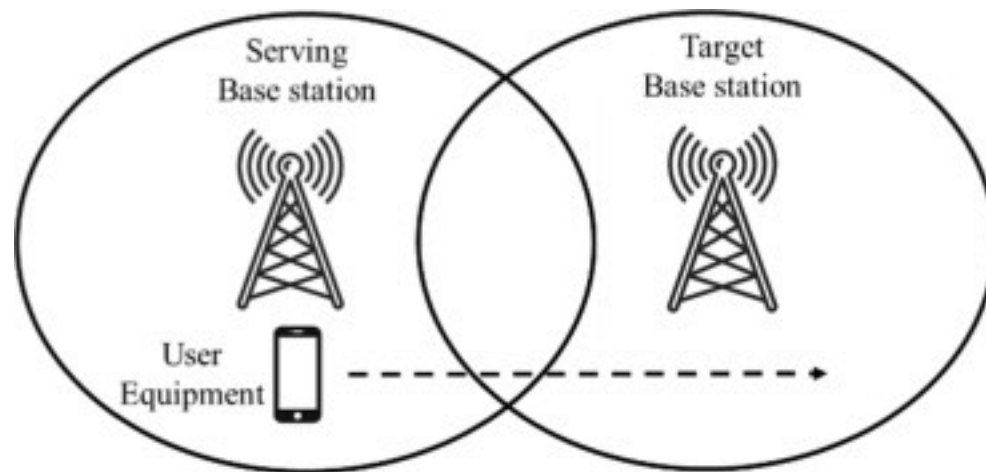
Location Updates

- The cells overlap and usually a mobile station can 'see' several transceivers (BTSEs)
- The MS monitors the identifier for the BSC controlling the cells
- When the mobile station reaches a new BSC's area, it requests an location update
- The update is forwarded to the MSC, entered into the VLR, after confirmation from the HLR, the old BSC is notified and an ACK is passed back



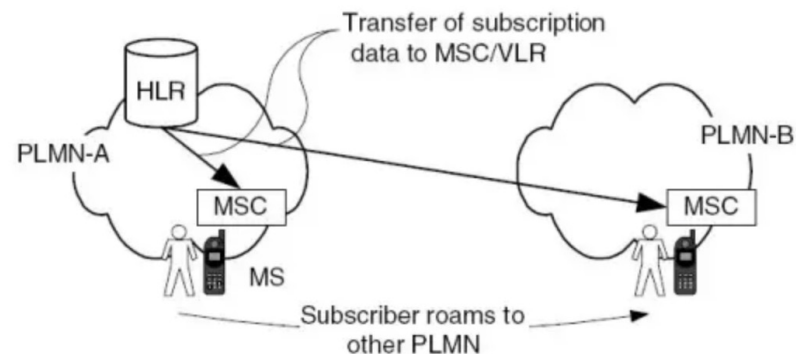
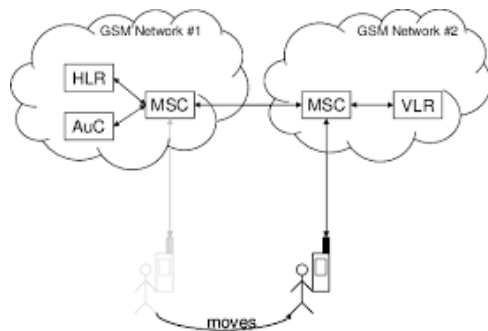
Handoff (Handover)

- When a call is in process, the changes in location need special processing
- Within a BSS, the BSC, which knows the current radio link configuration (including feedbacks from the MS), prepares an available channel in the new BTS
- The MS is told to switch over to the new BTS
- This is called a hard handoff
 - In a soft handoff, the MS is connected to two BTSes simultaneously and passes seamlessly to the new one



Roaming

- When a MS enters another operators network, it can be allowed to use the services of this operator
 - Operator to operator agreements and contracts
 - Higher billing
- The MS is identified by the information in the SIM card and the identification request is forwarded to the home operator
 - The home HLR is updated to reflect the MS's current location



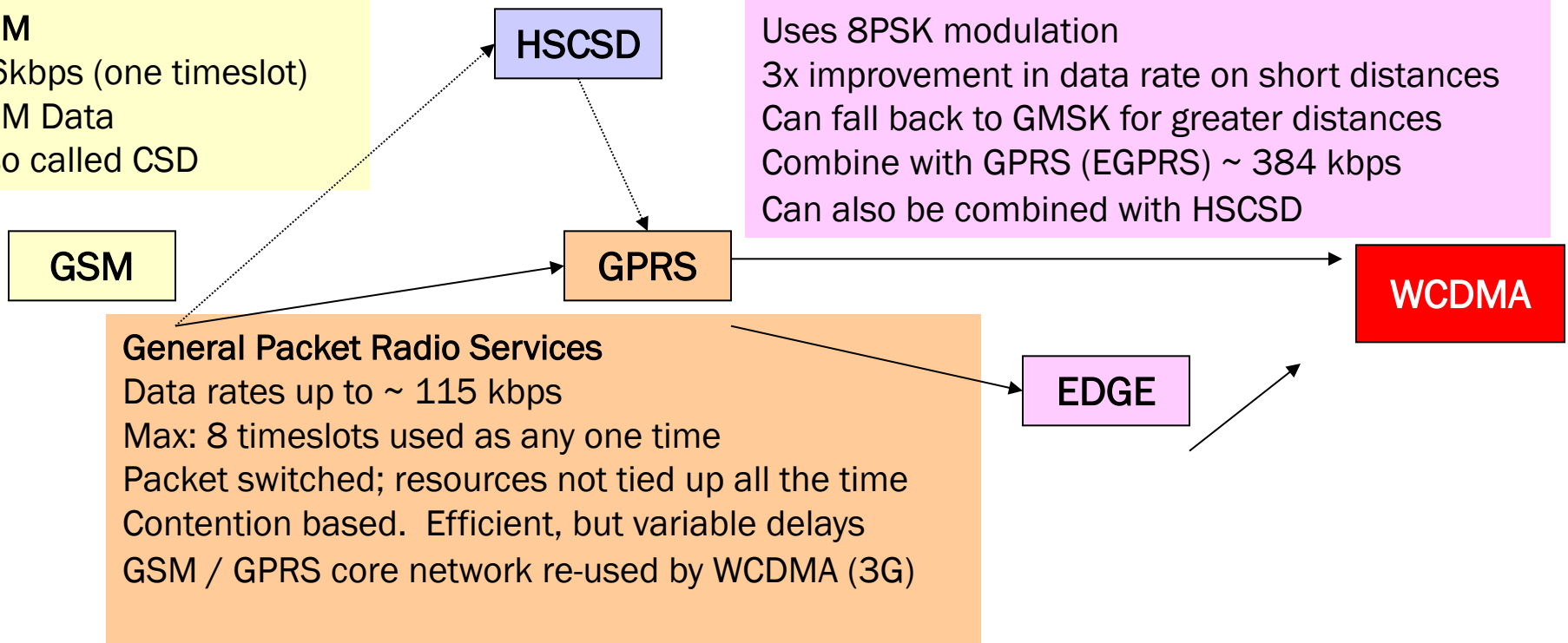
GSM Evolution to 3G

High Speed Circuit Switched Data

Dedicate up to 4 timeslots for data connection ~ 50 kbps
Good for real-time applications c.w. GPRS
Inefficient -> ties up resources, even when nothing sent
Not as popular as GPRS (many skipping HSCSD)

Enhanced Data Rates for Global Evolution

Uses 8PSK modulation
3x improvement in data rate on short distances
Can fall back to GMSK for greater distances
Combine with GPRS (EGPRS) ~ 384 kbps
Can also be combined with HSCSD



Service Evolution Roadmap

Improved performance, decreasing cost of delivery



A number of mobile services are bearer independent in nature

3G-specific services take advantage of higher bandwidth and/or real-time QoS

Broadband in wide area

Video sharing
Video telephony
Real-time IP multimedia and games
Multicasting

Multitasking
WEB browsing
Corporate data access
Streaming audio/video

MMS picture / video
xHTML browsing
Application downloading
E-mail
Presence/location
Push-to-talk

Voice & SMS

GSM
9.6
kbps

GPRS
171
kbps

EGPRS
473
kbps

WCDMA
2
Mbps

HSDPA
1-10
Mbps

CDMA
2000 1x

CDMA
2000-
EVDO

CDMA
2000-
EVDV

Typical average bit rates
(peak rates higher)

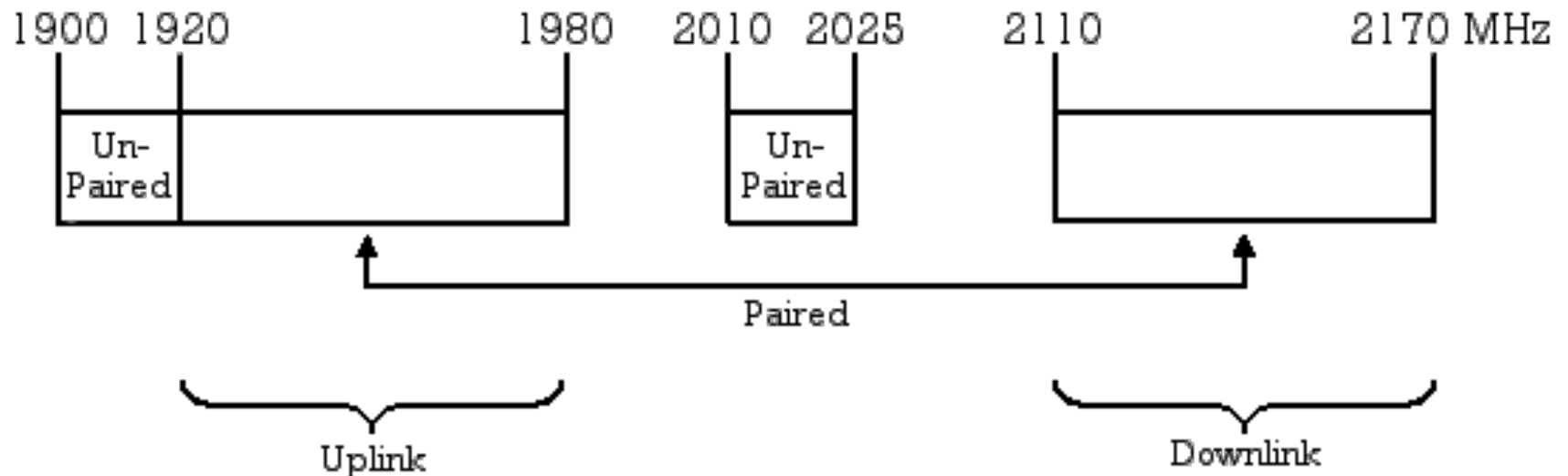
UMTS

- Universal Mobile Telecommunications System (UMTS)
- UMTS is an upgrade from GSM via GPRS or EDGE
- The standardization work for UMTS is carried out by Third Generation Partnership Project (3GPP)
- Data rates of UMTS are:
 - 144 kbps for rural
 - 384 kbps for urban outdoor
 - 2048 kbps for indoor and low range outdoor
- Virtual Home Environment (VHE)

UMTS Frequency Spectrum

– UMTS Band

- 1900-2025 MHz and 2110-2200 MHz for 3G transmission
- In the US, 1710–1755 MHz and 2110–2155 MHz will be used instead, as the 1900 MHz band was already used.



UMTS Network Architecture

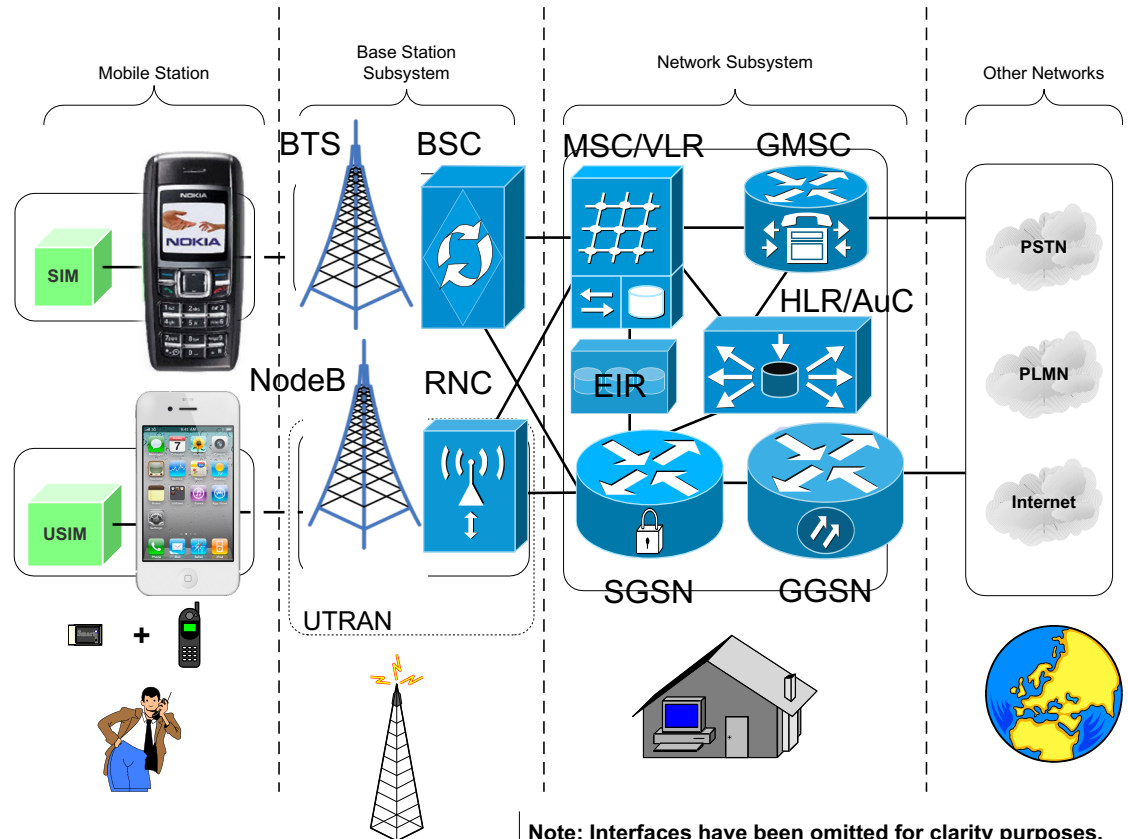
- Core Network subsystem: provide switching, routing and transit for traffic
- UMTS Terrestrial Radio Access Network (UTRAN): Provides the air interface access method for user equipment (UE).
- Base stations are referred to as Node-B and control equipment for Node-B is called as Radio Network Controller (RNC).

- Functions of Node-B:

- Air Interface Tx/Rx
- Modulation and Demodulation

- Functions of RNC:

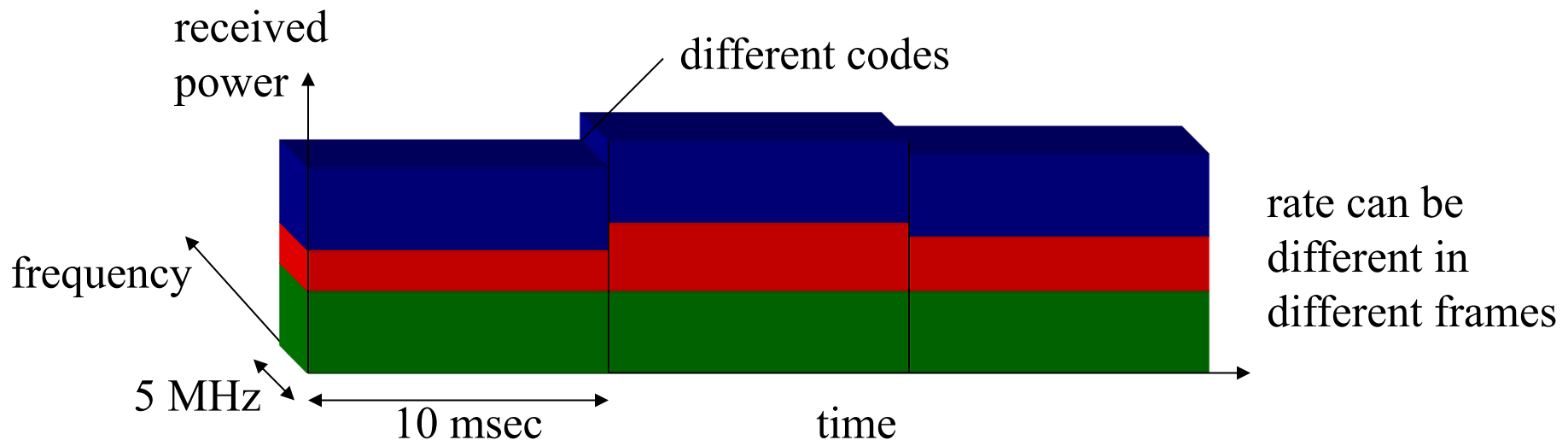
- Radio Resource Control
- Channel Allocation
- Power Control Settings
- Handover Control
- Ciphering
- Segmentation and reassembly



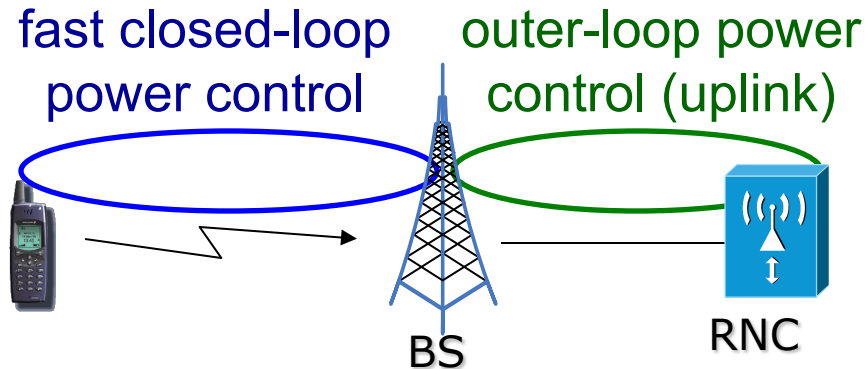
Note: Interfaces have been omitted for clarity purposes.

WCDMA

- W-CDMA uses the Direct Sequence-CDMA channel access method with 5 MHz wide channels
- Signals from different mobiles are separated based on **unique codes**
- **Transmission rate** can change between frames
 - Bandwidth $\geq 5\text{MHz}$
 - Multirate: variable spreading and multicode
 - Power control:
 - open power control
 - fast closed-loop power control
 - Frame length: 10ms/20ms (optional)



Power control in WCDMA



- Fast closed-loop power control: between MS and BS
 - Adjusts transmission power to achieve target signal quality (Signal-to-Interference Ratio, SIR)
 - Both uplink & downlink, frequency: 1500 Hz
- Uplink outer-loop power control: between BS and RNC
 - Adjusts target SIR to achieve given frame error rate (data: 10-20%, voice: 1%), frequency < 100 Hz
- Increase power when interference increases
- Diverge when signal qualities are infeasible

3.5G (HSPA)

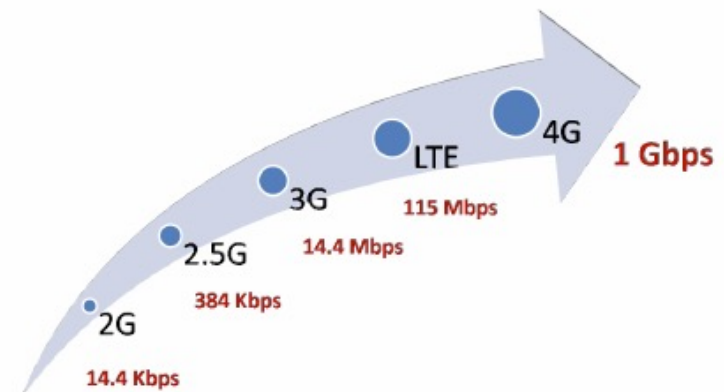
- High Speed Packet Access (HSPA) is an amalgamation of two mobile telephony protocols, High Speed Downlink Packet Access (HSDPA) and High Speed Uplink Packet Access (HSUPA), that extends and improves the performance of existing [WCDMA](#) protocols
- 3.5G introduces many new features that will enhance the UMTS technology in future. 1xEV-DV already supports most of the features that will be provided in 3.5G. These include:
 - Adaptive Modulation and Coding
 - Fast Scheduling
 - Backward compatibility with 3G
 - Enhanced Air Interface

4G (LTE)

- LTE stands for Long Term Evolution
- Next Generation mobile broadband technology
- Offers data transfer rates of 100 Mbps
- Based on UMTS 3G technology
- Optimized for All-IP traffic



2G – 4G Data download rates



• 2.5G speed is based on the maximum offered by EDGE
• 3G speed is based on the maximum offered by HSDPA

Advantages of LTE

- ▶ High network throughput
- ▶ Low latency
- ▶ Plug & Play architecture
- ▶ Low Operating Costs
- ▶ All-IP network
- ▶ Simplified upgrade path from 3G networks

for Network Operators

- ▶ Faster data downloads/uploads
- ▶ Improved response for applications
- ▶ Improved end-user experience

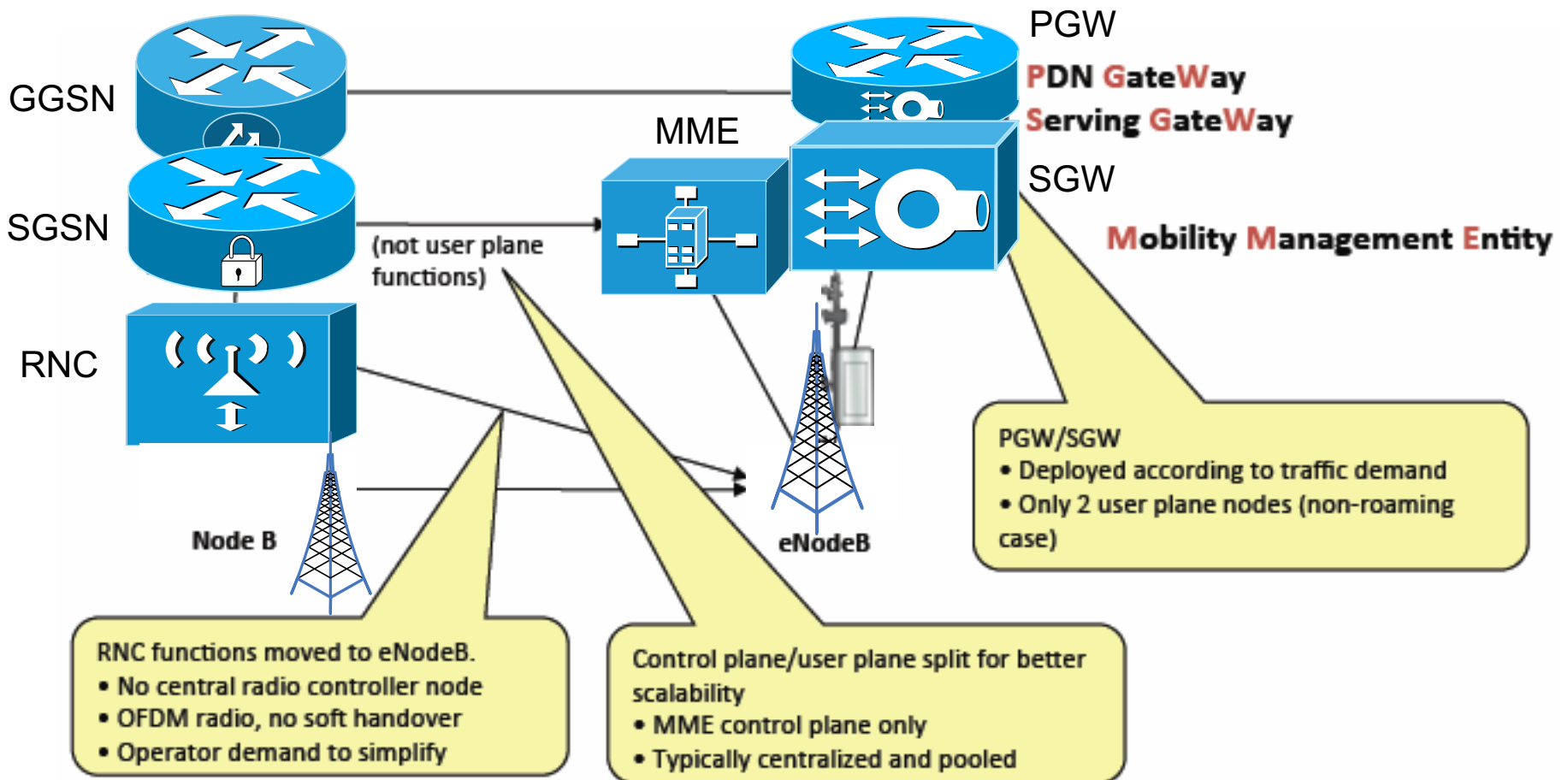
for End Users

Major LTE Radio Technologies

- Uses Orthogonal Frequency Division Multiplexing (OFDM) for downlink
- Uses Single Carrier Frequency Division Multiple Access (SC-FDMA) for uplink
- Uses Multi-input Multi-output (MIMO) for enhanced throughput
- Reduced power consumption
- Higher RF power amplifier efficiency (less battery power used by handsets)

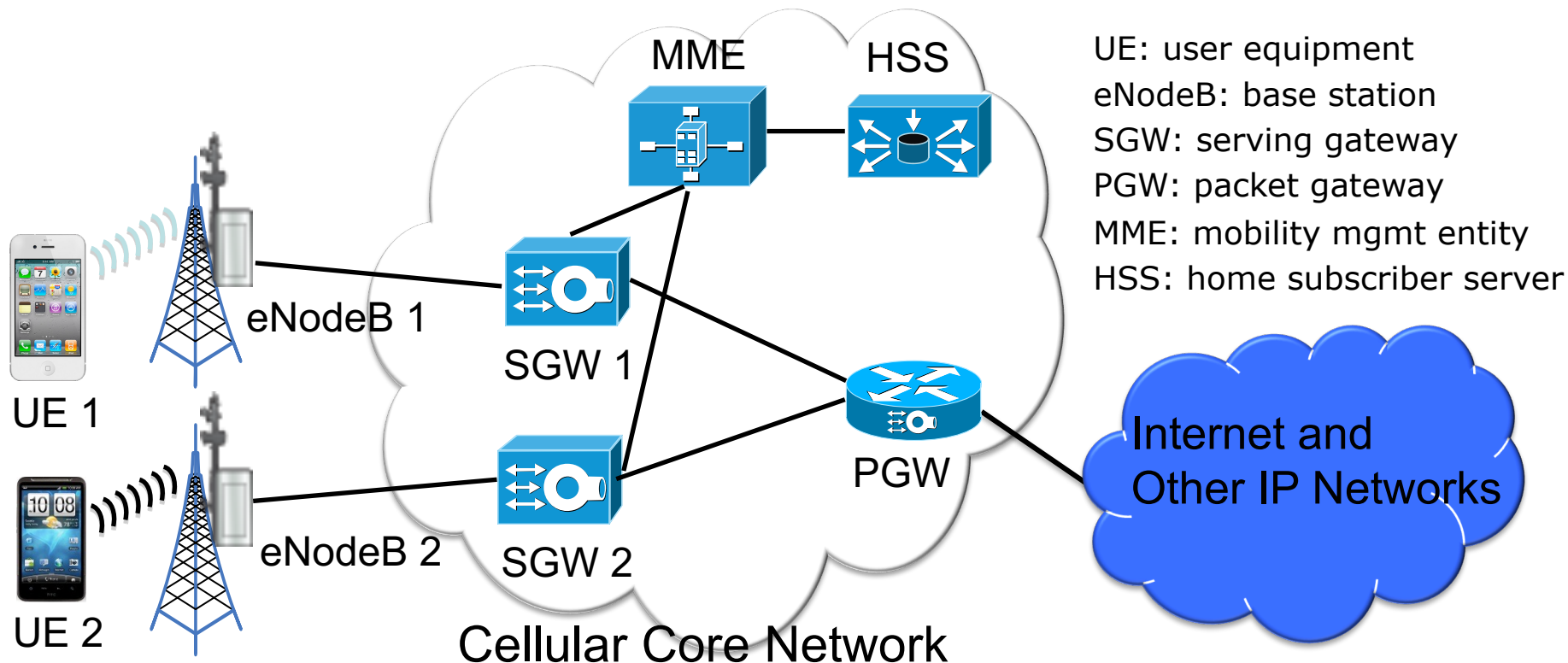
- LTE uses "all in one" approach, with everything over IP

- RNC functions moved on eNodeB
- SGW (Serving Gateway) replaced SGSN, PGW replaces GGSN
- Mobility management function move from SGSN to MME
- HLR/AUC collapses into HSS



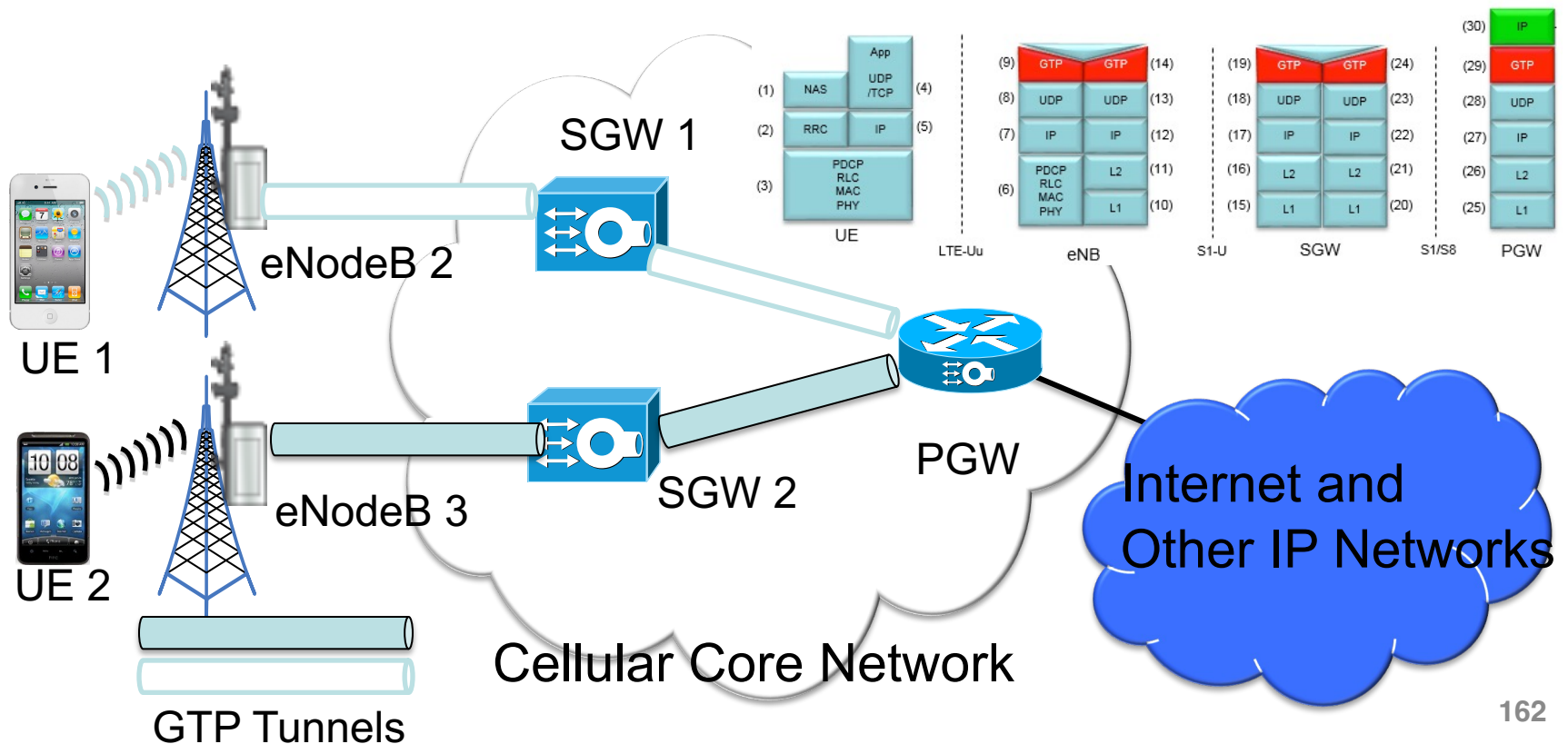
LTE Overall Architecture

- SGW is the anchor point for data connections between UE and PGW
- PGW interfaces the LTE network with the Internet
- MME is the key control node for managing mobility. Performs location tracking and UE authentication via HSS
- HSS manages subscription information and supports MME in user authentication



LTE Data Transfer

- GTP-tunneled connections transport UE data to the PGW
 - GPRS Tunneling Protocol (GTP) is an IP-based communications protocol used to carry general packet radio service (GPRS) within GSM, UMTS, LTE and 5G NR radio networks
 - MPLS LSPs can replace GTP tunnels in traffic-engineered backbones

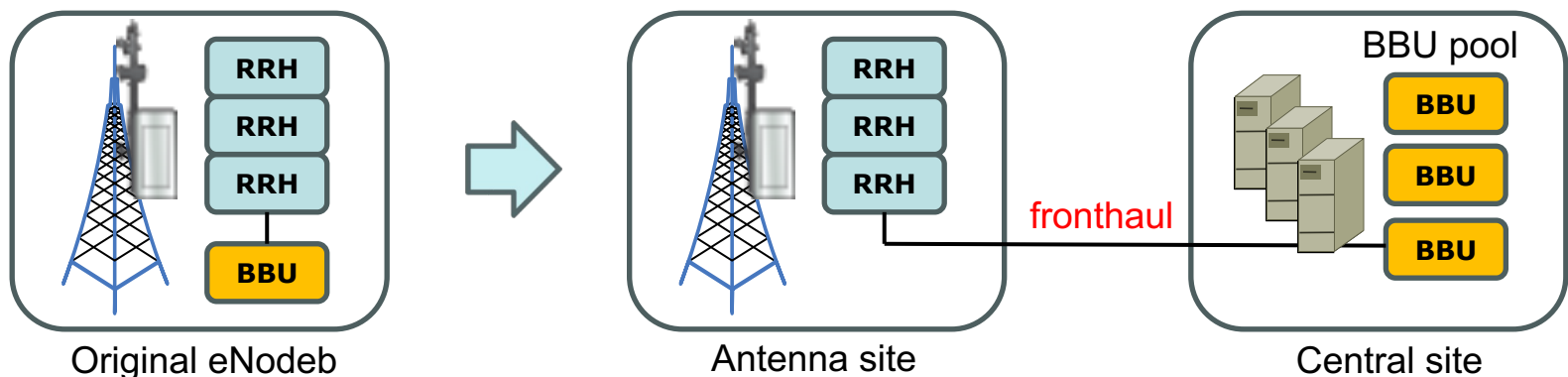


5G Architecture

- No More Cells
 - cell-centric systems do not match traffic variations and environments
- User-Centric Design
 - Decoupled signaling and data
 - Decoupled DL and UL
- New new RF bands
 - new RF bands (sub-1GHz, mmWave: 24/28 GHz, 30-90GHz)
 - Much spectrum available at higher frequencies
 - smaller cells and densification
 - use of licensed/unlicensed unshared/shared spectrum
- New antenna systems
 - Massive (16 or more, up to 256) MIMO
 - Replacement of macro BS with arrays of low-gain resonant antennas
- Paradigm shift to C-RAN
 - Decoupling between radio and baseband management functions

5G node split

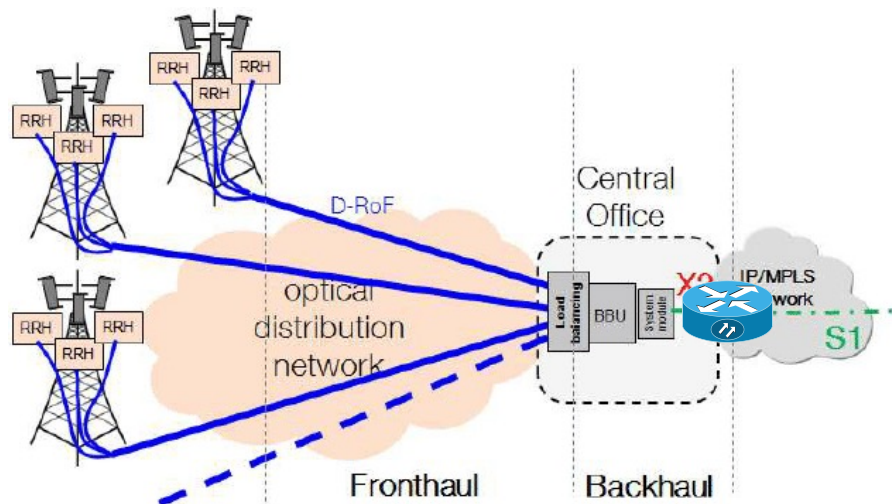
- A traditional base station (eNodeB) is made up of one BBU (Base Band Unit) and several Remote Radio Head (RRH).
 - The RRH is the RF circuitry mounted near the antenna
 - It performs all RF functions like transmit and receive functions, filtering, and amplification. It is connected to the BBU with an optical fiber
 - The BBU converts data into a digital signal and sends it to the RRH, which then converts it into an analog signal. It also handles MAC.
 - It is responsible for communication of physical interface to the core network.
- In 5G Base station architecture evolved from all-in-one base stations, including BBU and RRHs in a single site, to C-RAN architecture where RRH equipment at the cell sites (RRHs) connect via fronthaul network to centralized BBU resources



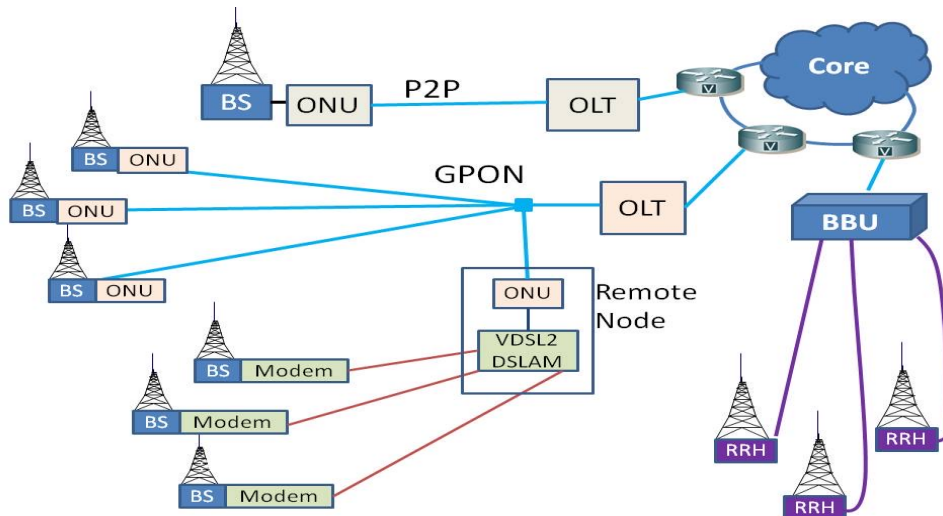
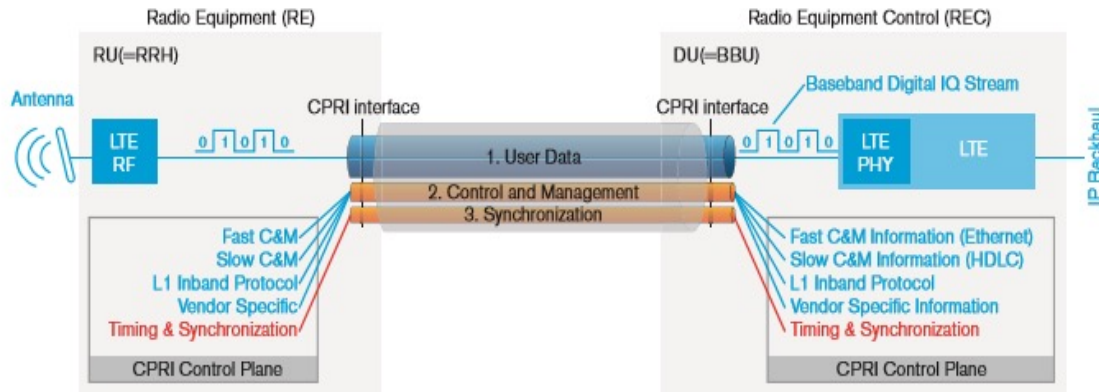
C-RAN architecture

- C-RAN brings Baseband units (BBUs), processing radio signals, from BSs to a central pooled location (maybe in the edge/cloud)
- Only physical radio functions and high speed sampling activities, the Remote Radio Head (RRH) facilities, are left at the BS side
- A Common Public Radio Interface (CPRI) enables the interworking through the fronthaul infrastructure between RRHs and BBUs
- An high speed MPLS-enabled fiber backhaul system enables high speed communication between the service hubs and to the cloud.

More scalability and agility in the management of new services !!!



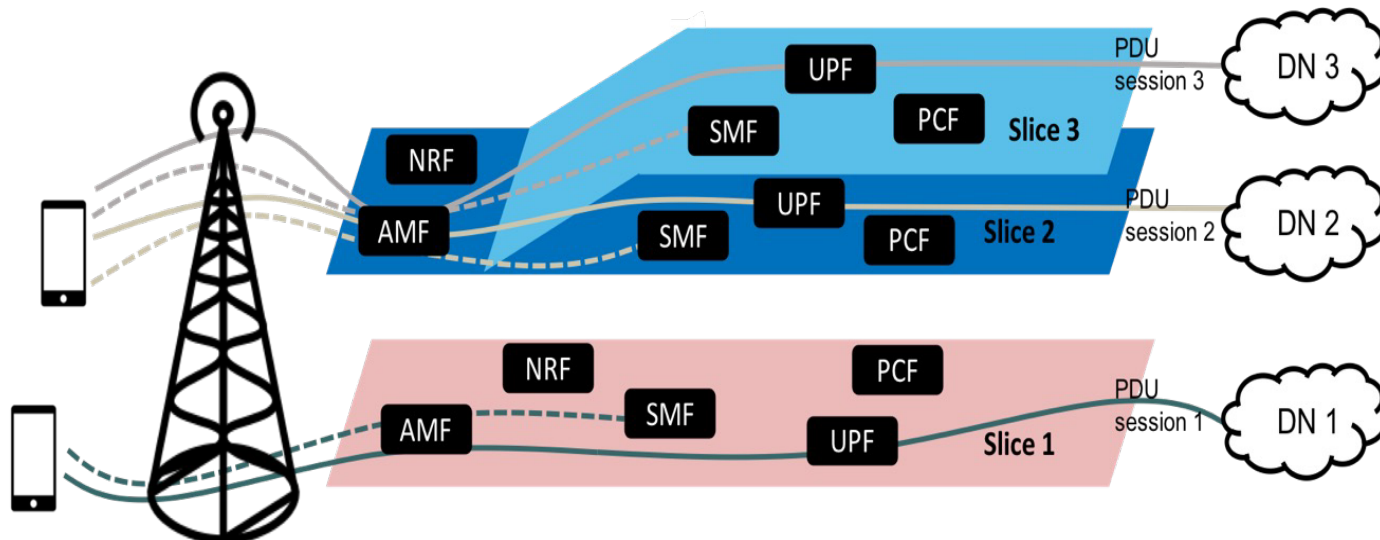
C-RAN Evolution



- C-RAN may evolve from BBU centralization to BBU virtualization in the cloud
- By decoupling BBU capabilities from physical HW we can achieve processing capacity on demand
- Cells may be engineered!
- Traffic-engineered (MPLS-GMPLS) optical transport backbones provide the needed guarantees for end-to-end services

Network Slicing

- support coexistence of traffic with wildly diverging requirements
- use network slicing to satisfy these divergent needs
- Slices are dynamically established strong-isolated networks
- Slices can support:
 - on-demand (i.e., dynamic) assignment of resources
 - bandwidth, forwarding tables, processing capacity, etc.
 - each slice acts as an end-to-end independent network
 - resources can be physical or virtual, but each slice acts as a strongly isolated network or cloud isolation of management, security, performance



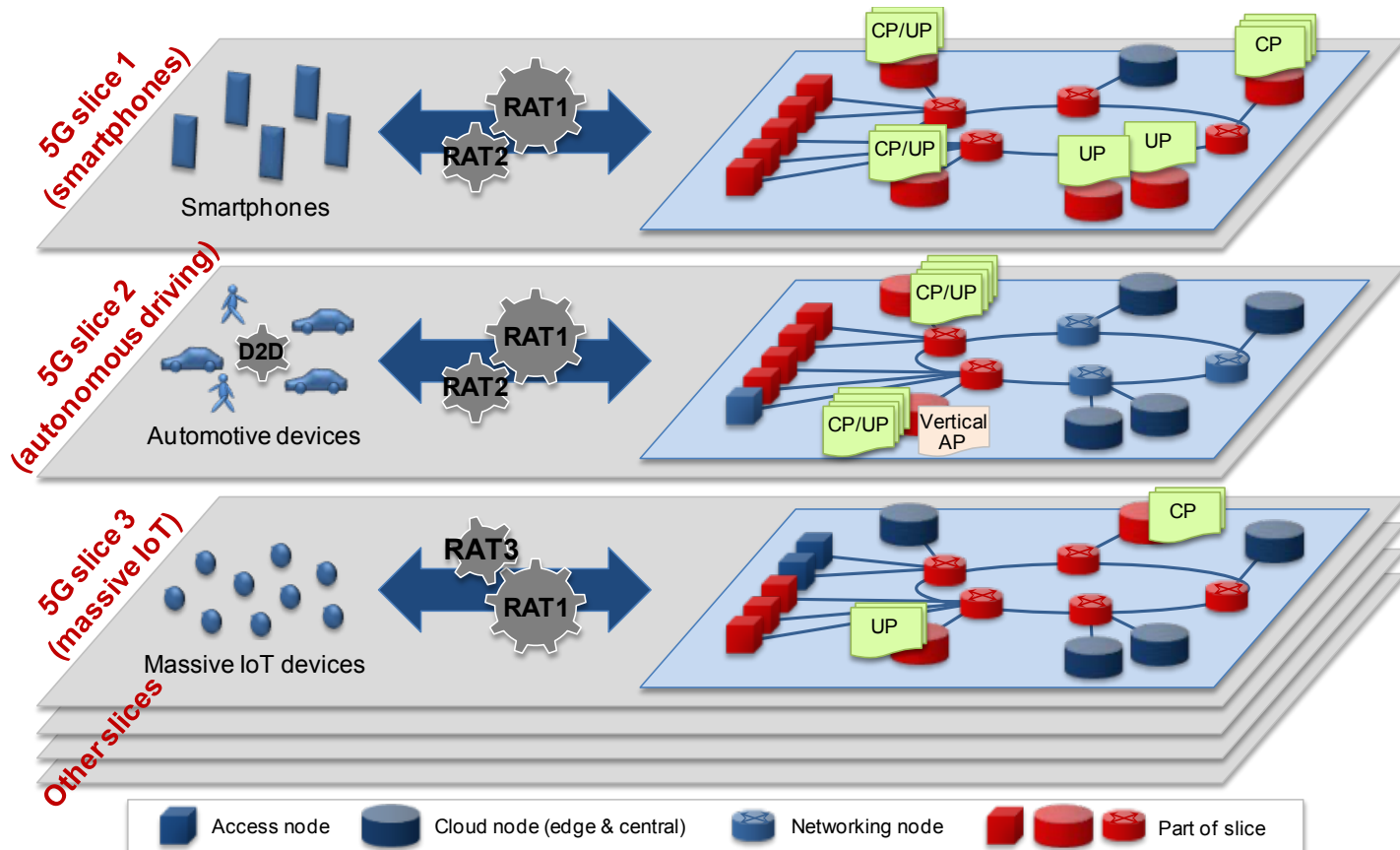
Network Slicing

– Usage examples

- enhanced mobile broadband
 - needs high data rates
 - doesn't need very low latency

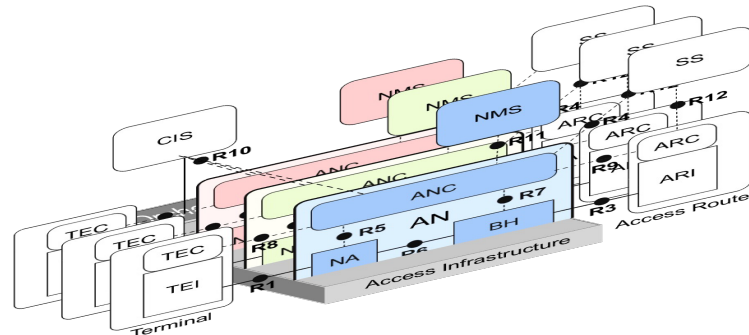
- massive IoT

- needs high connection density
- doesn't need high data rates



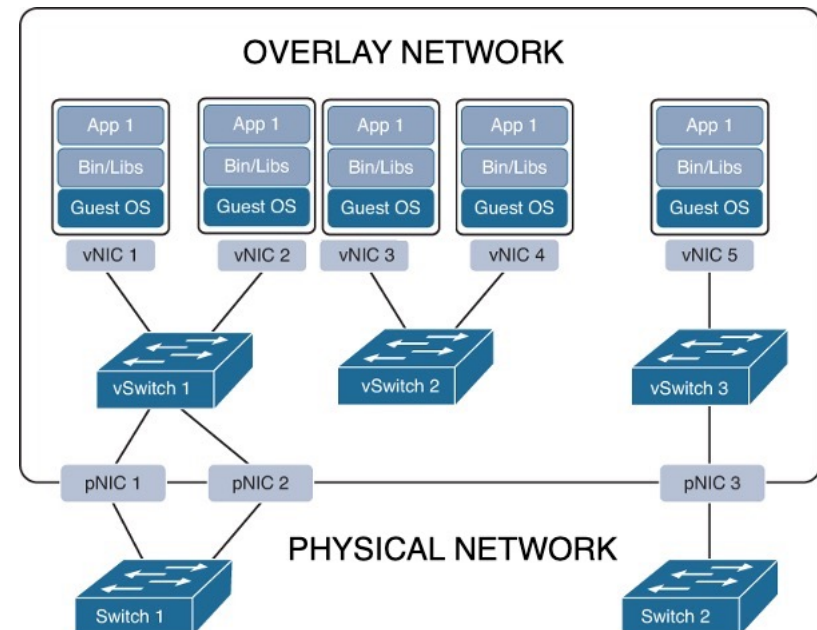
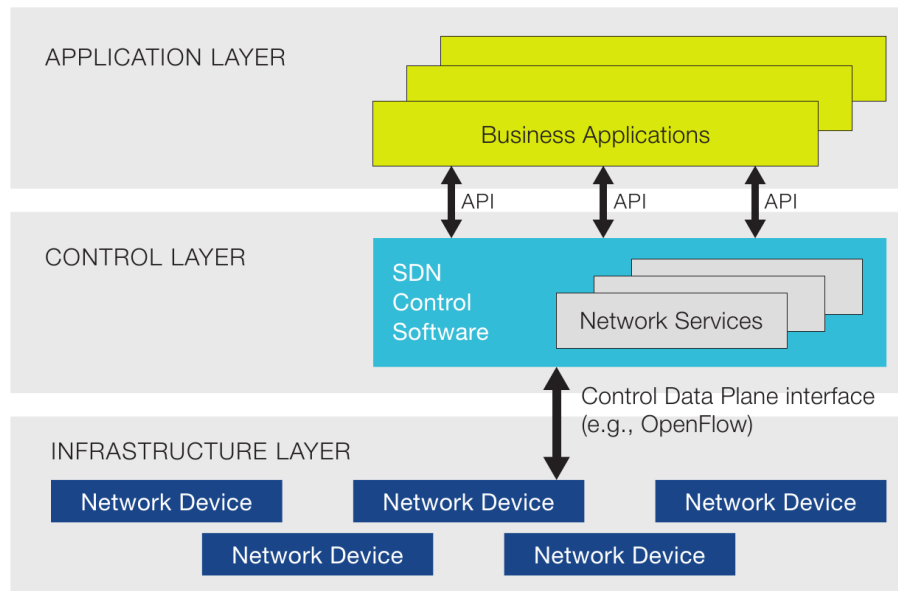
Implementing slicing

- Slicing requires programmability, flexibility, and modularity in order to create multiple virtual networks over a common network
- Different slices can be labeled using VLAN, VXLAN, IPv6 flow label, DSCP
 - but a slice is different from a separate physical network or a VPN because of dynamic set-up / release requirements and separate management
- SDN (*network softwarization*) are used to achieve slicing
 - APIs provided to specify requirements
 - SDN advocates replacing standardized networking protocols with centralized software applications that configure whitebox switches in the network
 - SR achieves SDN-like control over packet forwarding by adding routing information to packet headers
 - NFV advocates replacing hardware network elements with software running on whitebox servers that may be housed in data-centers or POPs or cell-sites
 - client/server with RESTful interfaces (CRUD operations)
 - use of global orchestrator for rapid yet highly optimized slice creation



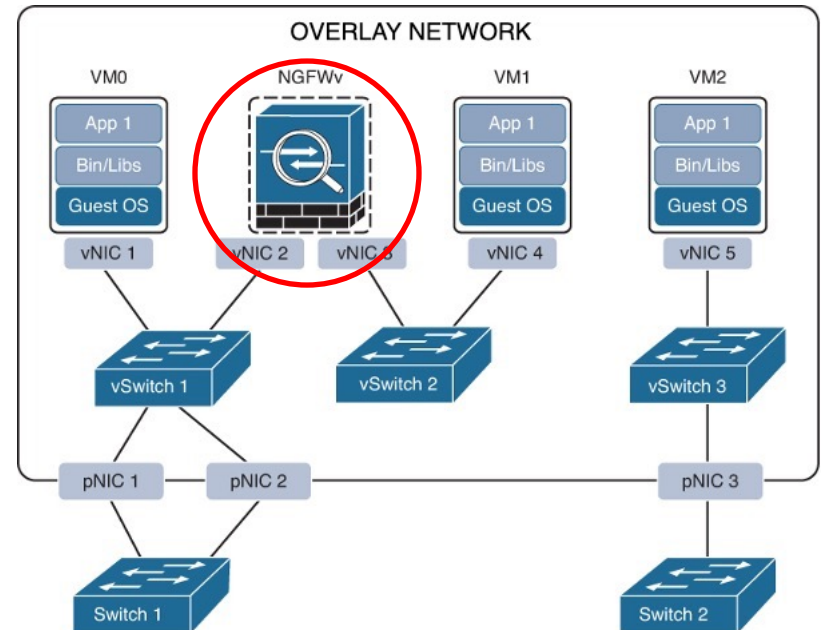
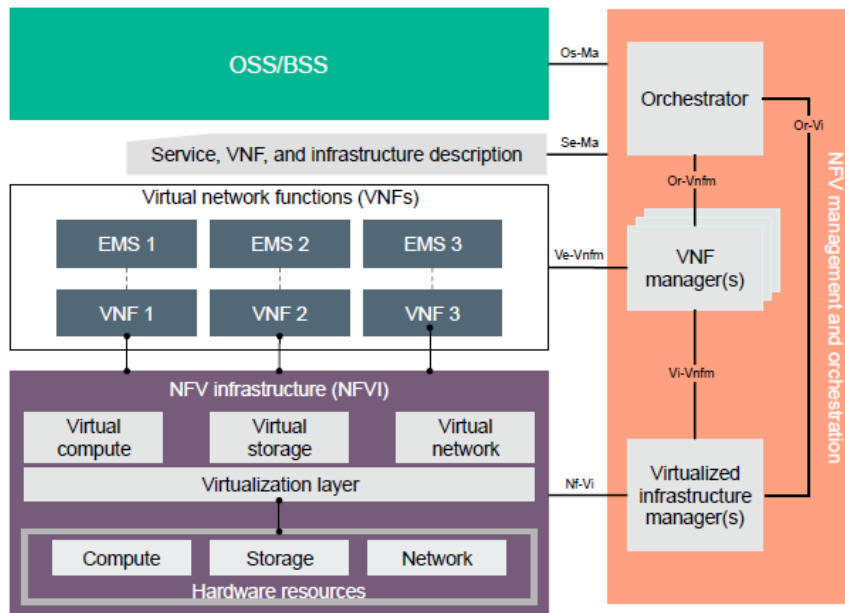
Software Defined Networks

- SDN transform a classic HW-based infrastructure into a new SW-based network.
- The concept is to create network HW abstraction under the control of the SW (network virtualization).
- The underlying physical infrastructure remains, but SDN allow to use software to create virtual networks on top of the physical one



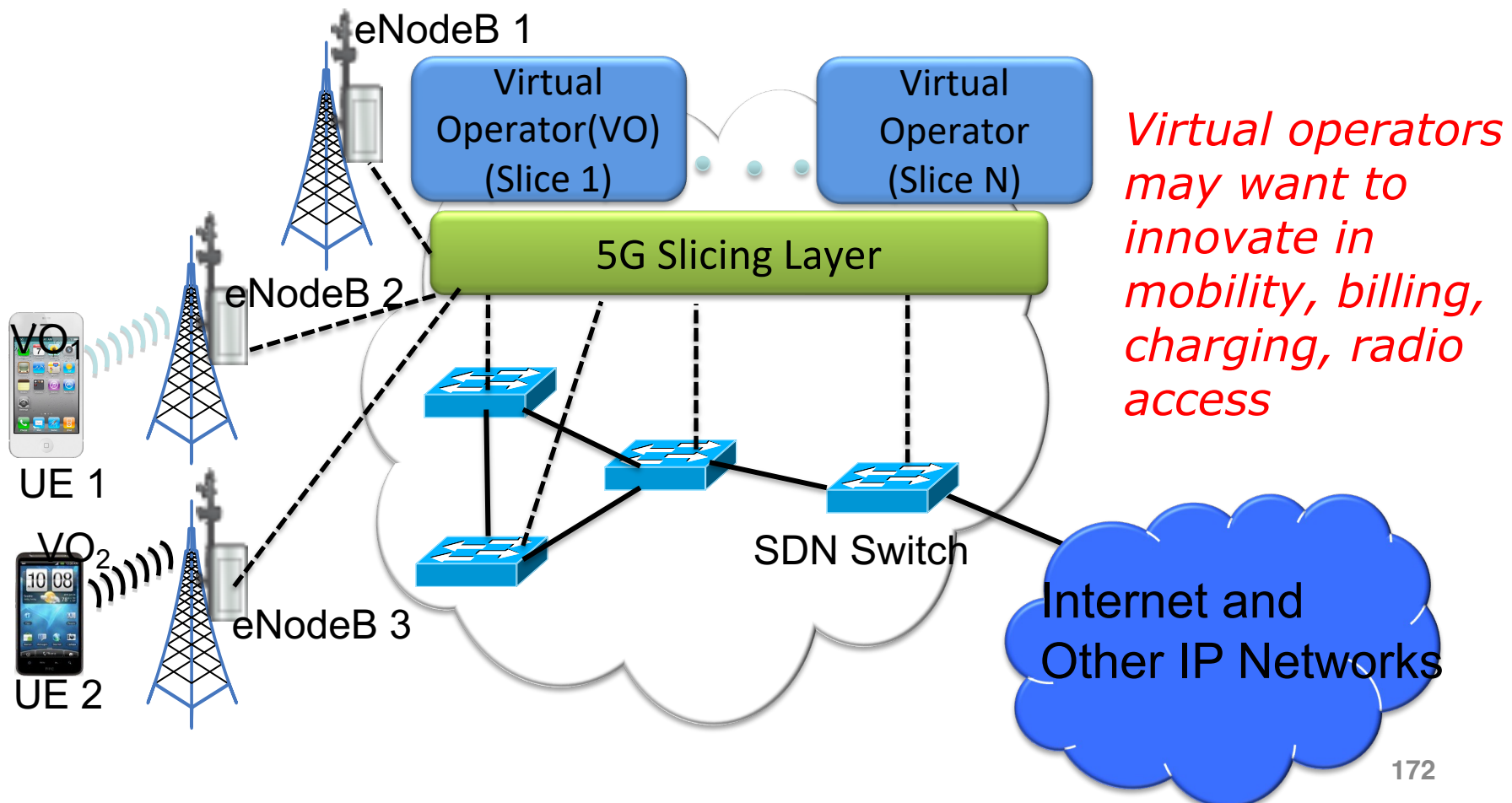
Network Functions Virtualization

- Network function virtualization (NFV) defines the standards for decoupling network functions from proprietary hardware by moving them to virtual machines, containers, etc.
- It also defines how to properly manage and orchestrate network functions, referring to the function performed by a physical device
- It becomes possible to "delocalize" the main network functions



Virtual Operators

- Flexible network virtualization by slicing
- Each operator has its own policies and resource management



Cellular IoT vs other LPWAN technologies

- 5G cellular has a combination of characteristics that makes it an better choice for IoT.
- Comparing with other LPWAN technologies like SigFox and Lora, Cellular IoT offers better performance in terms of:
 - global coverage,
 - More bandwidth and QoS,
 - Scalability
 - flexibility
- Most importantly unlicensed solutions can't guarantee reliability and security as they have to compete for bandwidth with other wireless networks.
- Cellular IoT has the advantage of reuseing existing network resource, which will help to reduce operational costs in areas like service provisioning, monitoring and billing.

5G in the IoT contest

- Smart grid, tactile, robotic, Automotive, domotic,...
- Massive data
- Different solutions:
 - Based on 2G, 3G, 4G (LTE-MTC)
 - Proprietary solutions: i.e. *LPWA (Low Power Wide Area) Networks* (LoRa, SigFox, e WMBus)
- Different approach for storage and elaborations
- Security
- Distributed management
- Distributed legend: Blockchain

Cellular IoT Use case evolution

- 5G opens a big opportunity for mobile industry and enables a huge number of new IoT use cases
- The use cases will be very diverse, from connecting billions of devices with very low energy consumption and low cost (massive IoT) to ultra reliability and very low latency (critical IoT)
- The most interesting scenarios are:

